Structure-Function analysis of the *Photorhabdus* Tc toxin complex



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Dedication

I dedicate this work to my family and community. May it be another step towards the achievement of our collective hopes and dreams.

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Table Contents

List of abbreviations xi
Abstract xiii
1. **Introduction** 1
1.1. Entomopathogenic nematodes 1
1.2. *Photorhabdus* spp 2
1.3. The *Photorhabdus* Tc toxins 3
1.3.1. Structure of TcA-like proteins 4
1.3.2. TcB and TcC-like proteins 8
1.4. Rho GTPases and the toxins that target them 13
1.4.1. Rho GTPases 13
1.4.2. Rho modifying toxins 16
1.5. Objectives of the project 22
2. **Experimental procedures** 25
2.1. **Materials** 25
2.1.1. Bacteria and culture media 25
2.1.2. Cell lines and growth media 25
2.1.3. Vectors and plasmids 26
2.1.4. Oligonucleotides 27
2.1.5. Antibodies 28
2.1.6. Cloning and expression work 29
2.1.7. Proteins and protein toxins 29
2.1.8. Radiolabeled substrates and isotopes 30
2.1.9. Other reagents 30
2.1.10. Kits 31
2.1.11. Laboratory equipment 31
2.1.12. Software 33
2.2. **Methods** 33
2.2.2. Polymerase chain reaction (PCR) 33
2.2.3. Agarose gel electrophoresis and gel extraction of DNA 33
2.2.4. Cloning and transformation 34
2.2.5. DNA sequencing 35
2.2.6. Protein expression and purification of Photorhabdus toxins 35
2.2.7. Purification of PrtA1 36
2.2.8. Sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) 37
2.2.9. Western blot 38

2.2.10. Cell cytotoxicity assays 39
2.2.11. Transepithelial electrical resistance (TEER) assays 39
2.2.12. Fluorescence-activated cell sorting (FACS) analysis 40
2.2.13. ADP-ribosylation of actin and RhoA 40
2.2.14. Glycosylation reaction 41
2.2.15. Membrane-release assays 41
2.2.16. Inhibition of cellular stress fibers 42
2.2.17. Bioluminescence assays 42
2.2.18. Data analysis and representation 44
3. **Results** 47
3.1. Intoxication by the *Photorhabdus* Tc toxin leads to cell rounding 47
3.2. TcdA1 is activated by proteolytic cleavage 48
3.2.1. TcdA1 is activated by a *Photorhabdus* metalloprotease (PrtA1) 48
3.2.2. Increased toxicity is specific to cleavage by metalloproteases 53
3.2.3. High pH further increases binding of cleaved-toxin 57
3.2.4. Cleaved toxin is no longer affected by bafilomycin A1 and dynasore 58
3.3. The *Photorhabdus* Tc toxin: an engineered peptide delivery nanomachine 60
3.3.1. The Tc toxin can deliver a foreign protein into target cells 60
3.3.2. Influence of the BC3 cocoon on the enclosed enzyme 62
3.3.3. Delivery of C3bot into target cells by the Tc toxin 64
3.3.4. Delivery of peptides through PTC depends on the size and charge of the cargo 65
3.3.5. The Type III secretion effector YopT is injected through PTC into target cells 66
3.3.6. BC3-chimeras of PaTox^{GTD} and Toxin B^{GTD} showed activity *in vitro* but were not delivered into cells 70
3.3.7. Quantification of the injection efficiency of the PTC3 nanomachine 72
3.4. The *Photorhabdus* TcdA1 binds to N-linked glycans on the cell surface 77
3.4.1. TcdA1 binds to fucosylated Lewis X and Y N-glycans 77
3.4.2. N-glycan deficient cells are resistant to intoxication by PTC3 81
3.4.3. Exogenous addition of Lewis X trisaccharides reduced intoxication kinetics 83
3.4.4. TcdA1 may interact with glycosaminoglycans (GAGs) 85
3.4.5. Sialic acid-, galactose-, fucose- and glycolipid-deficient cells were sensitive to PTC3 87
3.4.6. Exogenous addition of sialic acid, but not other monosaccharides, inhibited cell intoxication 88
3.4.7. TcdA1 may bind to a protein (co)receptor 92
4. **Discussion** 95
4.1. Activation of the TcA protein by metalloproteases leads to increased activity 95
4.2. The PTC3: a molecular delivery tool for foreign peptides 99

4.3. TcdA1 binds to cell surface N-glycans 102
5. **Suggestions for further research** 107
6. **Appendix** 109
7. **References** 123

List of abbreviations

A	alanine
ADP	adenosine-5'-diphosphate
amp	ampicillin
APS	ammonium persulfate
ATase	adenylyltransferase
ATP	adenosine-5'-triphosphate
BC3	TcdB2-TccC3 complex
bp	base pair
BSA	bovine serum albumin
C	cysteine
CS	chondroitin sulfate
CT	cholera toxin
CTxB	subunit B of cholera toxin
D	aspartic acid
DAPI	4’,6-diamidin-2-phenylindol
DexS	dextran sulfate
DMEM	Dulbecco’s modified Eagle medium
DMSO	dimethyl sulfoxide
DNA	deoxyribonucleic acid
dNTP	deoxynucleoside triphosphate
DTT	dithiothreitol
ECIS	electrical cell impedance sensing
EDTA	ethylenediaminetetraacetate
EPEC	enteropathogenic *Escherichia coli*
EPN	entomopathogenic nematode
et al.	et alii
ER	endoplasmic reticulum
EGFR	epidermal growth factor receptor
F	phenylalanine
FCS	fetal calf serum
FL	full-length
FPLC	fast protein liquid chromatography
Fuc	fucose
g	g-force
G	glycine
Gal	galactose
GalNAc	N-acetyl galactosamine
GAP	GTPase-activating protein
GDI	guanine nucleotide dissociation inhibitor
GDP	guanosine-5’-diphosphate
GEF	guanine nucleotide exchange factor
GFP	green fluorescence protein
GlcNAc	N-acetyl glucosamine
Glc	glucose
Gln	glutamine
GST	glutathione S-transferase
GT	glycosyltransferase
GT-A	glycosyltransferase type A
GTD	glycosyltransferase domain
GTP	guanosine-5’-triphosphate
GTPase	GTP hydrolyzing protein
h	hour/hours
H (His)	histidine
HEPES	2-[4-(2-hydroxyethyl) piperazine-1-yl] ethane sulfonic acid
HP	heparin
HPLC	high-performance liquid chromatography

HRP	horseradish-peroxidase	PTC	*Photorhabdus* toxin complex
HS	heparan sulfate	PTM	post-translational modification
IPTG	isopropyl-ß-D-thiogalactopyranosid	PVDF	polyvinylidenefluoride
K	lysine	Q	glutamine
kb	kilo base pairs	R	arginine
K_D	equilibrium constant	RNA	ribonucleic acid
kDa	kilo Dalton	RT	room temperature
L	leucine	S	serine
LB	lysogeny broth	SA	sialic acid
LPA	lysophosphatidic acid	SD	standard deviation
LPC	lysophosphocholine	SDS	sodium dodecyl sulfate
M	methionine	SEM	standard error of the means
M	molar	T	threonine
Man	mannose	Tc	toxin complex
min	minutes	TcA	toxin complex A
Mluc7	*Metridia* luciferase 7	T3SS	type 3 secretion system
MT	microtubule	TAE	tris acetate EDTA
MW	molecular weight	TBS	tris buffered saline
N	asparagine	TEMED	tetramethylethylenediamine
NS	not significant	TEER	transepithelial electrical resistance
PA	phosphatidic acid	Tris	tris(hydroxymethyl)-aminomethane
PAGE	polyacrylamide gel electrophoresis	UDP	uridine diphosphate
PBS	phosphate buffered saline	V	valine
PCR	polymerase chain reaction	W	tryptophan
PE	phosphatidylethanolamine	Xyl	xylose
PEG	polyethylene glycol	WB	Western blot
PI	phosphatidylinositol	WT	wild type
PIP	phosphatidylinositol phosphate	Y	tyrosine
PFA	*para*-formaldehyde		
PMSF	phenylmethylsulfonyl fluoride		
PS	phosphatidylserine		

Abstract

Photorhabdus are gram negative, bioluminescent, entomopathogenic bacteria. They are mostly found in symbiosis with entomopathogenic nematodes (EPNs). As pathogens, they produce a variety of toxins, among which are the large tripartite ABC-type *Photorhabdus* toxin complexes (PTCs), also known as Tc toxins. PTCs show efficacy against insect, mammalian and human cells. The complexes consist of three proteins, TcA, TcB and TcC, which combine for biological activity. TcA is the binding and translocation component. TcB links TcA to TcC, which contains the enzymatic component responsible for cytopathic and cytotoxic effects (TcChvr).

This study aimed to elucidate the structure-functional characteristics of the *Photorhabdus* Tc toxin. Specifically, we aimed to (1) elucidate the binding and activation characteristics of TcdA1, (2) to characterize the translocation mechanism of protein cargo through the complex, and (3) to determine and analyse putative receptors of the complex. In this regard, we show that the metalloproteases PrtA1 and collagenase can activate TcdA1. This activation leads to increased binding on both human and insect cells and increases intoxication kinetics 8 – 10-fold. Unlike uncleaved TcdA1, processed and thereby activated TcdA1 is not inhibited by the protease inhibitor MG132. Moreover, bafilomycin A1 and dynasore, which inhibit the action of unprocessed toxin, most likely by impairing uptake into cells, have no effects on the processed toxin.

We also show that the PTC-injection machinery can be used to deliver foreign peptides into target cells. Here, the native TcChvr was replaced with other toxic and non-toxic enzymes, or enzyme fragments, including the *Clostridium botulinum* C3 toxin, *Yersinia enterocolitica* effector, YopT and *Metridia longa* luciferase, Mluc7. The tripartite complexes of the toxin chimeras preserved enzyme activity and delivered the cargoes into target cells. Furthermore, Mluc7 enabled quantification of the efficiency of PTC. Overall, using these toxin chimeras, we determine the size and charge of the cargo optimal for efficient delivery through the injection machinery.

Finally, in the search for cell-surface receptors of PTC, a glycan microarray shows that the TcA protein (TcdA1) interacts with N-glycans, particularly Lewis X/Y antigens. This is confirmed using N-glycan deficient CHO-Lec1 cells, which are resistant to intoxication by PTC3. Additionally, exogenous addition of a Lewis X trisaccharide inhibits intoxication. Furthermore, the glycosaminoglycans, heparin and heparan sulfate, also interact with TcdA1. They too

inhibit intoxication. This data suggests that TcdA1-binding is multivalent and these sugars probably work cooperatively to facilitate binding and intoxication.

1. Introduction

1.1. Entomopathogenic nematodes

Entomopathogenic nematodes (EPNs) are a group of threadworms that target and kill insects through a symbiotic relationship with insect pathogenic bacteria. With help from the bacteria, they subdue and consume target insect prey (Gaugler, 2002; Boemare & Akhurst, 2006). They gained prominence after recognition of increased cases of resistance and the negative environmental impacts of chemical pesticides. This resulted in the discovery and development of *Steinernema glaseri* as a biological control agent in the early 20th century, thus opening up the space for extensive research on entomopathogenicity (Gaugler, 2002). Their ubiquity and efficiency in killing a variety of insect species validated their candidacy in biological control (Fischer-Le Saux *et al.*, 1999; Waterfield, Ciche and Clarke, 2009; Mulley *et al.*, 2015). Of major interest, have been the genera *Heterorhabditis* and *Steinernema*, the hosts of *Photorhabdus* and *Xenorhabdus* spp. respectively (Boemare & Akhurst, 2006).

The bacteria-EPN complex is very critical to the life cycle of both nematodes and bacteria. However, the bacteria have shown phenotypic heterogeneity, in that, they exist in two forms, primary (1°) and secondary (2°), which show a characteristic difference in their association with EPNs (Akhurst, 1980; Joyce and Clarke, 2003). The 1° bacteria are obligate symbionts which do not survive well without their nematode hosts, while the 2° are unable to associate with EPNs and are, therefore, putatively soil-dwelling (Akhurst, 1980; Eckstein *et al*., 2019; Han and Ehlers, 2001; Joyce and Clarke, 2003; Joyce *et al*., 2011; Smigielski *et al*., 1994). Nevertheless, the life cycle of the complex generally involves both symbiosis with each other and pathogenesis to insects (Boemare and Akhurst, 2006).

The infective stage of the EPNs (Infective juvenile (IJ)), either searches (cruiser species) or waits (ambusher species) for insect prey (Gaugler, 1993). The EPNs then penetrate the insect body, either by burrowing through the cuticle, or entering through natural openings (Bowen *et al*., 1998). Once in the haemocoel, they release bacteria from their gut by regurgitation (Ciche and Ensign, 2003). The bacteria in turn inhibit insect immunity, kill the insect and then begin to bio-convert its tissues as a nutrient source for the proliferation of the nematodes and themselves (Boemare and Akhurst, 2006; Hinchliffe *et al*., 2010). The bacteria also produce a variety of small molecule antibiotics to protect the cadaver from invading microorganisms (Bode, 2009). When

the resources are depleted, the bacteria and nematodes re-associate and leave the spent cadaver in search of new targets (Bowen *et al.*, 1998; Gaugler, 2002; Boemare and Akhurst, 2006).

1.2. *Photorhabdus* spp.

Photorhabdus species are gram-negative, bioluminescent bacteria of the family Enterobacteriacea. The genus contains three described species: *P. luminescens*, *P. temperata* and *P. asymbiotica*. All are known insect pathogens; however, *P. asymbiotica* has also been identitifed as a human pathogen (Ffrench-Constant *et al*., 2003; Fischer-Le Saux *et al*., 1999; Mulley *et al*., 2015). The genes responsible for both insect and human pathogenicity are encoded in the same pathogenicity island. It is therefore likely that many of the *Photorhabdus* virulence factors may be equally appropriate for insect and human infections (Hinchliffe *et al.*, 2010; Mulley *et al.*, 2015). Indeed, a set of *P. luminescens* toxin complexes efficient against insect pests (Bowen *et al*., 1998; Liu *et al*., 2003; Silva *et al*., 2002), were also found to intoxicate mammalian cell lines using a similar mechanism (Hinchliffe *et al*., 2010; Lang *et al*., 2010; Simon *et al*., 2014). Additionally, some strains of *P. luminescens,* as well as all strains of *P. asymbiotica* thrive at 37 °C (Mulley *et al*., 2015).

Photorhabdus produce a wide variety of virulence factors including Type III secretion (T3S) system-dependent effectors and secreted toxins (Hinchliffe *et al*., 2010). In an infection model, the bacteria were shown to first colonise the midgut, where they occupy a specific niche between the extracellular matrix and basal lamina of the midgut epithelium (Silva *et al*., 2002). Here, the bacteria use the toxins to destroy the epithelium, for example, the gut-active Toxin complex A (TcA) and the RTX-like metalloprotease PrtA1. This destruction leads to a cessation in feeding in the insect, ultimately resulting in starvation. Then, prior to insect death, the bacteria mainly colonize the haemocoel, or are associated with the gut. After death, they spread to every available tissue in the cadaver and begin bioconversion to generate a ready nutrient source for replication of both bacteria and nematode (Boemare and Akhurst, 2006; Hinchliffe *et al*., 2010; Silva *et al*., 2002).

As previously mentioned, two phenotypically different forms of *Photorhabdus* exist, 1° and 2°. The 2° form spontaneously arises from the 1° in prolonged *in vitro* culture, but may also be present in the insect cadaver at the tail-end of the infection cycle (Akhurst, 1980; Boemare and Akhurst, 1988; Eckstein *et al*., 2019). This phenomenon was previously referred to as phase variation (Boemare and Akhurst, 1988). However, the significant differences between the two

variants, particularly the irreversible unidirectional change (1° to 2°) and the genetic differences between the two, negates classical phase variation and supports phenotypic variation/heterogeneity (Davidson and Surette, 2008; Eckstein and Heermann, 2019; Forst and Clarke, 2002; Joyce *et al*., 2011). Both variants grow equally well and are equally virulent when injected into insect larvae. However, only 1° cells can colonize the intestinal tract of the IJ stage of EPNs and support nematode growth and development (Akhurst, 1980; Boemare and Akhurst, 1988; Eckstein *et al*., 2019; Joyce and Clarke, 2003).

The exact mechanism modulating the switch from 1° to 2° and the regulation thereof, remains unknown (Eckstein *et al*., 2019). However, one gene involved in this process has been identified. It is a homologue of the *Erwinia carotovora* gene hexA, which represses 1°-associated phenotypes, that promote the mutualistic interaction with EPNs (Joyce and Clarke, 2003). Interestingly, similar homologues of hexA have also been found in *Xenorhabdus* and other Enterobacteriaceae like *Escherichia coli* (Joyce and Clarke, 2003). The affected 1°-associated phenotypes include; bioluminescence, presence of crystal inclusion proteins and production of antibiotics, protease, lipase and catalase. Additionally, morphological characteristics of the bacterial colonies on solid medium, as well as pigmentation and dye absorption, also differ between the two variants (Eckstein *et al*., 2019; Forst and Clarke, 2002; Joyce and Clarke, 2003). Moreover, in contrast to 1° cells, genes involved in metabolism, stress response, motility, and chemotaxis were upregulated in 2° cells. In fact, 2° cells showed a stronger chemotactic response to different plant-root exudates, thus suggesting that the soil and/or rhizosphere could be an alternative environment for them (Eckstein *et al*., 2019).

1.3. The *Photorhabdus* Tc toxins

Among the deadly *Photorhabdus* toxins, are the high molecular mass (1.7 MDa) insecticidal toxin complexes (also known as Tc toxins or *Photorhabdus* toxin complexes (PTCs)), which are essential for toxicity and symbiosis (Bowen *et al*., 1998; Ffrench-Constant *et al*., 2003). Tc-like genes are widespread among bacteria and have been found in several species including the closely related *Xenorhabdus* spp. (Sergeant *et al*., 2003), as well as in bacteria not associated with nematodes like *Serratia entomophila* (Hurst *et al*., 2000), *Pseudomonas syringae* (Guttman *et al*., 2002) and *Yersina* spp. (Fuchs *et al.*, 2008; Piper *et al.*, 2019). Tc loci are also found in gram-positive bacteria such as *Paenibacillus* spp. (ffrench-Constant and Waterfield, 2005) and *Bacillus thuringiensis* (Blackburn *et al*., 2011). This indicates a crucial and conserved role of the toxins in the pathogenicity of these bacteria.

The PTCs are large tripartite ABC-type toxin complexes (ffrench-Constant and Waterfield, 2005). They are typically composed of three toxin types:- TcA type (tcaA/B, tcbA, tccA/B and tcdA), TcB type (tcaC and tcdB), and TcC type (tccC), which combine into a biologically active ABC complex (ffrench-Constant and Waterfield, 2005; Roderer and Raunser, 2019) (Fig. 1). They are produced in different isoforms to target a wide array of insect species (Boemare and Akhurst, 2006; ffrench-Constant and Waterfield, 2005; Forst *et al*., 1997). For instance, up to seven TcA- and TcC-type genes have been identified in *P. luminescence* strain TTO1 so far, whereas three TcB-type genes have been found (Roderer and Raunser, 2019). Additionally, some TcA genes exist as single genes encoding large proteins (2,100–2,600 residues), while other TcAs consist of two genes located next to each other on the genome (Roderer and Raunser, 2019). Furthermore, a mixture of A and BC complexes from different sources (intra- and cross-species) can form an active complex. For example, a single BC pair from *P. luminescence* strain W14 can combine with both tcdA- and tcaAB-encoded A proteins (Waterfield *et al*., 2005). Also, a BC complex from *Xenorhabdus* can be combined with TcdA from *Photorhabdus* (Sheets *et al*., 2011).

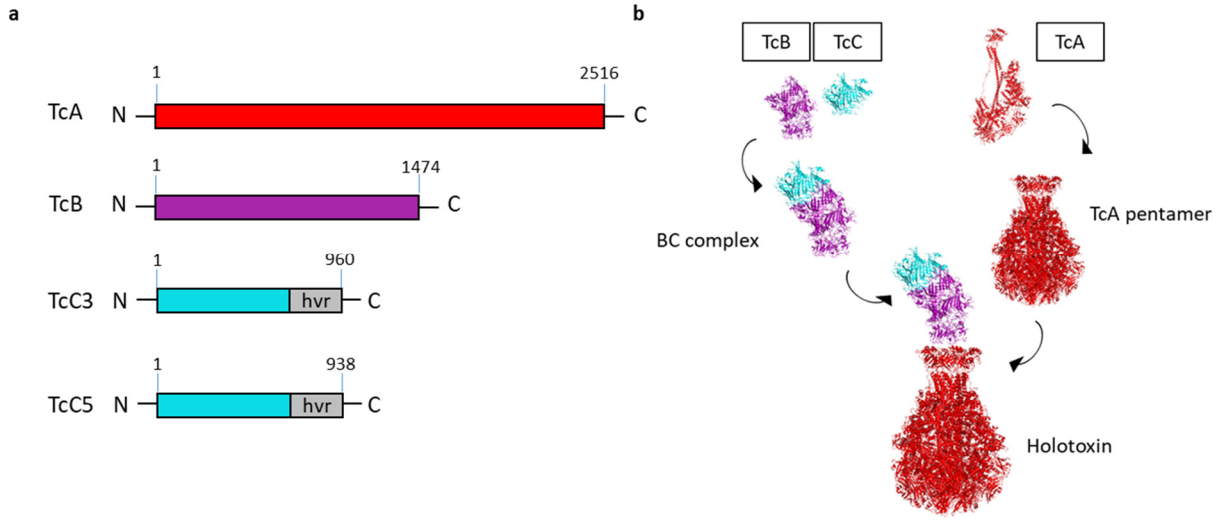


Figure 1: ***Formation of the ABC-type Photorhabdus toxin complex****. (**a**) Representation of the primary structure of the three Tc toxin components, TcA (red), TcB (purple), TcC (N-terminal (turquoise) and C-terminal hypervariable region (hvr) (grey)). Numbering based on proteins from P. luminescence W14. (**b**) Formation of the holotoxin complex. Five identical TcA proteins combine to form a homopentameric complex. On the other hand, a single TcB protein combines with a TcC protein to form the TcB-TcC (BC) complex. Finally, the two complexes combine to form the complete active holotoxin (Figure adapted from* Lang *et al.*, 2010*, Meusch et al., 2014).*

1.3.1. Structure of TcA-like proteins

Following excellent work by the group of Stefan Raunser and colleagues, the structures of the ABC proteins forming PTC have been solved and extensively described. TcA, the largest of the

three toxin complex proteins (~280 kDa) (Fig. 1), is the binding and translocation component (Sheets *et al.*, 2011; Lang *et al.*, 2013; Meusch *et al.*, 2014). It is a large (~1.4 MDa) complex formed by a structural combination of five identical A proteins (Fig. 1). The resultant homopentamer is a bell-shaped structure, composed of an α-helical central channel, surrounded by a shell (Gatsogiannis *et al*., 2013) (Fig. 2a, b). In general, each of the five protomers of TcA can be divided into four broad parts, namely, a TcB-binding domain (1), a linker domain (2), an α-pore-forming domain (3) and an α-helical shell domain (4) with four putative receptor-binding domains (RBDs A–D), and a neuraminidase-like domain (Meusch *et al*., 2014; Roderer and Raunser, 2019) (Fig. 2b).

The pore-forming domain of TcdA1 is a funnel-shaped structure, formed by the TcB-binding domain at the top and a tapered 200 Å-long α-helical central channel (Fig. 2e). The channel is closed at the bottom by conserved β-turn loops formed almost exclusively by hydrophobic residues. The outer surface of the lower end of the pore that forms the membrane-spanning part of the channel, is predominantly hydrophobic (Gatsogiannis *et al*., 2016) (Fig. 2d1, d2). A proline-rich relaxed spring-like peptide connects the pore-forming domain to the α-helical shell (Fig. 2c1, c2). This linker contracts upon cell contact and acidification, thereby pushing the pore forming-channel through the membrane in a syringe-like fashion (Gatsogiannis *et al*., 2016; Meusch *et al*., 2014).

Lipid bilayer experiments pointed to the TcdA1 channel as being water-filled and voltage-gated. TcdA1 also showed some cation selectivity due to negative point charges at the channel mouth. Therefore, it may presumably allow basic peptides to pass through (Lang *et al*., 2013). C2II, the binding component of *Clostridium botulinum* C2 toxin, and PA, the binding component of anthrax toxin, also form cation-selective channels (Schmidt *et al.*, 1994; Orlik *et al.*, 2005). Additionally, the TcdA1 channel also appeared to have a defined orientation when a protein was added to only one side of the membrane. Furthermore, it appeared to be asymmetric with respect to voltage dependence and could be blocked when the TcdB2-TccC3 complex was added on the same side (Lang *et al*., 2013).

The α-helical shell and neuraminidase-like domain together encase the channel and shield its hydrophobic tip (Fig. 2b). Structural similarity between the four putative RBDs and those of diphtheria and anthrax toxins, suggests their role in receptor binding (Meusch *et al*., 2014; Roderer and Raunser, 2019). However, they vary in structure and number in different TcAs. For

instance, TcdA2 and TcdA4 from *P. luminescens* have three RBDs, while TcA from *Yersinia pseudotuberculosis* has only one (Meusch *et al*., 2014). Perhaps, this explains the differences in host specificity of Tc toxins. Furthermore, in place of the RBDs, YenTcA from *Y. entomophaga* has chitinases, Chi1 and Chi2, associated with the TcA protein (Piper *et al.*, 2019; Roderer and Raunser, 2019).

Remarkably, TcdA1 forms pores *in vitro* at both pH 5 (endosomal pH) and pH 11 (insect midgut pH), on receptor-free artificial membranes (Gatsogiannis *et al*., 2013). Additionally, pore-forming activity at pH 11 was more than 100-fold greater than at pH 6. This activity at pH 11 may explain the oral toxicity of these toxins (Blackburn *et al*., 1998; Silva *et al*., 2002), however, it also suggests an alternative route for Tc toxin action (Roderer and Raunser, 2019). Interestingly, TcdA1 from *P. luminescens* embedded in lipid nanodiscs showed that the transmembrane helices at the bottom of the pore do not span the complete membrane upon insertion (Fig. 2a2, b2). Instead, extended loops connecting the helices interact with lipid head groups that reach into the space between the loops and stabilize the conformation of the TcdA1 pore in the membrane (Fig. 2d2) (Gatsogiannis *et al.*, 2016; Roderer and Raunser, 2019). The helices then rearrange in the membrane and open the initially closed pore in an iris-like fashion (Gatsogiannis *et al*., 2016; Roderer and Raunser, 2019). A pivot point region (G2123) in the downward-running helix at the tip of the channel swings outward, depending on the hinge, and thereby changes its conformation from a kinked to a continuous helix (Fig. 2d1, d2).

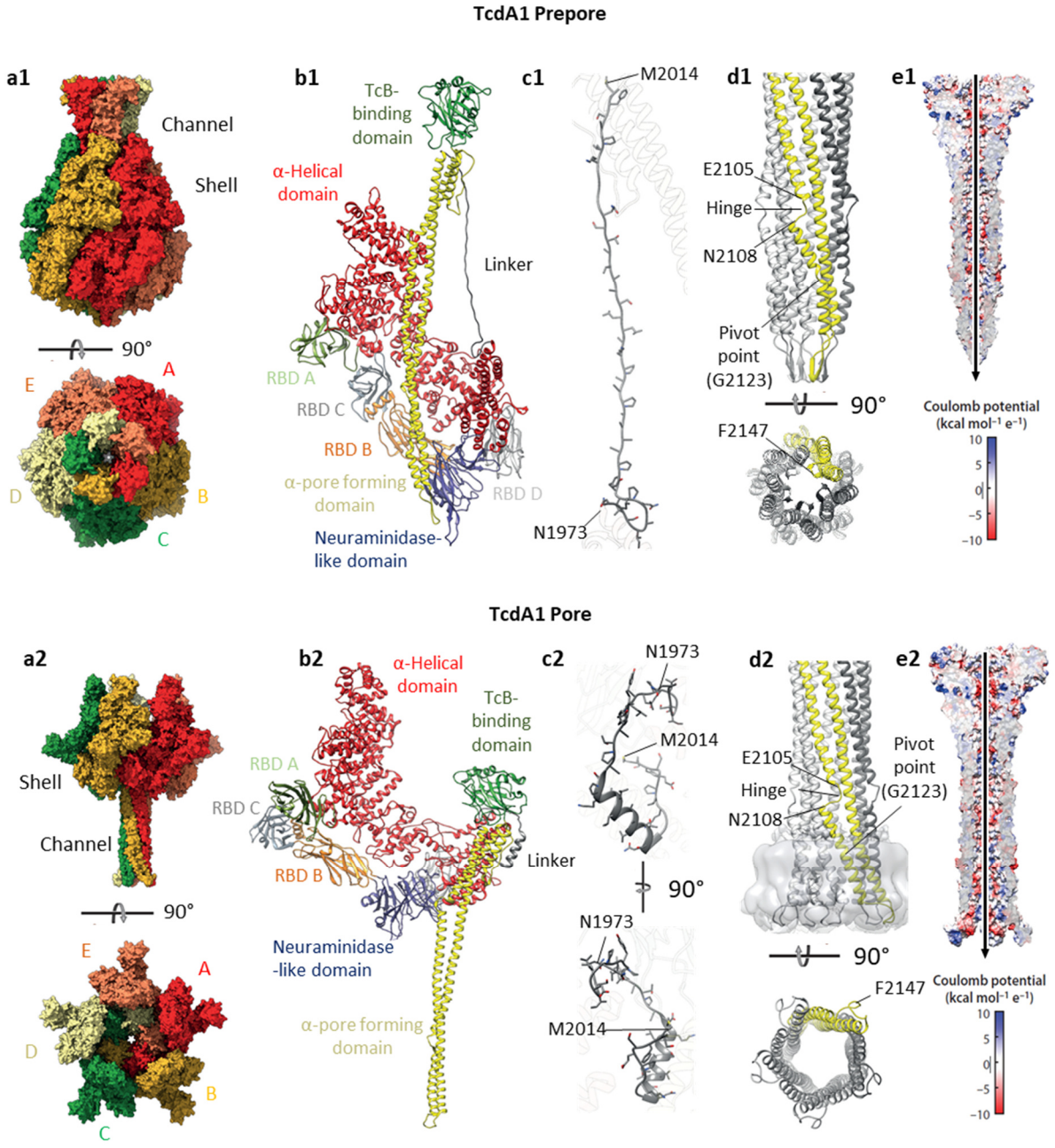


Figure 2: ***Structure of TcdA1 from P. luminescens in the prepore (a1 - d1) and membrane-embedded pore (a2 - d2) state***. *(**a1**, **a2**) Surface representation of the TcdA1 pentamer in side and top views. The protomers (A–E) are differently colored. (**b1**, **b2**) Ribbon representation of a single protomer showing the α-helical shell domain, the four putative receptor-binding domains (RBD A–D), the neuraminidase-like domain, the linker, the translocation channel, and the TcB-binding domain. (**c1**, **c2**) The proline-rich linker (N1973–M2014) connecting the α-helical shell domain to the channel in its stretched conformation (prepore) and condensed conformation (pore). (**d1**, **d2**) The transmembrane region in the side and bottom views. The positions of the tip (F2147), hinge (E2105, N2108), and pivot point (G2123) are indicated. The density of the nanodisc is shown in the side view of the pore. (**e1**, **e2**) Surface representation of the channel and the TcB-binding domain, coloured according to the Coulomb potential. The arrow shows the translocation direction of the toxic enzyme. (a1 – e1) PDB ID 1VW1. (a2 – e2) PDB IDs 5LKH and 5LKI (Figure adapted from Roderer and Raunser, 2019).*

1.3.2. TcB and TcC-like proteins

TcB and TcC (BC) combine to form a hollow ~280 kDa cocoon-like complex (Fig. 3a, b), which interacts with the TcA pentamer (Meusch *et al*., 2014; Roderer *et al*., 2019) (Fig. 4). The two proteins are usually expressed as a fusion protein. However, TcB can be singly expressed as soluble proteins while TcC cannot (Sheets and Aktories, 2017). Often, TcC is expressed fused to TcB and indeed in some bacteria like *Burkholderia* and *Pseudomonas*, they exist as gene fusions (Yang and Waterfield, 2013). TcB-like proteins are about 170 kDa in size and do not show any toxicity against insects when applied alone. In fact, they do not seem to play any other role apart from connecting TcC to the TcA pentamer. However, due to their involvement in the expression of TcC, they may act as a chaperon for the TcC protein (Sheets and Aktories, 2017).

Three crystal structures of BCs exist, (a) the fusion construct of *P. luminescens* TcdB2-TccC3, (b) the complex of YenB and YenC2 from *Y. entomophaga*, and (c) the complex of YenB and an RHS2 protein from *Y. entomophaga* (PDB ID 5KIS) (Busby, Hurst and Lott, 2017 (unpublished); Roderer and Raunser, 2019). All three reveal a large hollow cocoon-like structure with a stack of β-barrels that contain rearrangement hotspot (RHS) repeats (Busby *et al*., 2013). The N-terminus of TcB binds to the TcA pentamer and forms a β-propeller gate (Fig. 3a, b) that is positioned on top of the TcA pore-forming channel (Fig. 4a) (Busby *et al*., 2013; Meusch *et al*., 2014). The β-propeller gate opens when the BC cocoon binds to TcA forming a continuous joint channel running through the entire toxin complex (Meusch *et al*., 2014).

The inside of the cocoon is largely hydrophobic, mainly due to inward-facing tyrosines in YxY repeats of TcC (Busby *et al*., 2013; Meusch *et al*., 2014; Roderer and Raunser, 2019). This creates an inner environment unfavourable for protein folding, similar to that of chaperones. In fact, directly above the β-propeller gate at the bottom of the cocoon is a narrow 10.5-Å constriction site, which only permits the passage of helices, therefore, (semi-)unfolded peptides could pass through, but not tertiary structures, that is, fully folded proteins (Roderer and Raunser, 2019). The β-propeller gate is itself distorted, as two of its six blades, 3 and 4, are folded differently. The β-hairpin of blade 4 functions as a gatekeeper that closes the β-propeller at its centre (Fig. 3d, e).

TcC-like proteins are about 112 kDa in size and are important for the toxicity of the complex (Sergeant *et al.*, 2003; Meusch *et al*., 2014). TcC is divided into two domains, a highly conserved N-terminal TcB-binding domain and a hypervariable C-terminal domain (also known as hypervariable region (hvr)) (Fig. 1a). The hvr is encased in the BC cocoon and occupies its central

hollow chamber (Lang *et al.*, 2010; Sheets *et al.*, 2011). The hvr is itself a 30-35 kDa toxic enzyme, which modulates intracellular modifications, leading to cytopathic and cytotoxic effects in target cells (Lang *et al*., 2010). The hyperconserved N-terminal part of TcC, the RHS core domain, closes the BC cocoon at the top and also functions as an aspartyl autoprotease, to cleave off the hvr from the rest of TcC. The TcB β-propeller gate closes the bottom of the cocoon (Fig. 4b). TcC autoproteolyses when co-expressed with TcB to release the hvr, which then sits in the cocoon chamber until BC binds to TcA (Meusch *et al*., 2014).

Two highly conserved aspartic acid residues (D651 and D674 in *P. luminescens* TccC3) form the catalytic dyad of the TcC protease (Fig. 4c). Deletion/mutation of one or both of them results in a noncleavable and nontoxic complex (Fig. 4b, right panel) (Lang *et al.*, 2010). The cleavage site is between L678 and M679 in *P. luminescens* TccC3, on a loop that is traced through the cleft between the aspartates of the catalytic dyad. In other TcC homologs, L678 is mostly replaced by large hydrophobic residues. Therefore, in this respect, the TcC aspartyl protease resembles pepsin and renin, aspartyl proteases which cleave preferentially after hydrophobic residues (Dunn, 2001; Sparks *et al.*, 2014; Roderer and Raunser, 2019). The hvr is translocated with its C-terminus first. As seen in the non-cleavable cryo-EM structure of TccC3 from *P. luminescens*, the N-terminus remains attached to the cocoon, while the enzyme protrudes into TcA (Fig. 4b) (Gatsogiannis *et al*., 2018; Roderer and Raunser, 2019).

Two TcC hvrs from *P. luminescens* have been characterized, TccC3 and TccC5 (Fig 1a). Both are ADP-ribosyltransferases, with the TccC3hvr targeting actin and TccC5hvr targeting small GTPases (Lang *et al.*, 2010; Lang *et al.*, 2011; Pfaumann *et al.*, 2015) (Fig. 4c). ADP-ribosylation by the C3hvr occurs at threonine148 of actin, thus affecting actin polymerisation (Lang *et al*., 2011). Modified actin is no longer able to interact with the regulatory proteins thymosin β4, which sequesters actin in its monomeric form, or gelsolin, which caps and severs actin filaments. This subsequently results in increased actin polymerisation. The C5hvr, on the other hand, ADP-ribosylates small GTPases of the Rho family at glutamine63 of RhoA (or glutamine61 of Rac and Cdc42) (Lang *et al*., 2011; Pfaumann *et al*., 2015).

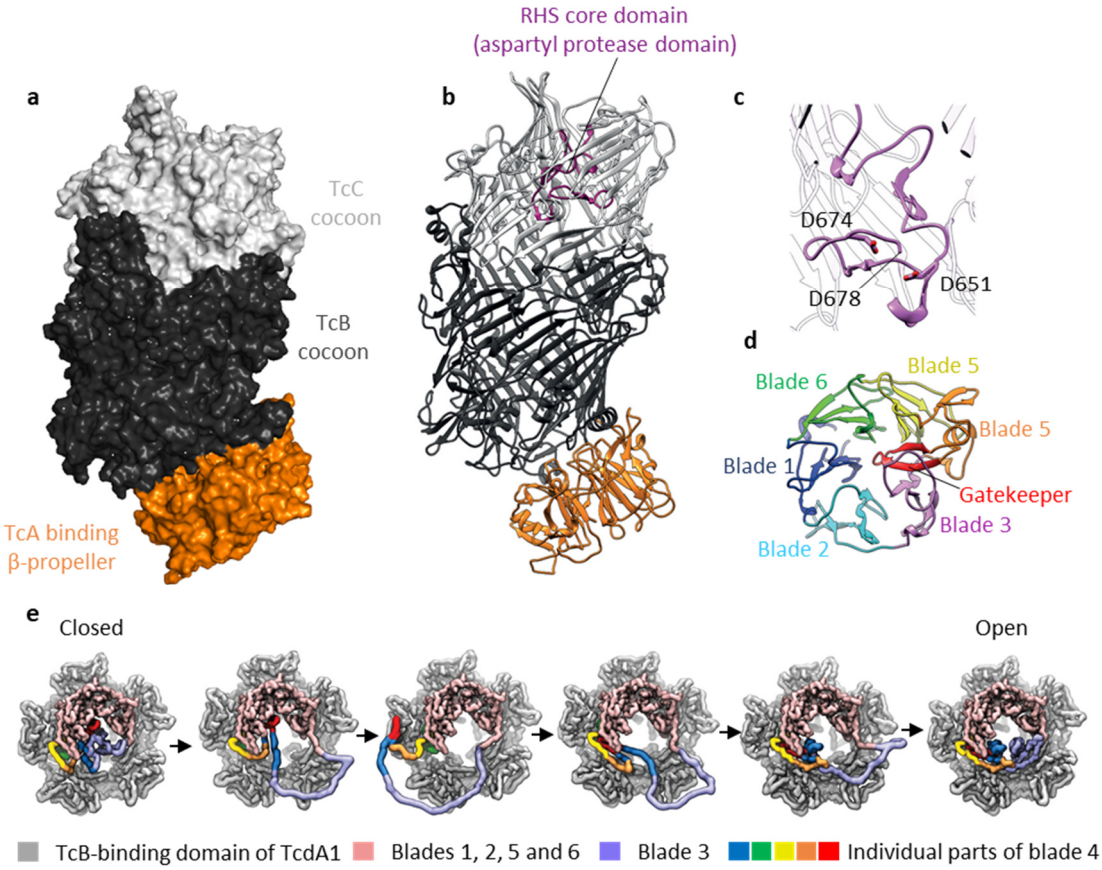


Figure 3: ***Structure of the P. luminescens TcdB2-TccC3 cocoon***. *(**a**) Surface representation of the BC fusion protein (PDB ID 4O9X). TcdB2 (TcA-binding β-propeller domain and cocoon) and TccC3 (cocoon) are shown in distinct colors. (**b**) Ribbon representation of the same structure as in (a), coloured similarly and showing the aspartyl autoprotease domain (coloured purple). The hypervariable region inside the cocoon is not resolved in the crystal structure. (**c**) Close-up view of the aspartyl protease domain of TccC3, formed by D651 and D674. (**d**) Top view of the TcdB2 β-propeller domain with the six blades individually coloured and the distorted blades 3 (purple) and 4 (multi-coloured) with the gatekeeper hairpin (red)* (Gatsogiannis *et al.*, 2018)*, which plugs the cocoon. (**e**) Snapshots of molecular dynamics simulations of transition intermediates of the BC cocoon from the closed (left), to the open conformation in the holotoxin (right). The TcB-binding domain ofTcdA1 and propeller blades are shown in distinct colors. (Figure adapted from Meusch et al., 2014 and Roderer and Raunser, 2019)*

Rho proteins are regulated by a guanine nucleotide cycle, where they cycle between an active GTP-bound form and an inactive GDP-bound form, after hydrolysis of GTP to GDP (Jaffe and Hall, 2005). This cycle will be expounded in the next section. However, Glutamine63/61 of Rho GTPases is essential for GTP hydrolysis and, therefore, modification affects their turning-off. The Rho GTPases are thus locked in a persistently active state (Lang *et al.*, 2011) and in the case of Rho, this results in actin polymerisation and stress fibre formation (Ridley and Hall, 1992; Lang *et al.*, 2011). Both TccC3 and TccC5, therefore, induce actin polymerisation and clustering (Fig. 4d),

which probably inhibits essential functions of immune cells such as phagocytosis and migration (Lang *et al*., 2011).

The mechanism of action of the Tc toxin is shown in Fig. 4c. It begins with toxin complex formation, which takes part in stages; first, the TcA oligomerizes to form a pentamer with a central pore-forming channel. Second, TcB and the N-terminal portion of TcC combine to form the hollow TcB-TcC cocoon, which houses the enzymatically active TcChvr. Third, the two complexes (A and BC) combine to form the full holotoxin. The complete toxin then binds to target cells, through a yet unknown receptor, and is taken up by receptor-mediated endocytosis. Acidification of the toxin-containing endosome, via an influx of protons, induces conformational changes in TcA, which then perforates the endosomal membrane in a syringe-like mechanism. Conformational changes in the transmembrane region of the channel lead to the opening of the pore in an iris-like fashion. This then allows the toxic TcChvr to translocate into the host cell cytosol, with the help of cytosolic host chaperones like the heat-shock protein Hsp90 and peptidyl prolyl cis/ trans isomerases (PPIases), including cyclophilins and FK506-binding proteins (FKBPs) (Lang *et al*., 2014). These chaperons help the enzymes to refold, after which they can modulate toxic activity (Fig. 4d).

In summary, the highly efficient actions of the *Photorhabdus* Tc toxins depend on productive assembly of toxin components that enable the following. (1) Toxic activity independent of bacteria-based delivery systems. The toxins are secreted, assembled and work independent of their host bacteria (Roderer and Raunser, 2019; Sheets and Aktories, 2017; Silva *et al*., 2002). (2) Potent and efficient delivery of different toxic enzymes into target cells (Lang *et al*., 2011; Meusch *et al*., 2014). (3) A completely closed and self-contained translocation mechanism that may protect the toxic enzymes from the external environment (Lang *et al*., 2013; Meusch *et al*., 2014). This greatly differs from other known toxins including anthrax (Collier, 2009), diphtheria (Auger *et al*., 2015) and *C*. *botulinum* C2 toxin (Barth and Aktories, 2011). (4) A selection of different TcAs for target-specific applications (Ffrench-Constant and Waterfield, 2005). Several TcAs have been identified in *Photorhabdus* that show differences in their RBDs suggesting target specificity (Meusch *et al*., 2014; Roderer and Raunser, 2019). (5) Interchangeability of ABC components to form suitable toxic combinations (Sheets *et al*., 2011; Waterfield *et al*., 2005). (6) Efficient intracellular toxic activity with active enzymes that work using different strategies, recall TccC3 and TccC5 (Lang *et al.*, 2010) (Fig. 4d). (7) Interactions with a variety of host cell membranes including those of insects, mammals and humans (Ffrench-Constant and Waterfield, 2005; Lang

et al., 2010; Hinchliffe *et al.*, 2010; Sheets *et al.*, 2011; Meusch *et al.*, 2014; Ost *et al.*, 2019). Possessing a variety of receptor binding domains (RBDs), the toxin could exploit either multiple receptors, or, ubiquitously distributed ones, thus allowing it to operate across species and genera.

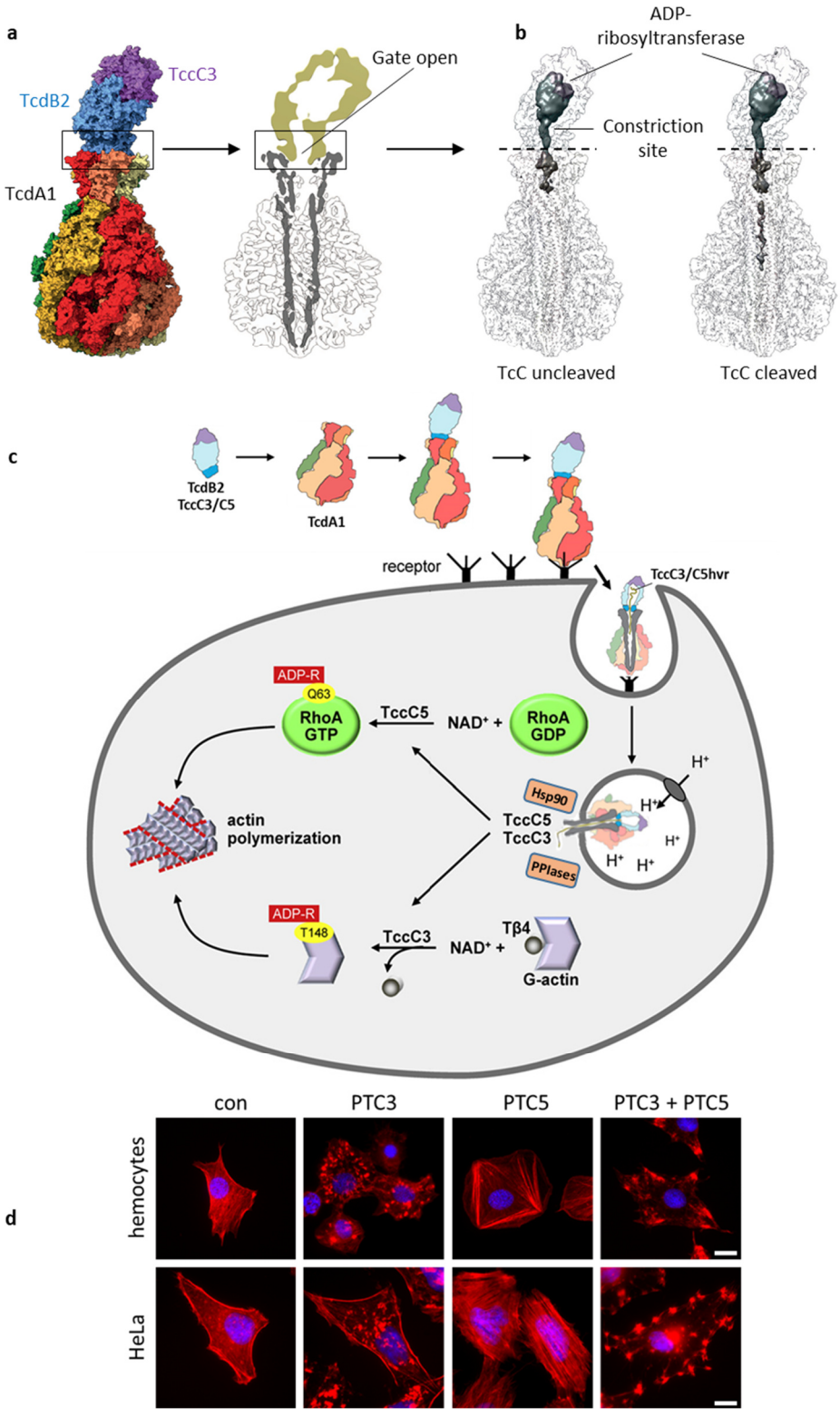


***Figure 4**: **Structure and mechanism of action of the Photorhabdus toxin complex**. (**a**) Surface representation of the overall structure of the holotoxin. The five protomers of TcdA1, TcdB2, and TccC3 are distinctly coloured. The boxes highlight the TcA-TcB interface with the open β-propeller gate of TcB. (**b**) The ADP-ribosyltransferase (or TcC hypervariable region) in the translocation channel of the holotoxin, cleavage-deficient holotoxin (left) and wild type (right). The dashed line indicates the border between TcdA1 and TcdB2. (**c**) Mechanism of action of the holotoxin. After holotoxin assembly, cell binding and receptor-mediated endocytosis, the holotoxin reaches endosomal compartments. There, upon acidification, TcA inserts into the membrane and forms stable pores through which the TcC enzyme components translocate into the cytosol. Here, they induce polymerization of actin, but have different targets. TccC3 ADP-ribosylates actin at threonine-148, leading to reduced binding of the actin sequestering protein thymosin-β4 to G-actin and hence actin polymerization. TccC5 ADP-ribosylates Rho proteins at glutamine-63, leading to their persistent activation, followed by stress fibre formation due to actin polymerization. Together, the enzymes cause massive clustering of F-actin. (**d**) G. mellonella hemocytes and HeLa cells intoxicated with the indicated Photorhabdus toxin complexes for 2 and 4 h respectively. After intoxication, the cells were fixed and actin fibres stained with tetramethyl rhodamine isothiocyanate–conjugated phalloidin and 4´,6´-diamidino-2-phenylindole (scale bar, 10 mm). (Figure adapted from* Lang *et al.*, 2010; *Aktories et al., 2012; Roderer and Raunser, 2019; Roderer et al., 2019)*

1.4. Rho GTPases and the toxins that target them

1.4.1. Rho GTPases

Like actin, Rho GTPases are key targets of the *Photorhabdus* toxin complex (PTC), particularly through the TccC5 ADP-ribosyltransferase, which was discussed above. Additionally, in this study, PTC was modified to deliver a variety of bacterial toxins, effectors, or their fragments, which modify Rho GTPases. It is therefore important to provide a description of these GTPases and the toxins that target them.

Rho GTPases are small GTP-binding proteins (GTPases) of the Rho family that have GTP hydrolysis activity. They form part of the Ras super-family and consist of 22 members including: Rho A, B, and C; Rac 1, 2, and 3, Cdc42, RhoD, Rnd1 and 2, RhoE/Rnd3, RhoG, TC10, and TCL, RhoH/TTF, Chp and Wrch-1, Rif, RhoBTB 1 and 2 and Miro-1 and 2 (Aspenström *et al*., 2004; Jaffe and Hall, 2005). They have molecular masses of 21–25 kDa and consist of a G-domain (GDP/GTP-binding domain) and a C-terminal variable region. This variable region is involved in membrane interaction through an isoprenyl moiety that is post-translationally added to a cysteine residue, which is part of the C-terminal CaaX box of Rho GTPases. As part of the Ras super-family, they have a similar G-domain topology with a central β-sheet of six strands, flanked by five α-helices. However, typical to Rho GTPases, is an insert helix (amino acids 124–136 of RhoA), which is not present in other Ras superfamily proteins (Aktories, 2015) (Fig. 5a). But similar to other regulatory

GTPases, they act as molecular switches cycling between an active GTP-bound state and an inactive GDP-bound state (Jaffe and Hall, 2005) (Fig. 5b).

The Rho GTPase cycle is regulated by three types of proteins (Aktories, 2015; Hodge and Ridley, 2016) (Fig. 5b). (a) Guanine nucleotide exchange factors (GEFs) (~80 proteins) that catalyse the exchange of GDP for GTP, thereby activating the GTPase (Rossman *et al*., 2005). (b) GTPase-activating proteins (GAPs) (~60 proteins) that stimulate the intrinsic GTPase activity, thereby inactivating it (Bernards, 2003). (c) Guanine nucleotide dissociation inhibitors (GDIs), that sequester the GDP-bound form of some GTPases in the cytosol, and prevent them from localizing to membranes, or being spontaneously activated by GEFs (Olofsson, 1999). However, some Rho GTPases can also be regulated atypically, through posttranslational modifications like phosphorylation, ubiquitination and sumoylation (Hodge and Ridley, 2016).

GTP-binding induces major structural rearrangements in the switch-I (residue 29–42) and switch-II (residues 62–68) regions of Rho GTPases. These regions allow interaction with regulators, as well as numerous effectors including kinases, actin regulators and adaptor proteins, that lead to changes in cell behaviour. In fact, a single Rho GTPase can activate a range of cellular responses, depending on the stimulus and cell type (Aktories, 2015; Hodge and Ridley, 2016). Activation (GTP-binding) takes place at the cell membrane, to which the GTPase is attached through the isoprenyl moiety on its carboxyl terminus (Wang and Casey, 2016). This active state is then terminated by hydrolysis of GTP, facilitated by certain GAPs. Later, GDIs bind to the Rho GTPases, and keep them in the cytosol through a close interaction with their isoprenylated tail (Fig. 5b). This also inhibits their activation (Aktories, 2011).

Some bacterial protein toxins exploit Rho GTPases to facilitate pathogenicity. One group covalently modifies Rho GTPases, affecting their regulatory functions and interfering with their corresponding host signalling pathways. Others non-covalently alter Rho GTPase activity by mimicking endogenous regulators of the GTPase cycle; this is the case for various bacterial effectors that are secreted via type III secretion (T3S) or type IV secretion (T4S) systems (Aktories, 2011). A key reason for targeting Rho GTPases is their role in host–pathogen interaction. They are master regulators of the cytoskeleton, and control stress fibre formation (Rho), membrane ruffling (Rac) and filopodia formation (Cdc42) (Aktories, 2015; Jaffe and Hall, 2005). Additionally, as switch proteins, they coordinate both extracellular and intracellular signals, as well as subsequent cellular responses. Moreover, they are crucial in immune cell migration and

signalling, phagocytosis, cytokine release and epithelial barrier functions. They may even play a role in sensing inflammation (Xu *et al*., 2014). It is therefore no wonder, that they are strategic targets for toxins (Aktories, 2015). A few examples of toxins that target Rho proteins are shown in Table 1 and discussed in the proceeding section.

Table 1: **Examples of bacterial toxins and effectors that target Rho GTPases**

Toxin or Effector	Source bacterium	Modification	Target	Target site	Reference
Covalent modifications					
Activators					
TccC5	*P. luminescence*	ADP-ribosylation	RhoA/B/C, Rac1/2/3 and Cdc42	Gln61/63	Lang *et al.*, 2010
CNF1, 2, 3	*Escherichia coli*	Deamidation	RhoA/B/C, Rac1/2/3 and Cdc42	Gln61/63	Schmidt *et al.*, 1997; Flatau *et al.*, 2000; Hoffmann *et al.*, 2004
CNFY	*Yersinia pseudotuberculosis*	Deamidation	RhoA	Gln63	Schmidt *et al.*, 1997; Flatau *et al.*, 2000; Hoffmann *et al.*, 2004
Inhibitors					
C3 (two strain variants)	*Clostridium botulinum*	ADP-ribosylation	RhoA/B/C	Asp41	Aktories, Weller and Chhatwal, 1987; Rubin *et al.*, 1988
C3	*C. limosum*	ADP-ribosylation	RhoA/B/C	Asp41	Aktories, Weller and Chhatwal, 1987; Rubin *et al.*, 1988
C3cer	*Bacillus cereus*	ADP-ribosylation	RhoA/B/C	Asp41	Aktories, Weller and Chhatwal, 1987; Rubin *et al.*, 1988
C3stau	*Staphylococcus aureus*	ADP-ribosylation	RhoA/B/C	Asp41	Aktories, Weller and Chhatwal, 1987; Rubin *et al.*, 1988
Toxins A and B	*C. difficile*	Glucosylation	Rho, Rac1, Cdc42, also TC10, RhoG (Toxin B)	Thr35/37	Just *et al.*, 1995
TcsL (lethal toxin)	*C. sordellii*	Glucosylation	Ras, Rac, RhoG, TC10, Rap, Ral	Thr35/37	Just *et al.*, 1996
TcsH (haemorrhagic toxin)	*C. sordellii*	Glucosylation	RhoA/B/C, Rac1, Cdc42, TC10 and others	Thr35/37	Genth *et al.*, 2014
TcnA (α-toxin)	*C. novyi*	N-acetyl-glucosaminylation	Rho, Rac, Cdc42	Thr35/37	Selzer *et al.*, 1996
TpeL	*C. perfringens*	N-acetyl-glucosaminylation	H-, K-, N- and R-Ras,	Thr35/37	Nagahama *et al.*, 2011

			Rap1B, Rap2A, Ral1		
PaTox	*P. luminescence*	N-acetyl-glucosaminylation	RhoA/B/C, Rac1/2/3 and Cdc42	Tyr32/34	Jank *et al.*, 2013
YopT	*Yersinia spp.*	Proteolytic cleavage	Rho-family proteins	Cys200 in RhoA	Shao *et al.*, 2003
LopT	*P. luminescence*	Proteolytic cleavage	Rho-family proteins	Cys200 in RhoA?	Brugirard-Ricaud *et al.*, 2005
Non-covalent modifications (mimics of host regulators)					
Activators					
SopE and SopE2	*Salmonella enterica subsp. enterica serovar* Typhimurium	GEF	Rac-proteins and Cdc42		Hardt *et al.*, 1998
BopE	*Burkholderia pseudomallei*	GEF	Rac-proteins and Cdc42		Upadhyay *et al.*, 2008
IpgB1, B2	*Shigella flexneri*	GEF	RhoA (IpgB2), Rac-proteins and Cdc42 (both)		Ohya *et al.*, 2005; Alto *et al.*, 2006; Klink *et al.*, 2010
Inhibitors					
YopE	*Yersinia spp.*	GAP	RhoA/B/C, Rac1/2/3 and Cdc42		Von Pawel-Rammingen *et al.*, 2000; Mohammadi and Isberg, 2009
ExoS and ExoT	*Pseudomonas aeruginosa*	GAP	RhoA/B/C, Rac1/2/3 and Cdc42		Goehring *et al.*, 1999; Krall *et al.*, 2000

CNF, cytotoxic necrotizing factor; Exo, exoenzyme, GEF, guanine nucleotide exchange factor; GAP, GTPase-activating protein

1.4.2. Rho modifying toxins

Covalent modification of Rho GTPases follows a diverse range of mechanisms including ADP-ribosylation, glycosylation, adenylylation, proteolysis, deamidation and transglutamination (Table 1). Unlike the non-covalently modifying effectors, many of the covalently-modifying toxins are released from the pathogen into the environment and enter their eukaryotic target cell through the cell's autonomous membrane translocation systems (Aktories, 2011, 2015). However, there are exceptions, and as will be discussed in the proceeding section, some secretion-system-dependent effectors covalently-modify Rho GTPases. Indeed, this phenomenon (delivery into target cells), forms the distinction between bacterial toxins (exotoxins) and effectors, which are virulence factors that are introduced into target cells by T3S or T4S systems (Aktories, 2011).

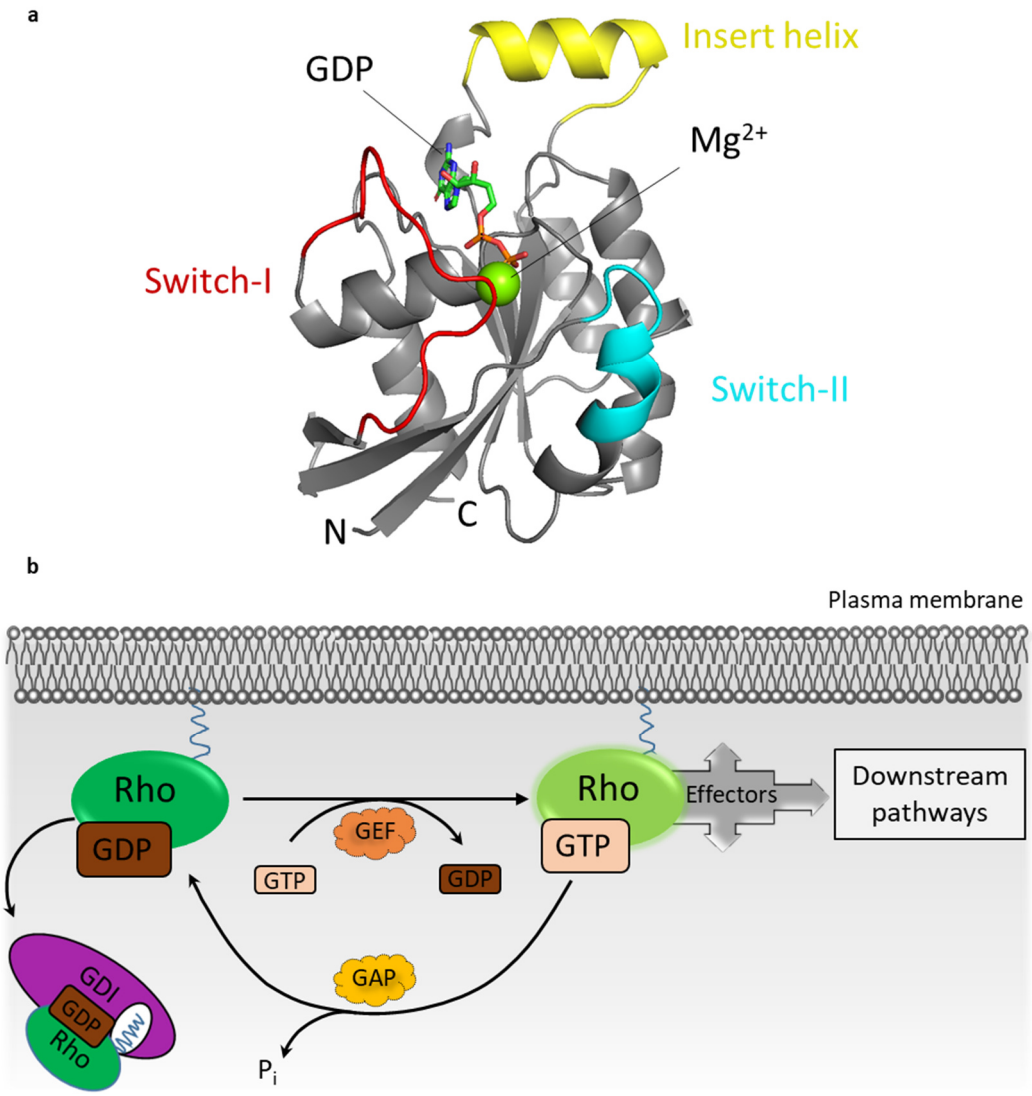


***Figure 5**: **Structure and regulation of Rho GTPases**. (**a**) A cartoon representation of the 3D structure (main chain only) of human RhoA (PDB ID 1FTN). The biologically relevant features are individually coloured and labelled. The figure was generated using PyMoL. (**b**) Regulation of the Rho GTPase cycle. Rho GTPases switch between an inactive GDP-bound state and an active GTP-bound state. Guanine nucleotide dissociation inhibitors (GDIs) prevent nucleotide exchange and keep the GTPases in the cytosol. Guanine nucleotide exchange factors (GEFs) activate the GTPases by inducing GDP–GTP exchange. Active GTP-bound proteins interact with and regulate the activity of endogenous effectors, which modulate downstream pathways. GTPase-activating proteins (GAPs) inactivate Rho-family proteins by facilitating GTP hydrolysis. Rho-family proteins are also isoprenylated, which allows membrane binding. (Figure adapted from* Wei *et al.*, 1997; Hodge and Ridley, 2016)

1.4.2.1. ADP-ribosylating toxins

ADP-ribosylation involves the transfer of an ADP-ribose moiety from nicotinamide adenine dinucleotide (NAD^+) onto substrates via N-, O-, or S- glycosidic linkages on target molecules. It is an ancient post-translational modification, catalysed by enzymes of the ADP-ribosyltransferase (ART) superfamily of proteins, which are found in viruses, bacteria and eukaryotes (Cohen and

Chang, 2018). A wide range of bacterial pathogens targeting insects, plants and animals, including humans, produce ART toxins. These toxins modify various eukaryotic substrates, including Rho proteins (C3 exoenzymes from *Clostridium botulinum* and TccC5 from *Photorhabdus luminescence*) (Fig. 6a), heterotrimeric G-proteins (cholera toxin from *Vibrio cholera* and pertussis toxin from *Bordetella pertussis*) and actin (C2 toxin from *C. botulinum* and TccC3 from *P. luminescence*) (Simon *et al*., 2014). Many of these modifications inhibit normal eukaryotic protein function; however, bacterial ART toxins that activate eukaryotic proteins to promote bacterial pathogenesis have also been identified (Simon *et al*., 2014).

Rho GTPases are a very good example of substrates that are both inactivated and activated via ADP-ribosylation by bacterial toxins (Table 1). One toxin that inactivates Rho GTPases by ADP-ribosylation is the *C. botulinum* C3 toxin. It was the first Rho-modifying toxin shown (Aktories *et al*., 1987; Rubin *et al*., 1988) and mainly targets RhoA, B and C at asparagine41 (Fig. 6b, Table 1), whereas Rac proteins and CDC42 are very poor substrates (Aktories, 2011). Seven C3-like toxins having ~35–90% sequence identity are known (Aktories, 2011; Simon *et al*., 2014). They are small (~25 kDa) proteins and consist of only an ART domain, without any specific modules for delivery into target cells. (Aktories, 2011). Modification at Asn41 leads to inactivation of Rho GTPases through tight binding to GDI (abrogated membrane localization) and reduced activation by GEFs. However, interaction with some effectors does not seem to be affected, even though Asn41 is in the switch one region (Aktories, 2015). This in turn leads to various cellular effects like destruction of stress fibres and redistribution of the actin cytoskeleton (Aktories, 2011).

In contrast to C3 toxin, ADP-ribosylation of Rho proteins by the *Photorhabdus* TccC5 toxin leads to its activation (Lang *et al.*, 2010; Pfaumann *et al.*, 2015). As was briefly discussed in the previous section, TccC5 modifies RhoA at Gln63 (Gln61 in Rac and Cdc42). This glutamine residue, found in the switch II region of the protein, is essential for the intrinsic and GAP-catalysed GTP hydrolysis (that is, the turn-off reaction) of the Rho-family GTPases (Aktories, 2011; Jaffe and Hall, 2005; Wei *et al*., 1997). Its modification leads to persistent activation of the proteins and subsequently, extensive formation of stress fibres in target cells, due to increased polymerization of actin. Together with its actin modifying counterpart, TccC3, they induce clustering of the actin cytoskeleton, which results in various effects like reduced phagocytosis of bacteria by insect haemocytes and rounding up of modified cells (Lang *et al.*, 2010).

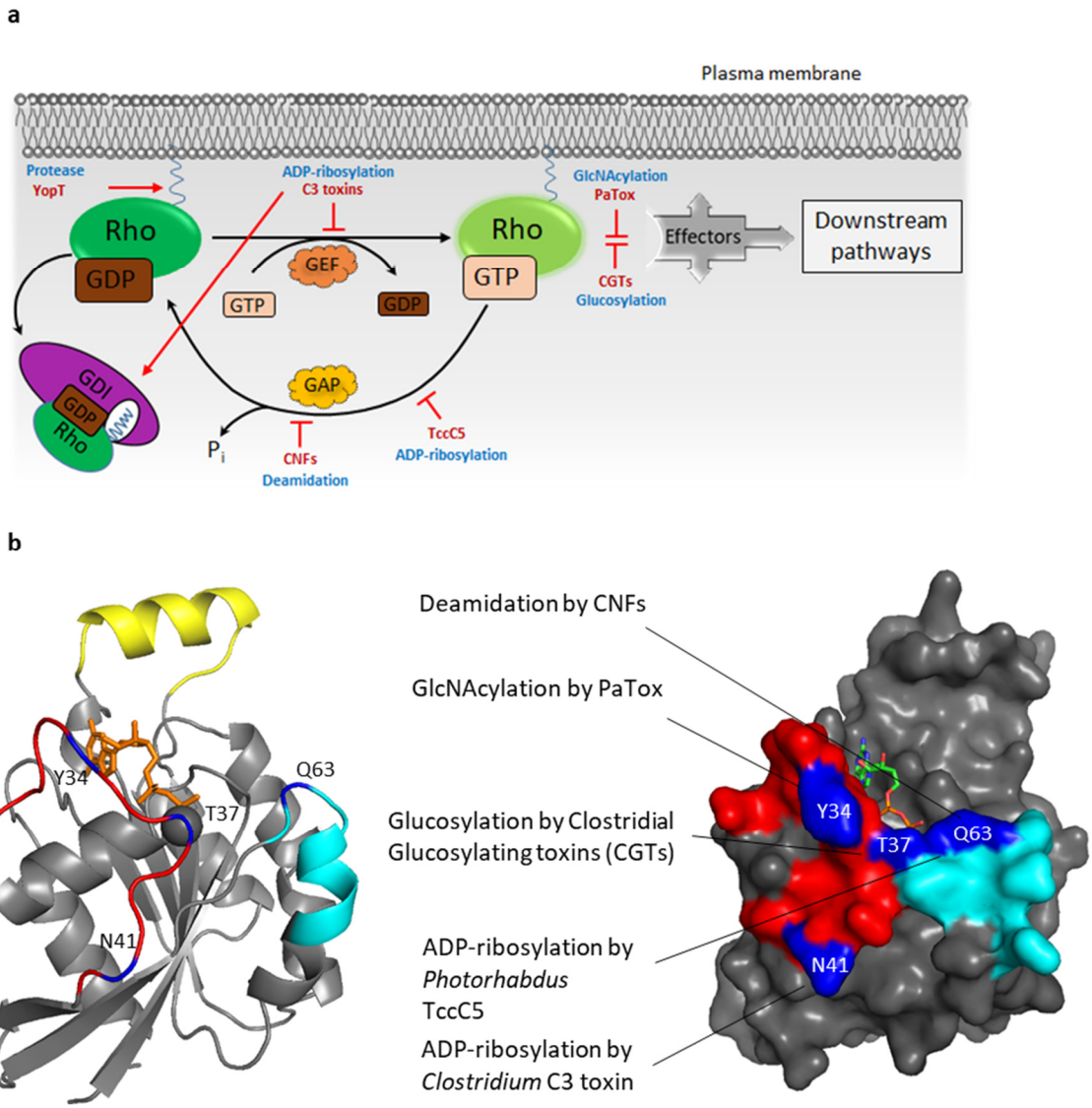


***Figure 6**: **Modification of Rho GTPases by bacterial toxins and effectors**. (**a**) Modification of the Rho GTPase cycle by various toxins and their subsequent functional consequences. Deamidation by CNFs result in persistent activation of the Rho GTPases by blocking GTP hydrolysis by GAPs. GlcNAcylation and Glucosylation by PaTox and CGTs, respectively, inhibit activation of Rho and block interaction of GTPases with their respective effectors. ADP-ribosylation by Photorhabdus TccC5 and clostridial C3 toxin results in converse effects. TccC5 modification, like deamidation, leads to persistent activation of the Rho GTPases while modification by C3 inhibits activation of Rho proteins by direct modification and through increased binding of Rho to GDI. Yersinia toxin YopT and Photorhabdus LopT inactivate Rho proteins by proteolytic cleavage of the C-terminal cysteine residue of Rho proteins that carries the isoprenyl moiety. (**b**) Sites of RhoA modification by the aforementioned toxins. The 3D structure of RhoA (left panel) is as shown in Fig 4a (PDB: 1FTN), while the right panel is a surface representation of the same. Modification sites are highlighted in blue. Structures were created using PyMol. (Figure adapted from* Aktories, 2015; Hodge and Ridley, 2016)

1.4.2.2. Glycosylating toxins

Another covalent modification employed by bacterial toxins against Rho GTPases is glycosylation. This is the covalent addition of a sugar moiety to the GTPase and several toxins are known to catalyse mono-O-glucosylation or N-acetylglucosaminylation (Aktories, 2011). The prototypes of this group are the large clostridial glucosylating toxins (CGTs). This toxin family includes the *Clostridium difficile* toxins A and B (also known as TcdA and TcdB, respectively) which are perhaps the most studied (Di Bella *et al*., 2016). Others are the *C. sordellii* lethal toxin (TcsL) (Just *et al*., 1996) and hemorrhagic toxin (TcsH) (Genth *et al*., 2014); *C. novyi* α-toxin (TcnA) (Selzer *et al*., 1996) and *C. perfringens* toxin TpeL (Nagahama *et al*., 2011) (Table 1). Glycosylation occurs at the highly conserved threonine35 (Rac-proteins and CDC42) and threonine37 (Rho A, B and C) in the switch I region of Rho proteins (Fig. 6b). This modification blocks Rho and Ras signalling by inhibiting the active conformation of the GTPases. Modified GTPases are no longer activated by GEFs, are GAP insensitive and also remain inactive, even when bound to GTP and are membrane bound.

Also in the group of glycosylating toxins, is the *Photorhabdus asymbiotica* toxin (PaTox), which harbours a glucosyltransferase domain (GTD) homologous to that of the CGTs (Jank *et al*., 2013). PaTox modifies Rho proteins by GlcNAcylation using UDP-GlcNAc as a sugar donor. Glycosylation is at tyrosine-34 of RhoA (Tyr32 of Rac and Cdc42). Tyr34, like Thr37 (modification site of CGTs) is in the switch I region of Rho GTPases (Fig. 6b) and its modification therefore leads to similar effects as the CGTs. However, while CGTs favour the inactive GDP-bound RhoA, PaTox prefers the active, GTP-bound form ((Jank *et al*., 2013; Aktories, 2015).

1.4.2.3. Rho protein deamidases

Most of the covalent modifications discussed above lead to inactivation of Rho GTPases. Only ADP-ribosylation by the *Photorhabdus* TccC5 activates the proteins. However, activation is also achieved through deamidation and many toxins that deamidate Rho GTPases activate them. Deamidation is the removal or conversion of an amide functional group in the side chain of an asparagine or glutamine (Shire, 2015). The prototypes of Rho-activating toxins are *Escherichia coli* cytotoxic necrotizing factors 1-3 (CNFs 1-3) and the CNF-like *Yersinia pseudotuberculosis* CNFy, which are associated with intestinal, urinary tract and soft tissue infections, as well as with neonatal meningitis (Aktories, 2015) (Table 1).

CNFs are single chain ~115 kDa proteins consisting of N-terminal cell binding and translocation domains, and a C-terminal catalytic domain (Lemichez *et al*., 1997). Like TccC5, they target glutamine63 of RhoA (Gln61 of Rac and Cdc42), which they deamidate, resulting in glutamic acid. Gln61/63 is in the switch II region of Rho proteins and is essential for hydrolysis of bound GTP (Schmidt *et al*., 1997) (Fig. 6b). Modified Rho proteins are, therefore, locked in an active state that is resistant to inactivation by GAPs (Fig. 6a). Ultimately, this modification has major effects on the actin cytoskeleton and induces stress fibres, formation of lamellipodia and filopodia, as well as multinucleation and spreading of cells (De Rycke *et al.*, 1996; Schmidt and Aktories, 2000; Hoffmann and Schmidt, 2004; Aktories, 2015). Additionally, persistent activation of Rho proteins has secondary impacts on their cellular fate. For example, in most cells, CNF1-activated Rac gets ubiquitinated and, thereafter, proteosomally degraded. Furthermore, activated RhoA is also degraded in a cell-type-specific manner. However, proteasomal independent inactivation also occurs (Doye *et al*., 2002; Hoffmann and Schmidt, 2004; Lerm *et al*., 2002).

1.4.2.4. Proteolysis of Rho GTPases

Some bacterial effectors catalyse covalent modifications of Rho proteins like toxins. The *Yersinia* outer protein T (YopT), an effector of *Y. enterocolitica* is a good example. Delivered into target cells by a T3S system, it has strong effects on the morphology of cells, causing rounding up and arborisation (Iriarte and Cornelis, 1998; Zumbihl *et al*., 1999). YopT cleaves Rho proteins directly upstream of the C-terminal cysteine residue, which is post-translationally geranylgeranylated (Shao *et al*., 2002). This cleavage removes the isoprenyl anchor, which is crucial for membrane binding, thus causing the release of Rho proteins from membranes into the cytosol and hence their inactivation (Fig. 6a). This in turn results in massive redistribution of the actin cytoskeleton (Shao *et al*., 2003). YopT requires the isoprenyl group and the C-terminal polybasic region of Rho proteins for substrate recognition, but the nucleotide-binding state of the GTPase does not seem to be important (Aktories, 2011). It acts on various Rho GTPases, but seems to prefer RhoA in HeLa and HUVEC cells (Aepfelbacher *et al*., 2003), but RhoG and Rac proteins are also targeted in COS1 cells. Cleaved Rac translocates into the nucleus, where it is degraded (Lanning *et al*., 2004). YopT and the related *P. luminescence* protease LopT, belong to a specific group of papain-like cysteine proteases that possess a catalytic site composed of cysteine, histidine and aspartate and are inhibited by the protease inhibitor E64 (Aktories, 2011).

1.4.2.5. Modulation of host GTPases by noncovalent modification

As previously mentioned, many T3S and T4S effectors modulate the GTPase cycle by mimicking host regulators. One group which includes the *Salmonella enterica* subsp. *enterica* serovar Typhimurium T3S effectors, SopE and SopE2 (Hardt *et al.*, 1998), and BopE from *Burkholderia pseudomallei* (Upadhyay *et al.*, 2008), ape the Rho activating GEF proteins. Through a GAGA motif, they bind GDP-loaded Rho proteins between the switch I and II regions, thus destabilizing the binding of $Mg^{2+}$and GDP. This in turn leads to GDP release and subsequently, binding of GTP, which is more abundant than GDP. This in turn induces a conformational change in the GTPase to yield the active form (Hardt *et al*., 1998; Upadhyay *et al*., 2008). Alternatively, instead of mimicking GEFs, other bacteria produce actual GEFs. These include proteins like, IpgB1 and IpgB2 from *Shigella flexneri* among others, which induce structural changes in Rho proteins similar to SopE (Alto *et al*., 2006; Klink *et al*., 2010; Ohya *et al*., 2005).

On the other hand, some effectors harbour GAP domains for inactivating Rho proteins. Like GAPS, YopE from *Yersinia pseudotuberculosis*, ExoS and ExoT from *Pseudomonas aeruginosa* contain an arginine finger. This is an arginine residue that combines with a highly conserved glutamine on the GTPase, to properly position GTP and facilitate the nucleophilic attack by a water molecule, to allow the GTP hydrolysis reaction and hence inactivation of the GTPase (Vetter and Wittinghofer, 2001).

1.5. Objectives of the project

The aforementioned features of the remarkable Tc toxin, present an opportunity for its use in clinical therapy, biological control and in further study of toxin activity and cell biology. For instance, the modular composition of the toxin complex allows loading and injection of various *Photorhabdus* toxins into mammalian cells. This makes it a possible vehicle for transport and injection of diverse enzymes and peptides of foreign origin. Additionally, the enclosed holotoxin structure provides a conduit that carries the active enzyme, sheltered from the external environment, into the cytosol. This would enable a choice of target-specific peptides to be injected into cells, as well as the study of toxins and effectors that lack intrinsic binding and uptake capabilities. However, as has become evident, a major drawback to the application of this toxin, is the fact that its receptor(s) and precise binding and uptake mechanism are yet to be elucidated. Other than expanding our knowledge base on intoxication by Tc toxins, knowledge of the receptors could help us to; (1) find ways to combat new pathogens like *P. asymbiotica* that

may use these toxins, or similar ones, in their pathogenicity. (2) To modify these toxins to achieve greater specificity to particular targets, for use in biocontrol and clinical therapy. (3) Mixing and matching Tc complexes from different sources, to increase their target host range and hence their potential for use. These possibilities and questions inspired this study.

In general, we aimed to elucidate the structure-functional characteristics of the *Photorhabdus* Tc complex. More specifically, we aimed to: (1) Elucidate the binding and activation characteristics of TcdA1. It had been shown before, that some bacterial toxins or toxin components like the anthrax toxin protective antigen (PA) and the binding component C2II of *C. botulinum* C2 toxin required cleavage by a protease in order to be activated (Klimpel *et al.*, 1992; Barth *et al.*, 2000). This raised the question of whether the *Photorhabdus* TcA proteins required, or could be activated by proteolytic cleavage via the bacteria's own proteases or host-cell surface proteases. If so, what would it mean for the mechanism of action of the Tc toxin?

The next objective (2) was to develop and analyse novel *Photorhabdus*-based Tc toxin complexes to characterize the translocation mechanism of protein cargo through the complex. The presence of multiple TcC isoforms in *Photorhabdus* (Ffrench-Constant and Waterfield, 2005; Roderer and Raunser, 2019) and characterization of TccC3 and TccC5 (Lang *et al.*, 2010) indicated the possibility of translocating a variety of peptides through the toxin complex. This raised the question of the requirements of a suitable cargo for the translocation system.

The third objective (3) was to determine and analyse putative receptors for the *Photorhabdus* toxin complex. As was mentioned previously, the receptors for the PTC remain unknown. What seems likely is that, based on the variety of RBDs, the toxins could bind multiple receptors (Roderer and Raunser, 2019). It is also likely that they could bind ubiquitously available receptors, which could account for their wide host range (Hinchliffe *et al*., 2010).

2. Experimental procedures

2.1. Materials

2.1.1. Bacteria and culture media

E. coli TG1 (Lucigen, Middleton, USA)

E. coli BL21 codon$^+$ (Stratagene, LaJolla, USA)

E. coli BL21 (DE3) (Stratagene, LaJolla, USA)

E. coli Oneshot Top10 (Invitrogen, now Thermofisher Scientific, Darmstadt, Germany)

E. coli strains were cultivated in LB (lysogeny broth) medium (Roth, Karlsruhe, Germany) supplemented with appropriate antibiotics (**Error! Reference source not found.**2).

Table 2: Antibiotics

	Supplier	***Final concentration***
Kanamycin	Roth (Karlsruhe, Germany)	30 µg/ml
Ampicillin	Roth (Karlsruhe, Germany)	100 µg/ml
Chloramphenicol	Sigma (Taufkirchen, Germany)	50 g/ml

2.1.2. Cell lines and growth media

Table 3: Cell lines and growth media

Name	***Cell type***	***Medium***	***Medium Supplement***
HeLa	Cervical cancer cells	DMEM (Dulbecco's Modified Eagle Medium)	10% FCS 1% penicillin /streptomycin 1% non-essential amino acids 1% sodium pyruvate
HeLa-EXT2	EXT2$^{-/-}$, Exostosin-2 deficient	DMEM	Same as for HeLa cells
CaCo-2	Colon adenocarcinoma cells	DMEM	Same as for HeLa cells
HT29	Colon adenocarcinoma cells	McCoy's 5A	10% FCS 1% penicillin /streptomycin
CHO-K1	Chinese hamster ovary cell K1	DMEM	Same as for HeLa cells
CHO-Lec1	GlcNAcT-I $^{-/-}$, N-glycan deficient mutant	DMEM/F12	10% FCS 1% penicillin /streptomycin
CHO-Lec2	Mutant *Slc35a1*, sialic acid deficient	DMEM	Same as for HeLa cells
CHO-Lec8	Mutant *Slc35a2*, galactose deficient	DMEM	Same as for HeLa cells

CHO-C6	Chinese hamster ovary cell C6	DMEM/F12	10% FCS 1% penicillin /streptomycin
CHO$^{\Delta pgsA\text{-}745}$	XT-I and XT-II$^{-/-}$, Xylosyltransferase deficient	DMEM/F12	10% FCS 1% penicillin /streptomycin
LADII	Leukocyte adhesion deficiency II cells	DMEM	Same as for HeLa cells
MEB4	Mouse B16 melanoma cells	DMEM	Same as for HeLa cells
GM95	Mouse B16 melanoma cells GlcT-1$^{-/-}$, glycolipid GM3 deficient	DMEM	Same as for HeLa cells
Schneider S2 cells	*Drosophila* S2 cells	Schneider S2 medium	10% heat-inactivated FBS

2.1.3. Vectors and plasmids

Vectors for expression in *E. coli*

pET-28a (Merck Millipore, Darmstadt, Germany)

pET-21a (Merck Millipore, Darmstadt, Germany)

pCR®-Blunt II-TOPO® (Invitrogen, now Thermofisher Scientific, Darmstadt, Germany)

Vectors for mammalian protein expression

pEGFP-C1, pEGFP-N1 (Clontech, Mountain View, CA, USA)

Table 4: Plasmids

Name	***Description***	***Reference***
pET-28a-TcdA1	6His-TcdA1	Lang *et al.*, 2013
pET-28a-BC3	6His-TcdB2-TccC3	Lang *et al.*, 2010
pET-21a-TccC3hvr	TccC3hvr	Lang *et al.*, 2010
pET-21a-TccC5hvr	TccC5hvr	Lang *et al.*, 2010
pET-28a-BC3^{E943A}	6His-TcdB2-TccC3 ^{E943A} (uncleavable hvr)	Pfaumann *et al.*, 2015
pET-28a-PrtA1	PrtA1	Bowen *et al.*, 2003
pEGFP-MgatI	MgatI	Eckhardt *et al.*, 2000

The following plasmids were created by modifying pET28a-BC3

Name	***Description***	***Reference***
pET-28a-BC3-C3bot	C3bot (without signal sequence) in TcdB2-TccC3$_{N\text{-}term}$	This work
pET-28a-BC3-YopT$^{\Delta 74}$	YopT$^{\Delta 1\text{-}74}$ in TcdB2-TccC3$_{N\text{-}term}$	This work
pET-28a-BC3-YopT$^{\Delta 30}$	YopT$^{\Delta 30}$ in TcdB2-TccC3$_{N\text{-}term}$	This work
pET-28a-BC3-YopT^{full}	Full YopT in TcdB2-TccC3$_{N\text{-}term}$	This work
pET-28a-BC3-PaTox^{G}	Glucosyltranferase domain of PaTox in TcdB2-TccC3$_{N\text{-}term}$	This work
pET-28a-BC3-ToxB^{GTD}	Glucosyltranferase domain of Toxin B in TcdB2-TccC3$_{N\text{-}term}$	This work

pET-28a-BC3-Mluc7	Metridia longa luciferase (without signal sequence) in TcdB2-TccC3$_{N\text{-}term}$	This work
pET-28a-BC3^{E943A}-C3bot	C3hvr^{E943A}-C3bot in TcdB2-TccC3$_{N\text{-}term}$	This work

The following plasmids were created by modifying pET28a- BC3-YopT$^{\Delta 74}$

Name	***Description***	***Reference***
pET-28a-BC3-YopT$^{\Delta 74}$-Lys3	YopT$^{\Delta 1\text{-}74}$ + 3 additional C-terminal lysines in TcdB2-TccC3$_{N\text{-}term}$	This work
pET-28a-BC3-YopT$^{\Delta 74}$-Lys6	YopT$^{\Delta 1\text{-}74}$ + 6 additional C-terminal lysines in TcdB2-TccC3$_{N\text{-}term}$	This work

2.1.4. Oligonucleotides

Table 5: Primer list

Name	***Sequence (5'→3')***	***Application***
5'-SalI in C3hvr	CCTGATGGATTAATGCCAACAATTGTC GACCGCATAGCAGCAC	Introduction of SalI restriction site after linker of C3hvr
3'-SalI in C3hvr	GTGCTGCTATGCGGTCGACAATTGTTG GCATTAATCCATCAGG	Introduction of SalI restriction site after linker of C3hvr
5'C3bot-SalI-new	GGGGTCGACGCTTATTCAAATACTTAC CAGGAGTTTAC	Sequencing of BC3-C3bot (5' SalI)
3'C3bot-XhoI	GGGGCTCGAGTTATTTAGGATTGATA GCTGTGCCCATCATT	Sequencing of BC3-C3bot (3' XhoI)
5'YopT-d47-SalI	GGGGTCGACGCTAACCAACGAAGCAG CTTTACCTTC	Sequencing of BC3-YopT$^{\Delta 74}$ (5' SalI)
3'YopT-d47-XhoI	GGGGCTCGAGTTAAACCTCCTTGGAG TCAAATGTTAACA	Sequencing of BC3-YopT$^{\Delta 74}$ (3' XhoI)
5'-YopT1-d30-SalI-new	GGGGTCGACATTGGCGCACACCGAGT GAAAGTGGAA	Sequencing of BC3-YopT$^{\Delta 30}$ (5' SalI)
3'YopT1-full-XhoI	GGGGCTCGAGTTAAACCTCCTTGGAG TCAAATGTAAAAAC	Sequencing of BC3-YopT$^{\Delta 30}$ (3' XhoI)
5'-YopT1-full-SalI	GGGGTCGACATGGACAGTATTCACGG ACACTACCATA	Sequencing of BC3-YopT^{full} (5' SalI)
3'YopT1-full-XhoI	GGGGCTCGAGTTAAACCTCCTTGGAG TCAAATGTAAAAAC	Sequencing of BC3-YopT^{full} (3' XhoI)
5'-YopT+Lys	AAGAAGAAGTAACTCGAGCACCACCA CCACCACCAC	Introduction of three lysines at the 3' end of BC3-YopT$^{\Delta 74}$
3'-YopT w/o Lys	AACCTCCTTGGAGTCAAATGTTAACAC G	Introduction of three lysines at the 3' end of BC3-YopT$^{\Delta 74}$
5'-YopT+6Lys	AAGAAGAAGAAGAAGAAGTAACTCG AGCA	Introduction of three lysines at the 3' end of BC3-YopT$^{\Delta 74}$-Lys3
3'-YopT w/o Lys	AACCTCCTTGGAGTCAAATGTTAACAC G	Introduction of three lysines at the 3' end of BC3-YopT$^{\Delta 74}$-Lys3
5'Patox-SalI	GGGGTCGACGAATGGTTTAAGCACTC AGAAACGGGC	Sequencing of BC3-PaTox^{G} (5' SalI)
3'Patox-XhoI	GGGGCTCGAGTCATGGTGCCAGAGGC GTTACTATTG	Sequencing of BC3-PaTox^{G} (3' XhoI)

5'ToxB-SalI	GGGGTCGACATGAGTTTAGTTAATAG AAAACAGTTAGAAAA	Sequencing of BC3-ToxB^{GTD} (5' SalI)
3'ToxB-XhoI	GGGGCTCGAGTTAAAGAGAACCTTCA AAATAATTCCTTTTATATT	Sequencing of BC3-ToxB^{GTD} (3' XhoI)
5'-SacI in C3hvr	CCTGATGGATTAATGCCAACAATTGAG CTCCGCATAGCAGCAC	Quickchange of 5' SalI restriction site to SacI after linker of C3hvr
3'-SacI in C3hvr	GTGCTGCTATGCGGAGCTCAATTGTTG GCATTAATCCATCAGG	Quickchange of 5' SalI restriction site to SacI after linker of C3hvr
EcoRI_MgatIF	GGGGCGAATTCATGCTGAAGAAGCAG	Sequencing of MgatI (Modified from Eckhardt *et al.*, 2000)
KpnI_MgatIR	CGGGGTACCATTCCAGCTAGGATCA	Sequencing of MgatI (Modified from Eckhardt *et al.*, 2000)

2.1.5. Antibodies

Table 6: Antibody list

Antibody (catalog number)	***Source***	***Supplier***	***Dilution factor (application)***
TcdA1 polyclonal	Mouse	Dr. Y. Belyi, Gamaleya research center (Moscow, Russia)	1:7000 (WB)
TcdA1ii	Mouse	Dow AgroScience (Joel Sheets)	1:4000 (WB)
TcdA1iii	Mouse	Dow AgroScience (Joel Sheets)	1:4000 (WB)
C3bot	Rabbit	Dr. C. Schwan, University of Freiburg	1:10000 (WB)
YopT polyclonal	Rabbit	Prof. G. Schmidt, University of Freiburg	1:10000 (WB)
PaTox^{G}	Rabbit	Dr. T. Jank, University of Freiburg	1:3000 (WB)
Toxin B^{GTD}	Mouse	Dr. T. Jank, University of Freiburg	1:10000 (WB)
EGFR-D381 (4267S)		Cell Signaling Technology (Frankfurt, Germany)	1:1000 (WB)
RhoA-67B9 (21175)		Cell Signaling Technology (Frankfurt, Germany)	1:1000 (WB)
Rac1/2/3 (2465S)		Cell Signaling Technology (Frankfurt, Germany)	1:1000 (WB)
GST	Rabbit	VWR (Darmstadt, Germany)	1:10000 (WB)
His-Tag (2365)	Rabbit	New England Biolabs (Frankfurt, Germany)	1:1000 (WB)
Mouse IgG HRP-linked	Rabbit	New England Biolabs (Frankfurt, Germany)	1:3000 (WB)
Rabbit IgG HRP-linked	Goat	Cell Signaling Technology (Frankfurt, Germany)	1:3000 (WB)
Avidin D HRP-linked		Biozol (Eching, Germany)	

2.1.6. Cloning and expression work

2.1.6.1. DNA and protein markers

GeneRuler 1kb Ladder (Thermo Fischer Scientific, Darmstadt, Germany)

PageRuler Prestained protein ladder (Thermo Fischer Scientific, Darmstadt, Germany)

Spectra multicolour high range protein ladder (Thermo Fischer Scientific, Darmstadt, Germany)

Unstained low molecular weight protein ladder (Thermo Fischer Scientific, Darmstadt, Germany)

2.1.6.2. Enzymes

Table 7: Enzymes

Enzyme	***Supplier***
Restriction endonucleases*	Thermo Fischer Scientific (Darmstadt, Germany)
T4 DNA ligase*	New England Biolabs (Frankfurt, Germany)
Pfu Ultra II polymerase*	Agilent technologies (Frankfurt, Germany)

*All enzymes were used following the manufacturer's instructions and with the supplied buffers.

2.1.6.3. Other enzymes

Table 8: Other enzymes

Enzyme	***Supplier***
Lysozyme	Carl Roth (Karlsruhe, Germany)
Clostridium histolyticum collagenase type IA	Sigma (now Merck, Darmstadt, Germany)
Trypsin	Sigma (now Merck, Darmstadt, Germany)
Chymotrypsin	Sigma (now Merck, Darmstadt, Germany)
Tritirachium album Proteinase K	Serva Electrophoresis (Heidelberg, Germany)
Vibrio cholera Neuraminidase	Sigma (now Merck, Darmstadt, Germany)
PNGase	New England Biolabs (Frankfurt, Germany)

2.1.7. Proteins and protein toxins

Table 9: Purified proteins and toxins used in this study

Name	***Source/Supplier***
TcdA1	Recombinant His-tagged protein from *E. coli* (Lang *et al*., 2013)
TcdB2-TccC3	Recombinant His-tagged protein from *E. coli* (Lang *et al.*, 2010)

TccC3hvr	Recombinant His-tagged protein from *E. coli* (Lang *et al.*, 2010; Pfaumann *et al.*, 2015)
TccC5hvr	Recombinant His-tagged protein from *E. coli* (Lang *et al.*, 2010; Pfaumann *et al.*, 2015)
BC3-C3bot	Recombinant His-tagged protein from *E. coli*
BC3-YopT$^{\Delta 74}$	Recombinant His-tagged protein from *E. coli*
BC3-YopT$^{\Delta 74}$-Lys3	Recombinant His-tagged protein from *E. coli*
BC3-YopT$^{\Delta 74}$-Lys6	Recombinant His-tagged protein from *E. coli*
BC3-YopT$^{\Delta 30}$	Recombinant His-tagged protein from *E. coli*
BC3-YopT^{full}	Recombinant His-tagged protein from *E. coli*
BC3-PaTox^{G}	Recombinant His-tagged protein from *E. coli*
BC3-ToxB^{GTD}	Recombinant His-tagged protein from *E. coli*
BC3-Mluc7	Recombinant His-tagged protein from *E. coli*
BC3^{E943A}-C3bot	Recombinant His-tagged protein from *E. coli*
Toxin B	Recombinant 6His-tagged protein purified from *B. megaterium* (Just *et al*., 1995)
C2IIb	Recombinant His-tagged protein from *E. coli* (Barth *et al*., 2000)
C2I	Recombinant His-tagged protein from *E. coli* (Barth *et al*., 2000)
PrtA1	Recombinant His-tagged protein from *E. coli*

2.1.8. Radiolabeled substrates and isotopes

UDP-[^{14}C]-glucose (Biotrend, Cologne, Germany)

UDP-[^{14}C]-N-acetyl-glucosamine (Biotrend, Cologne, Germany)

[^{32}P]-α-NAD^{+} (Perkin Elmer, Cologne, Germany)

^{86}Rubidium (Perkin Elmer, Cologne, Germany)

2.1.9. Other reagents

Table 10: Lab reagents

Substrate	***Supplier***
Bafilomycin A1	Enzo Life Sciences (Loerrach, Germany)
Biotin-NAD^{+}	Tocris (Wiesbaden-Nordenstadt Germany)
Bovine serum albumin, molecular biology grade	New England Biolabs (Ipswich, MA, USA)
Bovine serum albumin Fraction V (BSA)	Roth (Karlsruhe, Germany)
Bradford RotiQuant	Roth (Karlsruhe, Germany)

Coelenterazine H (Biotium)	Hölzel Diagnostika (Cologne, Germany)
cOmplete Protease Inhibitor Cocktail	Sigma (now Merck, Darmstadt, Germany)
Coomassie Brilliant Blue R250	Sigma (now Merck, Darmstadt, Germany)
Cycloheximide	
Dimethyl sulfoxide (DMSO)	Sigma (now Merck, Darmstadt, Germany)
DNAse I	Roche diagnostics (Mannheim, Germany)
dNTP mixture	Bioline (Luckenwalde, Germany)
Dynasore hydrate	Sigma (now Merck, Darmstadt, Germany)
Fluoresceinyl-aminomethyldithiolano-phalloidin	Enzo Life Sciences (Loerrach, Germany)
Isopropyl-β-D-1-thiogalactopyranoside (IPTG)	Bioline (Luckenwalde, Germany)
Membrane lipid strips	Echelon Biosciences (Salt Lake City, USA)
Midori Green	Biozym, Hessisch Oldendorf, Germany
Mowiol 40-88	Sigma (now Merck, Darmstadt, Germany)
Ni-IDA resin	Macherey-Nagel (Düren, Germany)
Paraformaldehyde (PFA)	Roth (Karlsruhe, Germany)
Phalloidin–Tetramethylrhodamine B isothiocyanate	Sigma (now Merck, Darmstadt, Germany)
Phenylmethylsulfonyl fluoride (PMSF)	Sigma (now Merck, Darmstadt, Germany)
Phosphatidylinositol 3-phosphate diC8 (PI3P)	Echelon Biosciences (Salt Lake City, USA)
PIP strips	Echelon Biosciences (Salt Lake City, USA)
Polyethylene glycol (PEG) 4000	Sigma (now Merck, Darmstadt, Germany)
Sialic acid (Neu5Ac)	Biosynth (Staad, Switzerland)
SignalFire ECL reagent	New England Biolabs (Ipswich, MA, USA)
Triton X-100	Sigma (now Merck, Darmstadt, Germany)
Tween 20	Sigma (now Merck, Darmstadt, Germany)

2.1.10. Kits

Substrate	***Supplier***
Dual Luciferase reporter assay system	Promega (Mannheim, Germany)
Qiagen plasmid MIDI kit	Qiagen (Hilden, Germany)
RotiPrep plasmid minikit	Roth (Karlsruhe, Germany)

2.1.11. Laboratory equipment

Table 11: Lab equipment

Name	***Supplier***
Äkta Purifier System	GE Healthcare, Munich, Germany

Amersham imager 600	GE Healthcare, Munich, Germany
AxioCam HRc	Carl Zeiss Microscopy GmbH, Jena, Germany
Axiovert 25	Carl Zeiss Microscopy GmbH, Jena, Germany
Axiovert 40CFI	Carl Zeiss Microscopy GmbH, Jena, Germany
Axiovert 200M	Carl Zeiss Microscopy GmbH, Jena, Germany
BD FACSCalibur	BD Biosciences, California, USA
Centrifuge 5424	Eppendorf, Hamburg, Germany
Centrifuge 5418R	Eppendorf, Hamburg, Germany
Countess II FL cell counter	Thermo Fischer Scientific (Darmstadt, Germany)
ECIS Z Theta	Applied BioPhysics, Munich, Germany
Endohm-12	World Precision Instruments, Friedberg, Germany
Gel Doc™ EZ Imager	Bio-Rad Laboratories GmbH, Munich, Germany
Heraeus Megafuge 1.0	Thermo Scientific, Waltham, USA
Infinite M200 microplate reader	Tecan, Männedorf, CH
LE-80 Ultracentrifuge	Beckman Coulter, Krefeld, Germany
Lionheart FX automated microscope	BioTek, Vermont, USA
LSM 800 microscope	Carl Zeiss Microscopy GmbH, Jena, Germany
Mini Protean II	Bio-Rad Laboratories GmbH, Munich, Germany
Mini Trans-Blot® Cell	Bio-Rad Laboratories GmbH, Munich, Germany
MP220 pH-Meter	Mettler Toledo, Giessen, Germany
NanoDrop ND-1000	Peqlab Biotechnologie GmbH, Erlangen, Germany
PD 10 columns	GE Healthcare Life Sciences, USA
PeqStar XS thermocycler	VWR International GmbH, Darmstadt, Germany
Pierce G2 Fast blotter	Thermo Fischer Scientific (Darmstadt, Germany)
PowerPac™ 300	Bio-Rad Laboratories GmbH, Munich, Germany
PowerPac™ basic	Bio-Rad Laboratories GmbH, Munich, Germany
Sorvall RC-5B Plus	Thermo Scientific, Waltham, USA
Sorvall lynx 4000	Thermo Scientific, Waltham, USA
Thermomixer compact	Eppendorf, Hamburg, Germany
Thermomixer pro	Eppendorf, Hamburg, Germany
Typhoon FLA 7000	GE Healthcare Life Sciences, USA
Ultrospec 2100pro spectrophotometer	Amersham biosciences (now GE Healthcare Life Sciences, USA)
Vortex-Genie 2	Scientific Industries, Bohemia, USA
Wet-blot System	Bio-Rad Laboratories GmbH, Munich, Germany

2.1.12. Software

Name	*Name*
Axiovision SE 64 Rel. 4.9	Multi Gauge V3.0
BioEdit	PyMOL 2.0
ClustalX 2.0	ImageJ/FiJi
Flowing Software 2	Image Reader LAS-3000
GENtle v1.9.4	Maxquant
Graphpad Prism 5	Metamorph
i.control 1.6	

2.2. Methods

2.2.2. Polymerase chain reaction (PCR)

DNA amplification was performed using Pfu Ultra II polymerase (Agilent technologies). Cycle conditions were adapted according to the specific template. However, the reaction mixture generally contained 50 ng of template DNA, 0.25 mM of each deoxynucleotide (dNTP), 0.5 µM of each primer (forward and reverse), 10x reaction buffer, 1 U polymerase and water added to achieve 50 µl total volume. The reaction was then run in a PeqStar XS thermocycler (VWR International) as shown below.

Step	*Temperature*	*Time*
Initial denaturation	95 °C	2 min
	95 °C	20 sec
30 Cycles	55-60 °C	20 sec
	72 °C	15-30 sec per kb
Final extension	4-10 °C	5-10 min
Hold	72 °C	

2.2.3. Agarose gel electrophoresis and gel extraction of DNA

DNA fragments were separated based on size in an agarose gel electrophoresis system, bands of interest were identified and then extracted from the gel. To prepare a 0.8-1.0 % agarose gel, an appropriate amount of agarose was dissolved in hot TAE buffer (Tris-Acetate-EDTA) (see below) in a glass flask. The solution was then allowed to cool and Midori green (1:50 000) was added, before being mixed briefly and poured into an electrophoresis tray with an appropriate comb. Subsequently, the gel was allowed to solidify at room temperature (RT),

immersed into TAE buffer, samples loaded and then run for up to 60 min at a constant voltage of 100 V. To determine the molecular sizes of the DNA fragments in the gel, an appropriate DNA ladder was loaded alongside the samples in the gel. Additionally, a 6X DNA loading buffer (Sigma) was mixed with the samples before loading. After electrophoresis, the gel was visualized under UV light and an image acquired using the Gel Doc EZ (Bio-Rad). Bands of interest were then cut out from the gel and DNA extracted using a Wizard SV Gel and PCR clean-up kit (Promega) according to the manufacturer's protocol.

50x Tris–Acetate-EDTA buffer (TAE)

	Final concentration
Tris-HCl	2 M
EDTA	50 mM
Glacial acetic acid	5 % (v/v)

2.2.4. Cloning and transformation

DNA restriction was performed with commercially available restriction enzymes at 37 °C according to the manufacturer's instructions. Digested fragments were separated by agarose electrophoresis and appropriately sized bands purified from the gel as previously described. Then, ligation into chosen plasmids was performed in 20 µl of 1X T4 ligation buffer containing 5 U of T4 DNA ligase and the maximal amount of DNA fragments in an insert to vector ratio of 7:1. The ligation mixture was incubated at 22 °C for 1 h and then used for transformation of *E. coli* TG1 competent cells. Alternatively, PCR fragments were cloned into pCR-Blunt II-TOPO plasmids (ThermoFischer) and these plasmids used to transform Oneshot Top10 *E. coli* according to the manifacturer's protocol. After sequencing, correct plasmids were then transferred to BL21 codon+ *E. coli* for protein expression.

For bacterial transformation, 50-100 µl of the chosen competent bacteria were mixed with 100-400 ng of plasmid DNA at 4 °C and incubated for 20-30 min. Then, a heat shock was performed by incubation of the DNA and cells mixture for 90 sec at 42°C. Then, the cells were transferred to ice for 2 min and then incubated for 1 h at 37 °C and 700 rpm of shaking in 250 µl of pre-warmed SOC medium. Afterwards, the bacterial cells were centrifuged at 4000 rpm for 3 min before 250 µl of the medium was removed. The cells were then resuspended in the remaining medium and plated on LB plates with the appropriate antibiotic(s).

2.2.5. DNA sequencing

Purified plasmid DNA or PCR products were sequenced to confirm their identity. Sequencing was performed as a commercially available service by GATC biotech (Cologne, Germany) following the dideoxynucleotide chain termination method (Sanger, 1975).

2.2.6. Protein expression and purification of Photorhabdus toxins

Photorhabdus proteins were expressed and purified as previously described (Gatsogiannis *et al*., 2013). Briefly, *E. coli* BL21 (DE3) cells were transformed with *P. luminescens* TccC3hvr as described in the previous section and used to inoculate LB culture medium containing relevant antibiotics. Then, protein expression was induced by the addition of IPTG to a final concentration of 75 µM. For *P. luminescens* TcdA1 or TcdB2-TccC3 (BC3) fusion proteins, *E. coli* BL21-Codon$^+$ cells were used and protein expression was induced with 25 µM IPTG. For TcdA1, a 200 ml preculture was first prepared, from which 20 ml was used to inoculate 1 L main cultures. For the main culture, bacteria were first grown 3-5 h at 37 °C up to OD_{600} = 0.6 before expression of the protein was induced and the bacteria further incubated overnight at 22 °C. For expression of BC3 and BC3-fusion proteins, main cultures were inoculated directly from plates and induced immediately after inoculation. Then, the bacteria were incubated overnight at 28 °C.

After the overnight incubation, all cells were harvested and resuspended in lysis buffer (shown below). They were then sonicated 10 × 15 sec with 15 sec rests in between at 0.7 cycles and 80% power. Afterwards, cell lysate was centrifuged for 45 min, 14000 rpm at 4 °C in a Sorvall lynx 4000 centrifuge (Thermo) to remove cell debris. Then, the supernatant was incubated for 2 h at 4 °C with Ni-IDA beads (150 mg/litre of main culture), and later loaded onto empty PD-10 columns for affinity chromatography. His-tagged proteins were washed (see buffer below), eluted (see buffer below) and the protein-containing fractions were pooled. Then, Bradford assay was performed before the samples were dialyzed (see buffer below) overnight. Later, the dialysed samples were run on an SDS-PAGE gel together with a molecular ladder. A 1 µg/µl sample of βγ actin was run alongside the samples to help determine the concentration of the proteins by densitometry. Purified samples were stored at - 80 °C in a storage buffer consisting of dialysis buffer, supplemented with 25 % glycerol.

Lysis buffer (*10 ml per 1l main culture*) **Wash buffer (100ml per *1l main culture*)**

	Final concentration		***Final concentration***
Tris-HCl (pH 8)	20 mM	Tris-HCl (pH 8)	20 mM
NaCl	500 mM	NaCl	500 mM
DTT	1 mM	Tween-20	0.05 %
EDTA	500 µM	Imidazole	20 mM
Glycerol	10 %	Glycerol	5 %
DNase I	5 µg/ml		
Lysozyme	1 mg/ml		
PMSF/cOmplete	1 mM		

Elution buffer (*1ml per 1l main culture*) **Dialysis buffer (*5l*)**

	Final concentration		***Final concentration***
Tris-HCl (pH 8)	20 mM	Tris-HCl (pH 8)	50 mM
NaCl	500 mM	NaCl	100 mM
Tween-20	0.05 %	Tween-20	0.05 %
Imidazole	500 mM	Glycerol	5 %
Glycerol	5 %		

2.2.7. Purification of PrtA1

BL21 competent cells were transformed as previously described. They were then inoculated into 200 ml of LB medium and a preculture was done overnight at 37 °C, 800 rpm. From this, 20 ml was used to inoculate 1 L of main culture supplemented with 50 µM $ZnCl_2$. The main culture was first grown to an OD of 0.6 before being induced with 0.2 mM IPTG and allowed to grow for 6 h at 30 °C. The cells were then harvested and lysed as previously described. All buffers used for this purification were supplemented with 50 µM $ZnCl_2$ and 5 mM $CaCl_2$. Being in inclusion bodies, the supernatant was discarded after centrifugation and the pellet washed with 10 ml/L of IB-wash buffer (below) before being centrifuged again. Afterwards, the pellet was solubilized for 1 h at 4 °C in 8 ml/L of urea-solubilization buffer (see below) and centrifuged a third time. Later the solubilized protein was incubated with Ni-IDA beads, loaded onto PD-10 columns and washed in buffer also supplemented with 6 M urea. The protein was then eluted in buffer also supplemented with 6 M urea and dialysed in buffer also supplemented with 0.5 M arginine without urea.

IB-Wasch buffer		**Urea-solubilization buffer**	
	Final concentration		***Final concentration***
Tris-HCl (pH 8)	20 mM	Tris-HCl (pH 8)	20 mM
NaCl	500 mM	NaCl	500 mM
Triton X-100	1 %	Urea	6 M
$ZnCl_2$	50 µM	$ZnCl_2$	50 µM
$CaCl_2$	5 mM	$CaCl_2$	5 mM

2.2.8. Sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE)

A method modified from Laemmli (1970) was used for separation of proteins by size under denaturing conditions in a discontinuous polyacrylamide gel. Samples were mixed with SDS-PAGE loading buffer in a ratio of 1:5 and incubated at 95 °C for 5 min to denature the proteins. Polyacrylamide mini gels were cast in a commercial system (Bio-Rad, Munich, Germany) following the manufacturer's instructions. First, the gel cast assembly was filled up to 80% volume with separation gel solution (see below) and overlaid with a small amount of isopropanol. The gel was allowed to polymerize at RT. Afterwards, the isopropanol was removed and the stacking gel solution (shown below) was added. Combs were inserted immediately, gels allowed to polymerize and before storage at 4 °C. Gel lanes were loaded with up to 30 µl of sample mixed with 1X SDS-PAGE sample buffer (see below) or with 10 µl of a protein ladder. Electrophoresis was conducted under a constant electric current of 35 mA per gel for the time required for the dye front to run out. To visualize proteins, gels were either subjected to Coomassie staining (see below) or a Western blot was performed as described in the proceeding section.

SDS-PAGE gel casting solutions

	Separation gel (12.5 %)		***Stacking gel (7 %)***	
Component	***Stock concentration***	***Volume***	***Stock concentration***	***Volume***
Water		4.8 ml		20.2 ml
Acrylamide	30 %	31.3 ml	30 %	5.65 ml
Tris (tris(hydroxymethyl)-amino methane)	0.75 M, pH 8.8	37.5 ml	0.625 M, pH 6.8	6.5 ml
SDS (sodium dodecyl sulfate)	20 %	375 µl	20 %	150 µl
APS (ammonium persulfate)	10 %	750 µl	10 %	300 µl

TEMED (N,N,N',N'-tetramethylethylenediamine)	10 %	300 µl	10 %	300 µl

SDS-PAGE sample buffer (5x)

	Final concentration
Tris (pH 6.8)	100 mM
SDS	10 % (w/v)
Glycerol	34 % (v/v)
Bromophenol blue	0.01 % (w/v)
1,4 Dithiothreitol (DTT)	100 mM

Coomassie solutions

Staining solution		***Destaining solution***	
	Final concentration		***Final concentration***
Methanol	45 % (v/v)	Methanol	5 % (v/v)
Acetic acid	10 % (v/v)	Glacial acetic acid	10 % (v/v)
Coomassie Blue R250	0.25 % (w/v)		

2.2.9. Western blot

To perform a Western blot analysis, proteins separated by SDS-PAGE were first transferred to a PVDF membrane over an electric current. For this, the membranes were pretreated with methanol for 30 sec and then placed in transfer buffer (20 % (v/v) Methanol, 25 mM Tris-HCl, pH 7.4 and 190 mM glycine). A transfer sandwich was then assembled with the membrane facing the anode and gel facing the cathode, sandwiched between two sheets of Whatman paper in transfer buffer. The assembly was then positioned in the blotter subjected to a constant voltage. Two forms of blots were performed, either a wet blot (140 v for 95 min) in a Mini Trans-Blot® Cell (Biorad), or a semi-dry blot (25 v, 10 min) in a PierceG2 fast blotter (ThermoScientific).

To visualize the blots, immunodetection was performed. Blotted membranes were first blocked at RT for 1 h, to minimize unspecific binding of antibodies by incubation with blocking solution containing 5 % (w/v) fat-free milk powder or 5% bovine serum albumin in TBS-T (50 mM Tris-HCl, pH 7.4, 150 mM NaCl and 0.1% Tween-20). The blocked membrane was then incubated overnight with the primary antibodies in TBS-T buffer at 4 °C on a shaking platform. The membranes were then washed three times for at least 10 min with TBS-T before

incubation for 1 h at RT with an HRP-conjugated secondary antibody. After three more washing steps with TBS-T, fluorescence was induced using 1x SignalFire ECL Reagent (NEB) and detected using an Amersham imager 600 (GE healthcare).

2.2.10. Cell cytotoxicity assays

All human and mammalian cell lines were cultivated in their appropriate media (See Table 2) and cultured at 37 °C and 5 % CO_2. Drosophila S2 cells were cultured at 27 °C without CO_2. For cytotoxicity experiments, cells were seeded in culture dishes and incubated in hunger medium (0.5 % FCS) together with the respective toxins. Where pharmacological inhibitors were used, the cells were preincubated with the inhibitors for the stated duration before intoxication. After the indicated incubation periods, cells were visualized using an Axiovert 40CFI microscope (Zeiss) with a Jenoptik Progress C10 CCD camera (Jena, Germany). The cytopathic effects caused by the toxins were analyzed in terms of morphological changes and quantified by counting the number of intoxicated cells.

2.2.11. Transepithelial electrical resistance (TEER) assays

For the TEER assay, CaCo-2 cells were seeded onto Millicell hanging cell culture inserts (Merck Chemicals, Darmstadt, Germany) and incubated 5-7 days with medium exchange every 3 days to allow for formation of a cell monolayer. Then, in an Endohm-12 (World Precision Instruments), a resistance system for electrophysiological readings of filter cups, an electrical current was passed through the cell monolayer and the subsequent transepithelial electrical resistance (TEER) was measured. When TEER values reached ~1800-2000 ohms/cm^2, assays were performed. Alternatively, instead of using cell culture inserts, the Electrical Cell-substrate Impedance Sensing (ECIS) system (Applied biophysics) was used. Here, cells were seeded onto 8-well 8W10E+ ECIS arrays (Ibidi GmbH, Martinsried, Germany) and cultured for 2 days. When TEER values remained constant at ~1000-1600 ohms (100% confluence), assays were performed. For culture inserts, ready monolayers were directly intoxicated, incubated at 37 °C and checked hourly for changes in resistance. On the other hand, ECIS arrays were first precooled on ice for 15 min before intoxication. The toxins were allowed to bind for 1 h on ice and then the arrays were transferred to 37 °C, where TEER was measured automatically. For each assay, the *Photorhabdus* toxin complex 3 (PTC3), consisting of 0.7 nM TcdA1 and 1.75 nM BC3 was used as a positive control. Subsequently, the resistance readings were tabulated and statistically analyzed.

2.2.12. Fluorescence-activated cell sorting (FACS) analysis

For analysis of toxin binding onto target cells, proteins labeled with DyLight488 (Thermo Scientific) according to the manufacturer's recommendations were used. First, HeLa cells cultured on 10 cm dishes were detached using 10 mM EDTA in PBS, washed twice with PBS and adjusted to 100,000–500,000 cells per ml. Second, labeled proteins were added and the cells incubated in the dark at 4 °C on a rotary shaker. Third, after the indicated incubation time, the cells were washed twice in PBS and subjected to FACS analysis using the BD FACSCalibur platform (BD Biosciences). Cell surface-bound fluorescence was detected with an argon ion laser (488 nm) and 530 nm band-pass filter.

2.2.13. ADP-ribosylation of actin and RhoA

In vitro ADP-ribosylation of actin was performed as previously described (Lang *et al*., 2017). Briefly, 1.25 nM of TccC3hvr, 3 nM of BC3 or 0.8 nM of CDTa, was incubated at 21°C with 1.9 µM of recombinant βγ-actin in ADP-ribosylation buffer (see below). After the indicated time periods, the reaction was stopped by addition of SDS-containing Laemmli buffer. Then, the samples were subjected to SDS-PAGE and radiolabeled actin was detected and visualized by autoradiography in a Typhoon FLA 7000 instrument (GE Healthcare). For *in vitro* ADP-ribosylation of RhoA, 60 nM of recombinant wildtype C3bot or BC3-C3bot fusion protein, was incubated at 37°C with 0.4 µM GST-RhoA in RhoA ADP-ribosylating buffer (shown below). After the indicated time points, the reaction was stopped and processed as previously described. In some instances, ADP-ribosylation reactions were performed post intoxication of cells. In these cases, 15-30 µg of cell lysate from the intoxicated cells was used as a substrate in the described reactions, without addition of unlabeled NAD^+ and BSA.

Actin ADP-ribosylation buffer

	Final concentration
Hepes buffer (pH 7.5)	5 mM
$CaCl_2$	0.1 mM
ATP	0.1 mM
NAD^+	150 µM
$[^{32}P]$-NAD^+	0.5 µCi per sample
NaN_3	0.5 mM

RhoA ADP-ribosylation buffer

	Final concentration
TEA	25 mM
$MgCl_2$	2 mM
DTT	1 mM
GDP	1 mM
BSA	2 µg
NAD^+	1 µM
$[^{32}P]$-NAD^+	0.5 µCi per sample

2.2.14. Glycosylation reaction

In vitro glycosylation of Rac1 was performed as previously described (Jank *et al*., 2013). Recombinant Rac1 (6 µM) was incubated at 30°C with 60 nM of either recombinant $PaTox^{G}$, BC3-$PaTox^{G}$, Toxin B^{GTD} or BC3-$ToxinB^{GTD}$ in a buffer A (PaTox) or buffer B (Toxin B) (see below) for indicated time points. As cosubstrates, 10 µM of UDP-[^{14}C]-GlcNAc and UDP-[^{14}C]-glucose were used for PaTox- and Toxin B-based toxins. A 20 µl total reaction volume was prepared each time and the reaction was stopped by addition of SDS-containing Laemmli buffer. Labelled proteins were analyzed by SDS-PAGE followed by phosphorimaging inTyphoon FLA 7000 reader (GE Healthcare).

10x Glycosylation buffer A

	Final concentration
HEPES (pH 7.5)	500 mM
$MgCl_2$	20 mM
$MnCl_2$	10 mM

10x Glycosylation buffer B

	Final concentration
HEPES (pH 7.5)	500 mM
$MgCl_2$	20 mM
$MnCl_2$	10 mM
KCl	1 M

2.2.15. Membrane-release assays

For *in vitro* membrane release assays a modified protocol by Sorg *et al.* (2001) was used. HeLa cells cultured in 10 cm dishes were washed once with phosphate buffered saline (PBS) and then detached by scrapping in 500 µl of lysis buffer (see below). The cells were then lysed by sonication and clarified by centrifugation at 1,000 × g for 10 min at 4 °C. Afterwards, membranes and cytosol were separated by ultracentrifugation at 32,000 rpm for 1 h at 4 ° C in an LE 80 ultracentrifuge (Beckman Coulter). Subsequently, the pellet, consisting of the membrane fraction, was retained while the supernatant was discarded. This pellet was then resuspended in lysis buffer and either treated with 0.5 µM of each BC3-YopT fusion protein for 30 min at 37 °C, or left untreated and just incubated. The samples were then ultracentrifuged again to separate the pellets (membranes) from the supernantants containing released Rho GTPases. Finally, samples were mixed with SDS-containing Laemmli buffer, separated by SDS-PAGE and used for Western blots.

For *in vivo* assays, HeLa cells in 10 cm dishes were first intoxicated with 20 mM of TcdA1 and BC3-$YopT^{full}$ for 4 h at 37 °C, followed by 1 h with 4 nM of CNF1. Afterwards, the cells were washed, lysed, clarified and the lysates ultracentrifuged as described above to separate

membranes (pellet) from cytosol (supernatant). The pellets were then immediately resuspended in Laemmli buffer, while the supernatants were precipitated by addition of trichloroacetic acid (1:9 ratio), incubated for 30 min at 4 °C, followed by centrifugation at 14,000 rpm for 5 min. The pellets from these precipitated supernatants were then resuspended in Laemmli buffer and together with the membrane fraction, run on SDS-PAGE and blotted as above.

Lysis buffer for membrane release assays

	Final concentration
Tris-HCl (pH 7.4)	50 mM
NaCl	150 mM
$MgCl_2$	5 mM
EDTA	1 mM
DTT	2.5 mM
PMSF	1 mM

2.2.16. Inhibition of cellular stress fibers

Four sets of Hela cells were cultured on coverslips overnight. Then, the cells were washed with PBS and incubated in starvation medium (DMEM with 0.5% FCS). Afterwards, two sets of cells were intoxicated for 4 h at 37 °C, with 20 nM of TcdA1 together with all the indicated BC3-YopT fusion proteins. Then, one set of YopT-treated cells and an untreated set were intoxicated with 4 nM of CNF1 for 1 h. After CNF1 intoxication, the cells were washed twice with PBS, fixed with 4% paraformaldehyde for 15 min, washed again with PBS and permeabilized with 0.15% (v/v) Triton X-100 for 10 min. Subsequently, the cells were incubated in the dark with *Phalloidin*–tetramethylrhodamine (Sigma) for 2 h at RT to fluorescently label actin filaments. After a wash and treatment with 70% and 100% ethanol respectively, the cells were dried and embedded with Mowiol supplemented with DABCO (1,4-Diazabicyclo[2.2.2]octane) and DAPI (4´,6-diamidino-2-phenylindole) for nuclear staining. Finally, samples were incubated for 24 h and the cells visualized using an Axiophot system (Zeiss, Oberkochen, Germany).

2.2.17. Bioluminescence assays

In vitro bioluminescence assays, were performed by adding the indicated concentration of BC3-Mluc7 to 200 µl of either passive lysis buffer (Promega), HeLa cell lysate, or indicated buffer. Then, this BC3-Mluc7 mix was aliquoted into a white plate and the plate transferred to

an infinite M200 microplate reader (Tecan). Here, 30 µl of Stop & Glo® solution (Promega), diluted 1:1 with Passive lysis buffer, was dispensed into each well and the resultant luciferase activity recorded. For *in vivo* assays, HeLa cells cultured in 10 cm dishes were detached, counted and 500,000 cells/ml taken in PBS for each treatment. The cells were incubated for 1 h at 37 °C with 10 nM of either BC3-Mluc7 only, or the full toxin complex PTC3-Mluc7 (TcA + BC3-Mluc7). Afterwards, the cells were collected by centrifugation for 5 min at 300 × g, washed and then lysed in 200 µl of passive lysis buffer on a rotary shaker for 30 minutes at RT. The protein concentration of the lysate was determined by Bradford assay, after which it was aliquoted into a white plate and luciferase activity detected as described.

For pH-dependent bioluminescence assays, HeLa cells were first incubated with 100 nM of Bafilomycin A1 (Enzo life sciences) for 30 min at 37 °C. Then, they were transferred to ice, precooled to 4 °C. Afterwards, BC3-Mluc7 only or PTC3-Mluc7 was added and incubated for 1 h in 1× Hank's balanced salt solution (HBSS) buffer supplemented with 20 mM Hepes, pH 7.5, to allow for toxin binding. This experimental setup ensured toxin binding at the cell surface without uptake into the cells. Then, to induce injection of Mluc7 into the cells by the surface-bound toxin complex, the pH of the surrounding medium was changed either to acidic pH 5 medium (medium with 0.5% FCS and 20 mM MES, pH 5), or neutral pH 7.5 medium (medium with 0.5% FCS and 20 mM Hepes, pH 7.5) as indicated. This pH change mimicked the endosomal pH required to induce pore formation and translocation of the payload into the cytosol. These cells in pH medium were incubated for 1 min at 37 °C before the medium was shifted back to pH 7.5. Afterwards, the cells were washed and treated with trypsin/EDTA for 2 min at 37 °C to degrade extracellular protein and to simultaneously detach them from the plate. Finally, the cells were harvested, lysed in passive lysis buffer and transferred into a white plate for detecting luciferase activity.

For bioluminescence microscopy, CaCo-2 cells were treated as in the pH-dependent assays described above without Bafilomycin A1. However, these cells were not detached by the trypsin/EDTA treatment and were thus able to be visualized by a Zeiss LSM 800 microscope. After intoxication with 10 nM of BC3-Mluc7 only or PTC3-Mluc7, 2 µg/ml of coelenterazine H (Biotium) diluted in PBS was added and image acquisition started. To allow simultaneous bioluminescence and DIC imaging, the 405-nm laser was set to 0.005 mW. Bioluminescence images (between 450 and 700 nm) were acquired with the GaAsP detector set to maximum

gain. Images were taken every 30 sec for a total duration of 30 min. As controls, substrate only, TcdA1 only and full toxins with $YopT^{\Delta 1-74}$ and $YopT^{full}$ were used.

2.2.18. Data analysis and representation

In general, samples measured multiple times constituted technical replicates. Biologically distinct samples including protein preparations and different cell lines or cultures measured in parallel were defined as biological replicates. Repeated experiments (biochemical reactions or assays) performed independently but in the same experimental conditions and setup, and with the same biological samples were regarded as independent experiments.

Graphpad prism 5 and Microsoft excel were used for the statistical analysis and presentation. Normal distribution of data points was assessed by the Shapiro-Wilk test. For comparison of groups of data, t-test and ANOVA analyses were performed. A P-value below 0.05 was generally considered a significant difference in the treatment groups and was indicated with a single asterix (*) in a graph. More statistically significant differences were indicated as follows, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$. Insignificant differences were indicated with “ns”. Error bars represent the standard error of the means (SEM) unless indicated otherwise.

For analysis of bioluminescence assays, relative luminescence units (RLU) were determined. For this, luminescence values were divided by the total protein concentration of the respective lysates. In order to estimate the efficiency of the *Photorhabdus* toxin complex (PTC) in toxin delivery. The bioluminescence of Mluc7 was used. As a selected cargo of the PTC, it could enable us to do the following; (1) to visualize translocation without relying on biochemical modifications of the cells, which were in most cases detrimental to the cells. (2) To quantify the amount of cargo delivered into live cells by the PTC. This quantification involved the determination of the number of luciferase molecules delivered into a single cell by the PTC.

The process of quantification was as follows. First, light units emitted from a single cell were calculated. For this, data from *in vitro* bioluminescence assays was used to perform a linear regression, which generated the concentration of BC3-Mluc7 (number of protein molecules) that produces one light unit (LU). Second, from the pH-dependent assays, luminescence values obtained from 1 nM of PTC3-Mluc7 were used to calculate the respective number of luciferase molecules delivered in total. Third, a separate quantification of the number of cells per well,

was used to narrow down to light units emitted from a single cell, and finally, from that, the number of luciferase molecules delivered per cell was calculated.

3. Results

3.1. Intoxication by the *Photorhabdus* Tc toxin leads to cell rounding

It was previously shown that intoxication of cells by the *Photorhabdus* toxin complex (PTC) leads to cell rounding as a result of toxin-induced acting clustering (Fig. 4d), which subsequently results in the collapse of the cell cytoskeleton (Lang *et al.*, 2010). Therefore, to study these effects of the toxin on cells, an intoxication assay was performed. HeLa cells were intoxicated with affinity-purified TcdA1 (TcdA1^{aff}) (see methods, sub-section 2.2.6 and Fig. 8 for TcdA1 nomenclature), together with the BC3 complex for 5 h at 37 °C. Pictures of the intoxicated cells were taken every hour and the number of rounded cells was quantified. As expected, intoxication resulted in cell rounding (Fig. 7a), and the number of rounded cells increased over time (Fig. 7b).

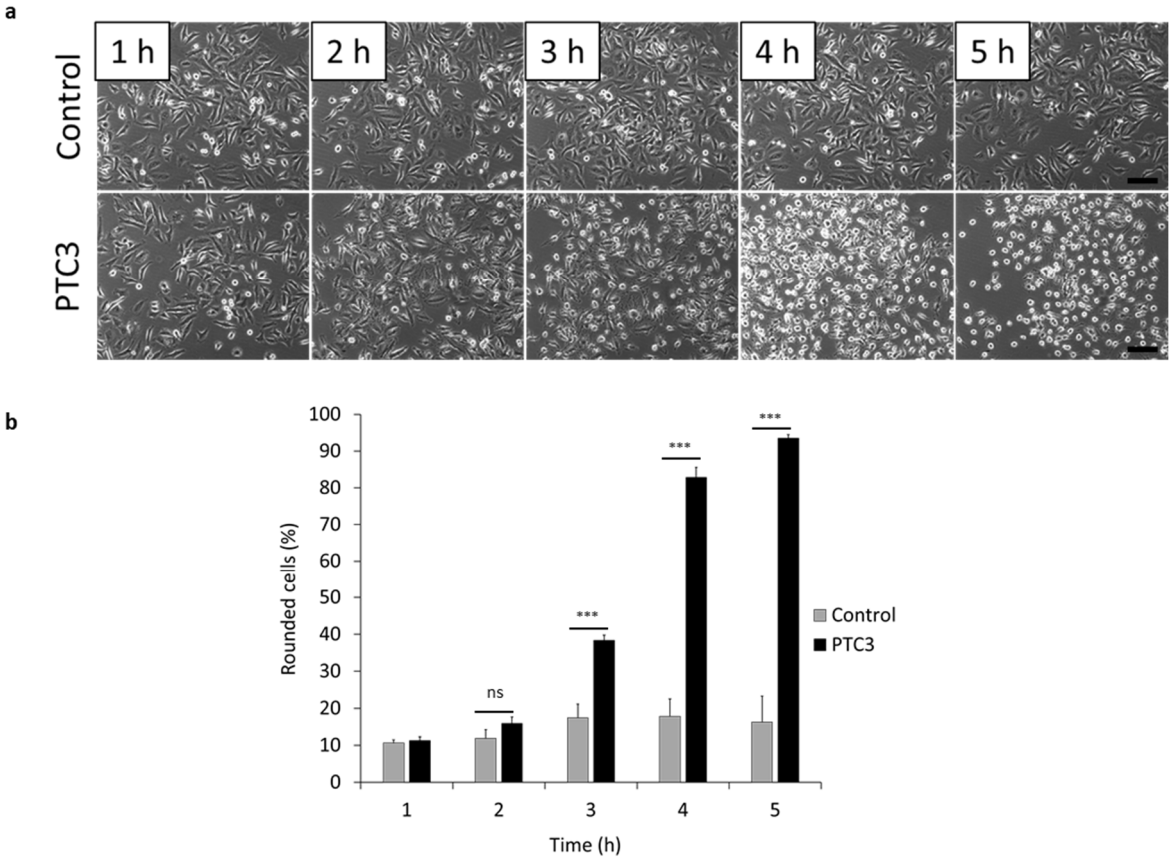


Figure 7: **Intoxication of HeLa cells with the Photorhabdus toxin complex 3 (PTC3)**. *HeLa cells were intoxicated for 5 h at 37 °C with PTC3 (TcdA1^{aff} (2 µg/ml) + BC3 (1 µg/ml)) in starvation medium. Then, pictures of the intoxicated cells were taken every hour (**a**) and the number of rounded cells was quantified as a percentage of total (**b**). Data presented as ± standard error of the mean (SEM) (n = 3). Significance was tested by using the Student's t test (P < 0.001 (***), ns = not significant). Scale bar is 80 µm.*

3.2. TcdA1 is activated by proteolytic cleavage

3.2.1. TcdA1 is activated by a *Photorhabdus* metalloprotease (PrtA1)

Bacterial toxins or toxin components including the anthrax protective antigen (Klimpel *et al*., 1992), *C. botulinum* C2II toxin (Barth *et al*., 2000) and iota-like binary toxins like *C. difficile* toxin (CDT), iota toxin and *C. spiroforme* toxin (CST) (Barth *et al*., 2004), are known to be processed by proteases. Additionally, it was previously shown that *P. luminescence* produces a protease that can cleave a TcA-like protein (Guo *et al.*, 1999). This protease had a molecular weight similar to the *Photorhabdus* metalloprotease PrtA1 (Bowen *et al*., 2003). Furthermore, cleaved TcdA1 was also mentioned in various other studies (Blackburn *et al*., 1998; Bowen *et al*., 1998; Liu *et al*., 2003). Moreover, *P. luminescence* was shown to secrete PrtA1 along with TcA proteins in target insects (Silva *et al*., 2002). Taken together therefore, it was likely that TcdA1 may be a substrate for PrtA1.

To study activation of TcdA1, a cleavage and purification protocol was designed. Affinity purified TcdA1 ($TcdA1^{aff}$) (see methods, sub-section 2.2.6 and Fig. 8) was treated with PrtA1 (Bowen *et al*., 2003). For this, 3 µg of TcdA1 was first incubated for 3 h with PrtA1 (9.26 nM) at 37 °C in dialysis buffer. Then, the cleaved toxin was separated from the enzyme by fast protein liquid chromatography (FPLC), in an Äkta purifier system (GE healthcare) in the same buffer without glycerol ($TcdA1^{FPLC/PrtA1}$). The Äkta purification involved a run of gel filtration/size-exclusion chromatography in a Superose 6 Increase column (GE healthcare, Freiburg, Germany). TcdA1 (~1,400 kDa, pentamer) was therefore separated from PrtA1 (~56 kDa). As a control, $TcdA1^{aff}$ incubated in the same conditions in buffer only without the enzymes, was purified by FPLC ($TcdA1^{FPLC}$). This procedure is summarised schematically in Fig. 8a. Alternatively, $TcdA1^{FPLC}$ was treated with PrtA1 and purified by FPLC a second time to remove the enzyme. This yielded $TcdA1^{FPLC/PrtA1post}$ (Fig. 8b).

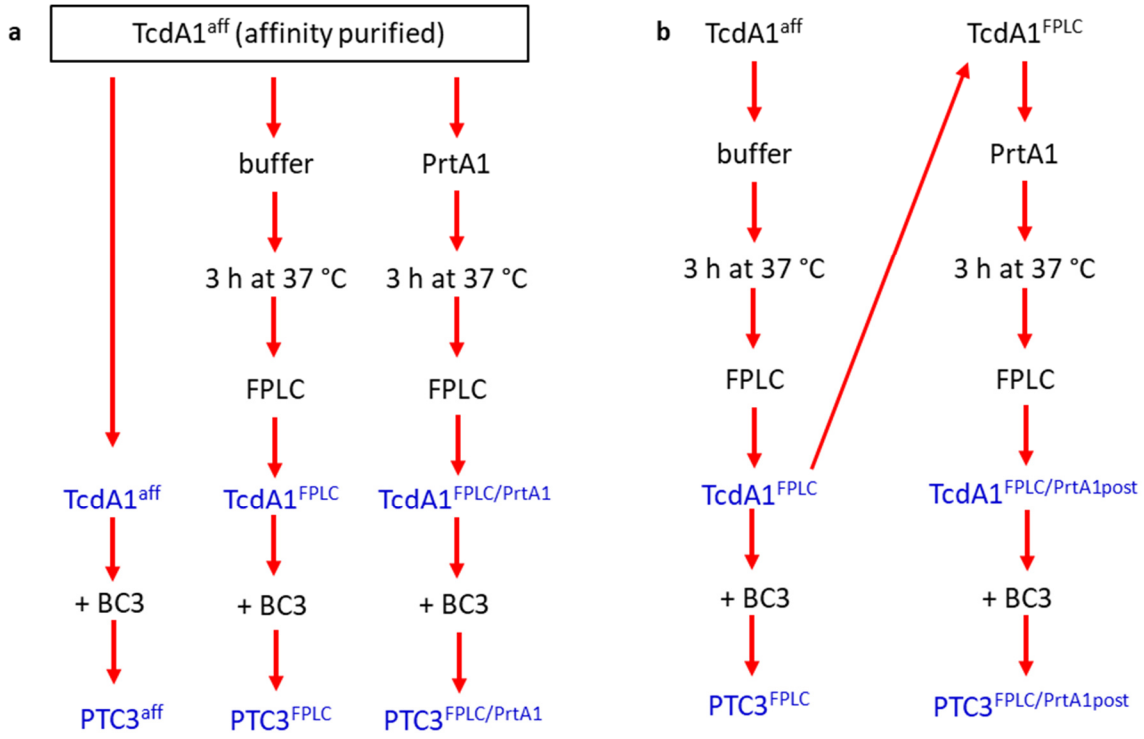


Figure 8: Schematic representation of the purification of enzyme-treated TcdA1*. (**a**) Affinity purified TcdA1 (TcdA1^{aff}) was either treated with the enzyme PrtA1, or buffer only and then further purified by FPLC. (**b**) FPLC-purified TcdA1 (TcdA1^{FPLC}) was treated with PrtA1 and then purified a second time to achieve TcdA1$^{FPLC/PrtA1post}$.*

To investigate cleavage of TcdA1 by PrtA1 and the effects thereof, affinity-purified TcdA1 (TcdA1^{aff}) (3 µg) was treated with increasing concentrations of PrtA1 for 3 h at 37 °C. Since PrtA1 itself has cytotoxic effects on mammalian cells (Bowen *et al*., 2003), it was separated from TcdA1 by gel filtration (FPLC) as described above (Fig. 8). As a control, uncleaved TcdA1^{aff}, was treated similarly to the cleaved toxin and also purified by FPLC. Subsequently, all toxins were analysed by SDS-PAGE. As shown in Fig. 9a, TcdA1 was cleaved into two fragments ~63 kDa and ~208 kDa. However, interestingly, both cleaved fragments eluted as one peak upon gel filtration (see Appendix 1).

To test the effect of this cleavage on toxin activity, cleaved toxin binding component (TcdA1$^{FPLC/PrtA1}$) and the BC3 component were added to cells. As shown in Fig. 9c and d, TcdA1$^{FPLC/PrtA1}$ caused a very rapid intoxication of cells, as evidenced by cell rounding. Additionally, MG132, a proteasome inhibitor (Tsubuki *et al*., 1996) which was previously shown to protect cells from the *Photorhabdus* toxin complex (PTC) (Ost *et al*., 2019), neither affected cleavage of TcdA1 (Fig. 9b), nor intoxication of cells by the PrtA1-treated toxin

(PTC3$^{FPLC/PrtA1}$) (Fig. 9c, d). However, as was previously observed, it blocked intoxication by PTC3^{aff} and PTC3^{FPLC}.

Interestingly, the control uncleaved TcdA1^{FPLC}, which was similarly purified by gel filtration like the cleaved toxin, showed a much slower intoxication than the non-FPLC-purified toxin (TcdA1^{aff}) (Fig. 9c, d). This intoxication was even slower after 5 h in the presence of MG132 (Fig. 9d). Furthermore, when TcdA1^{FPLC} was then treated with PrtA1, rapid intoxication was restored (Fig. 10a). Note that TcdA1^{FPLC} was treated with PrtA1 after the initial gel filtration and then filtered again after enzyme treatment (TcdA1$^{FPLC/PrtA1post}$) (Fig. 8b). A direct comparison of the two FPLC-purified toxins, TcdA1^{FPLC} and TcdA1$^{FPLC/PrtA1post}$, revealed an 8- to 10-fold increase in toxin potency due to protease treatment (Fig. 10b). This data therefore indicates that the TcA component of Tc toxin is strongly activated by proteolytic cleavage.

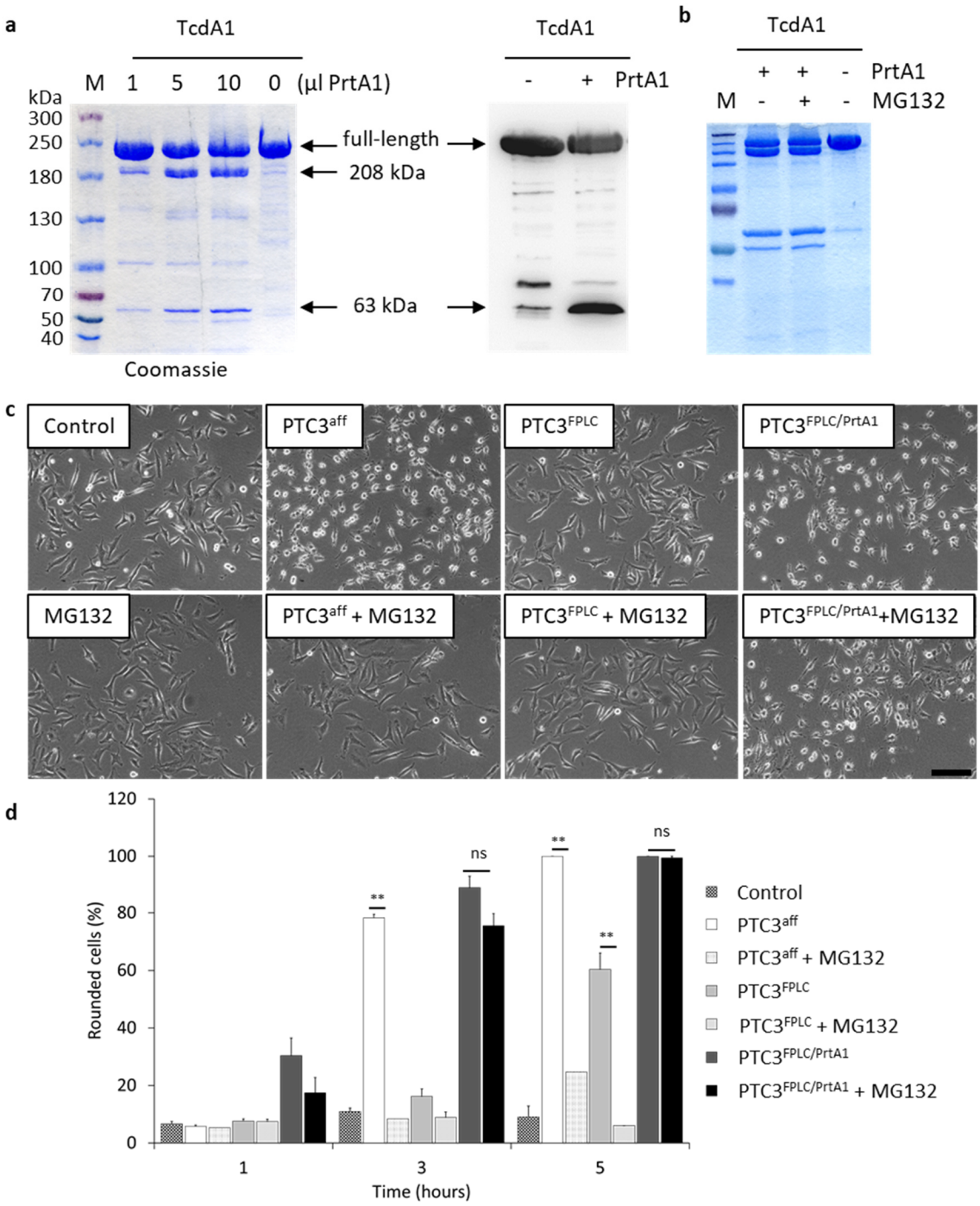


***Figure 9*: Cleavage of TcdA1 with the Photorhabdus metalloprotease PrtA1**. *(**a**) 3 µg of TcdA1^{FPLC} was incubated with increasing amounts of PrtA1 for 3 h at 37 °C. The cleaved products were then separated by SDS-PAGE and either stained by Coomassie (left panel) or detected via Western blot (right panel). M = molecular marker. (**b**) Protease cleavage of TcdA1 similar to (a) in the presence of MG132. (**c**) Protease cleaved toxins (3 µg/ml) were added to HeLa cells together with the BC3 component (1.5 µg/ml) and incubated for 5 h at 37 °C. (**d**) Rounded cells intoxicated in (c) were quantified as a percentage of total. Data presented as ± standard error of the mean (SEM) (n = 3). Significance was tested by using the Student's t test (P < 0.01 (**), ns = not significant). Scale bar is 80 µm.*

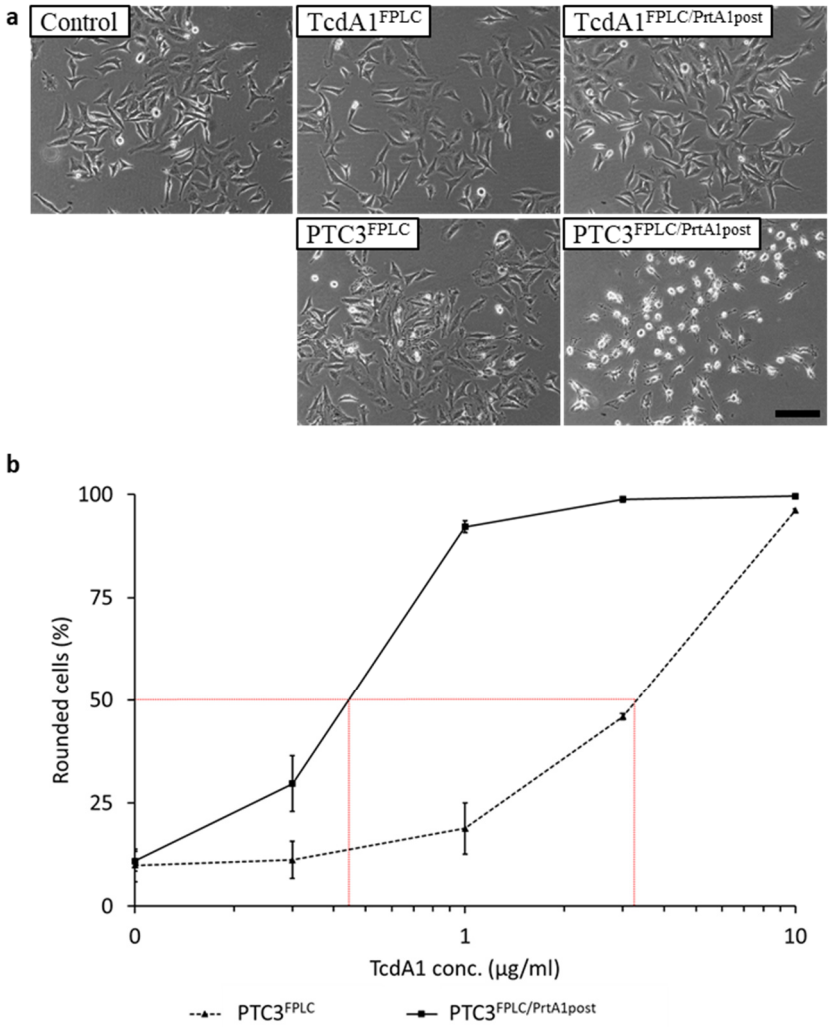


Figure 10: Cleavage of TcdA1^{FPLC} by PrtA1 increases toxicity*. (**a**) HeLa cells were treated with TcdA1^{FPLC} variants (1.5 µg/ml) + BC3 (0.75 µg/ml) for 4 h. Scale bar, 80 µm. (**b**) HeLa cells were treated with increasing concentrations of TcdA1 variants together with BC3 (always the half concentration as indicated for TcdA1). After 5 h, the percentage of rounded cells was determined from the pictures and plotted in logarithmic scale. Values are given as mean ± SD (n = 3). Red lines indicate the concentration of both TcdA1 variants required to achieve 50% cell rounding. For explanation of the nomenclature and purification of TcdA1 variants, see Figure 7 in materials and methods. Scale bar is 80 µm.*

A Maldi–TOF–MS analysis of the small PrtA1-cleaved ~63-kDa fragment on Fig. 9a was then performed to identify the protease cleavage site of TcdA1. This analysis revealed that the ~63-kDa fragment mainly represented the C- terminal α-helical shell of TcdA1 (amino acids 1762 – 1972) (Fig. 11). Interestingly, contained within this fragment was a small loop, which was unresolved in the TcdA1 crystal structure (1VW1) (Fig 11, right panel). This loop encompassed a putative cleavage site between amino acids 1937 - 1938 (Blackburn *et al*., 1998; Bowen *et al*., 1998; Liu *et al*., 2003).

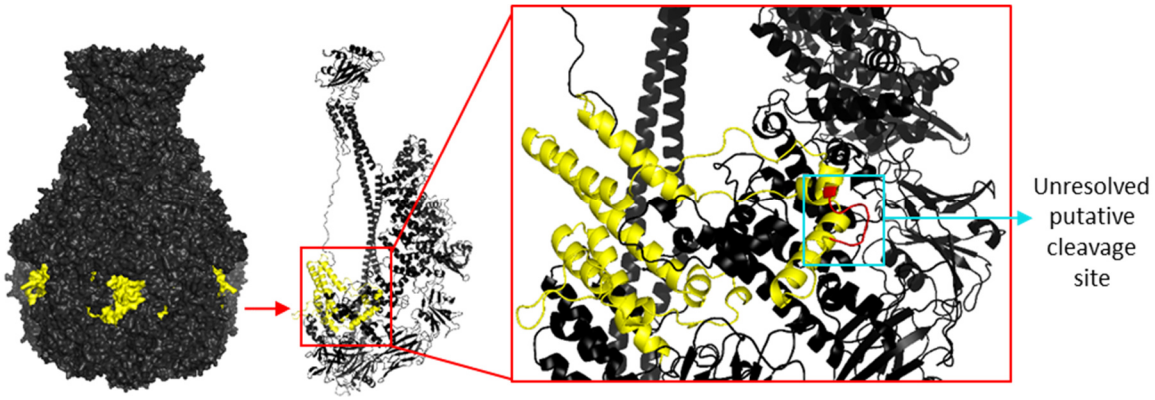


***Figure 11**: **Structure of TcdA1 showing the location of the cleaved 63 kDa fragment (yellow) after Maldi-TOF-MS analysis (PDB ID 1VW1)**. A small loop within the cleaved fragment (coloured red) that was unresolved in the crystal structure encompassed a putative cleavage site. Figure adapted from Meusch et al., 2014.*

3.2.2. Increased toxicity is specific to cleavage by metalloproteases

To investigate whether other proteases could have a similar effect on TcdA1, a cleavage experiment was performed as described above. However, this time, TcdA1^{FPLC} was treated with collagenase, trypsin and chymotrypsin. Trypsin (30 ng) and chymotrypsin (30 ng) were incubated with TcdA1^{FPLC} for 10 min each at RT, followed by 30 min on ice with trypsin inhibitor (60 ng). On the other hand, collagenase (150 ng) was incubated for 3 h at 37 °C with TcdA1^{FPLC}. Subsequently, all cleavage products were separated by SDS-PAGE (Fig. 12a). All tested enzymes cleaved TcdA1 to release the small, ~63 kDa fragment. However, chymotrypsin yielded multiple degradation bands.

Analysis of intoxication by processed toxin revealed that only collagenase treatment resulted in increased toxic activity (compare Fig. 12b and 12c). While trypsin-cleavage was similar to collagenase (Fig. 12a), it did not result in increased intoxication kinetics (Fig. 12b). Both collagenase and PrtA1 are metalloproteases, whereas trypsin and chymotrypsin are serine proteases. This data therefore suggests that TcdA1 activation is specifically performed by metalloproteases. A closer analysis of the processing by collagenase revealed that cleavage was completed at between 10 – 30 min (Fig. 13a, left panel) and like for PrtA1, MG132 did not affect collagenase cleavage (Fig. 13a, right panel).

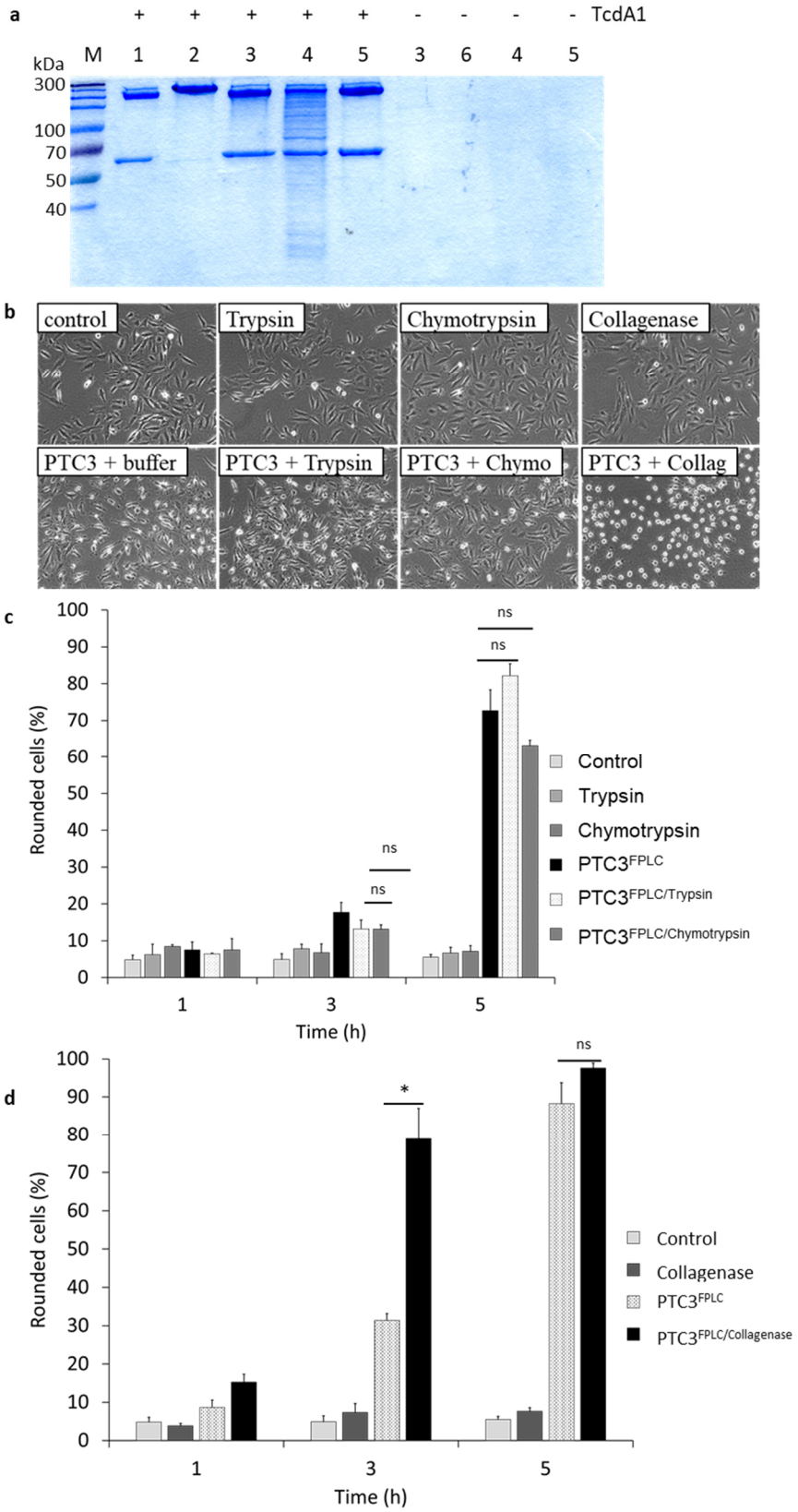


***Figure 12: TcdA1^{FPLC} cleavage by different enzymes**. (**a**) TcdA1^{FPLC} (3 µg) was treated with different enzymes before being separated on an SDS-PAGE gel and stained by Coomassie. Collagenase treatment was for 3 h at 37 °C, while trypsin and chymotrypsin were incubated for 10 min at room temperature (RT), before addition of trypsin inhibitor and a further incubation of 30 min at 4 °C. Sample 1 = PrtA1, 2*

*= TcdA1^{FPLC} alone, 3 = Collagenase (+/- TcdA1^{FPLC}), 4 = Chymotrypsin (+/- TcdA1^{FPLC}), 5 = Trypsin (+/- TcdA1^{FPLC}) and 6 = Trypsin inhibitor. (**b**) Hela cells intoxicated with each of the cleaved toxins combined with BC3 (1.5 µg/ml). As controls, trypsin, chymotrypsin, collagenase and TcdA1^{FPLC} alone, treated similarly, were used. Pictures were taken after 3 h at 37 °C. (**c**) Quantification of cells intoxicated with trypsin- or chymotrypsin-treated toxins from (b). Significance was tested by using the Student's t test (ns = not significant). (**d**) Quantification of cells intoxicated with collagenase-treated toxins from (b) (P < 0.05 (*), ns = not significant). All values are given as mean ± SD (n = 3).*

Due to the increased cell toxicity, the effect of cleavage on toxin binding was assessed. For this and further experiments, either PrtA1-cleaved (TcdA1$^{FPLC/PrtA1}$), or collagenase-cleaved (TcdA1$^{FPLC/Collag}$) toxins prepared as described above (Fig. 8), were used. A flow cytometry-based binding assay of Dylight-488-labeled TcdA1 variants (cleaved and uncleaved) on HeLa cells was performed. To prevent toxin uptake, binding was performed at 4 °C. Remarkably, a significant increase in binding to target cells by TcdA1$^{FPLC/Collag}$, compared to TcdA1^{FPLC}, was observed (Fig. 13b, blue arrow). A similar result was obtained when *Drosophila* S2 insect cells were used (Fig. 13c, blue arrow). Additionally, binding was not affected by MG132 (Fig. 13b).

To compare the effect of cleavage by PrtA1 and collagenase on cell binding, competitive flow cytometry-based binding assays were performed. Unlabelled collagenase-cleaved (Fig. 13d, left panel) and PrtA1-cleaved (Fig. 13d, right panel) toxins were added in increasing concentrations to compete with Dylight-488-labeled collagenase-cleaved TcdA1^{FPLC} in binding on HeLa cells. As expected, both unlabelled toxins were able to compete with the labelled collagenase-cleaved toxin, thus indicating that binding was not influenced by Dylight-488-labeling. Secondly, competitive binding of labelled collagenase-treated toxin with toxin processed by PrtA1, indicated that processing by collagenase and PrtA1 resulted in similar, but not identical, binding properties of cleaved-TcdA1. The PrtA1 curve was slightly different from the collagenase one. At low concentrations (below 10 µg/ml), the unlabelled collagenase-cleaved TcdA1 competed more potently with the labelled toxin, than the unlabelled PrtA1-cleaved TcdA1 (Fig. 13d). This indicated slightly different PrtA1-related binding properties.

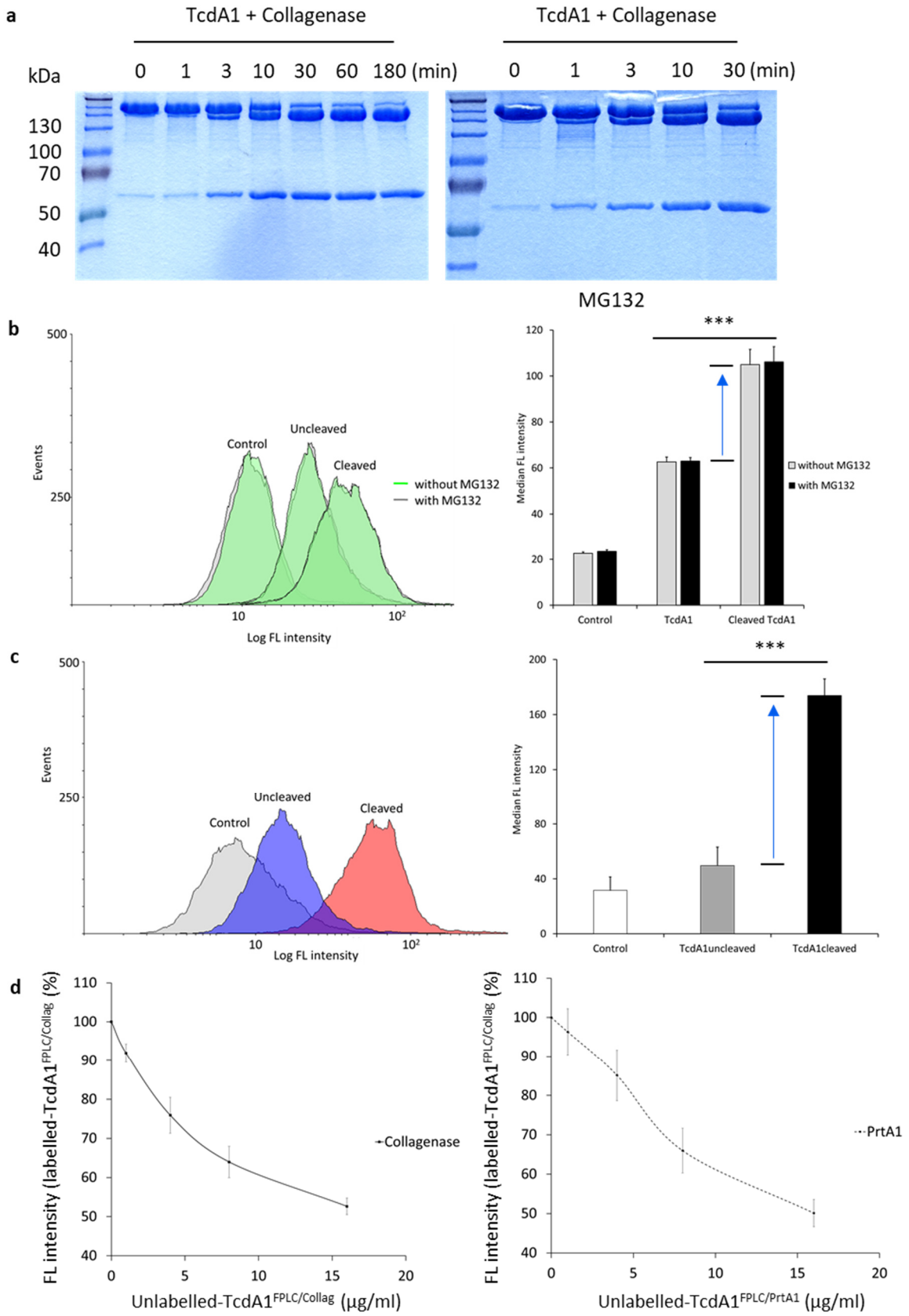


Figure 13: Cleavage of TcdA1 with collagenase increases cell binding. *(**a**) TcdA1^{FPLC} (3 µg) was treated with collagenase, with (right panel) or without (left panel) MG132, for the indicated time series at 37 °C, before being separated on an SDS-PAGE gel and stained by Coomassie. (**b**) Flow cytometry-based analysis of the binding of collagenase-cleaved or uncleaved Dylight-488-labeled TcdA1^{FPLC} (1 µg/ml) on*

*HeLa cells, in the presence or absence of MG132. Cells were pretreated with (grey line) or without (green line) MG132 (30 µM) for 1 hr at 37 °C then, precooled to 4 °C, before the toxins were allowed to bind for 15 min. The cells were then washed and analysed. The left panel shows one of the experiments. While the right shows median fluorescence intensity ± SEM of at least three independent experiments, P < 0.001 (***). (**c**) Flow cytometry analysis of the binding of Dylight-488-labeled TcdA1$^{FPLC/Collag}$ or TcdA1^{FPLC} (1 µg/ml) on Drosophila S2 insect cells. The left panel shows one of the experiments, while the right shows median fluorescence intensity ± SEM of at least three independent experiments, P < 0.001 (***). (**d**) Competitive flow cytometry-based analysis of binding of collagenase-cleaved Dylight-488-labeled TcdA1^{FPLC} (1 µg/ml) and increasing concentrations of unlabelled collagenase-cleaved (left panel) and PrtA1-cleaved TcdA1^{FPLC} (right panel).*

3.2.3. High pH further increases binding of cleaved-toxin

It was previously observed in bilayer experiments that pore-formation by TcdA1 increased at higher pH. In fact, pore-forming activity at pH 11 was more than 100-fold greater than at pH 6 (Gatsogiannis *et al*., 2013). Therefore, it was of interest to check whether the high level of binding observed for the cleaved toxin could be increased further by pH change. To this end, HeLa cells were resuspended in PBS (pH 7.5) and then the pH was changed to either pH 5 or pH 9 by addition of 228 mM of 2-ethanesulfonic acid (MES), or 8 mM of N-cyclohexyl-3-aminopropanesulfonic acid (CAPS) respectively. As a control, cells were left in PBS only. Then, Dylight-488-labeled TcdA1$^{FPLC/Collag}$ (1 µg/ml) was added and allowed to bind for 15 min at 4 °C, before washing and analysis. Interestingly, the toxin showed significantly higher binding at pH 9 than at pH 5 and 7.5 (Fig. 14a). This data indicated that binding of the cleaved TcdA1 could be increased even further at high pH. Note, Dylight-488 is photo-stable at between pH 4 and 9 (ThermoScientific manual); this confirms that the effect of pH was on the protein and not the dye.

To assess whether pH had any effect on cleavage of TcdA1 per se, uncleaved TcdA1^{FPLC} was incubated at 4 °C or 37 °C for 15 min in PBS adjusted to pH 5 or 9. Alternatively, 25 µg of CaCo-2 cell lysates adjusted to either pH 7.5 or pH 5, in the presence or absence of MG132, were used. In every instance, the incubation was terminated by addition of 5 µl of Laemmli buffer and the samples separated by SDS-PAGE, before a Western blot was performed using an antibody against the cleaved ~63 kDa fragment. As shown in Fig 14b and c, no specific pH-dependent cleavage of TcdA1 was observed. However, several degradation bands were observed at pH 5. Additionally, gel profiles of proteins in PBS (Fig. 14b) and cell lysate (Fig. 14c) were similar.

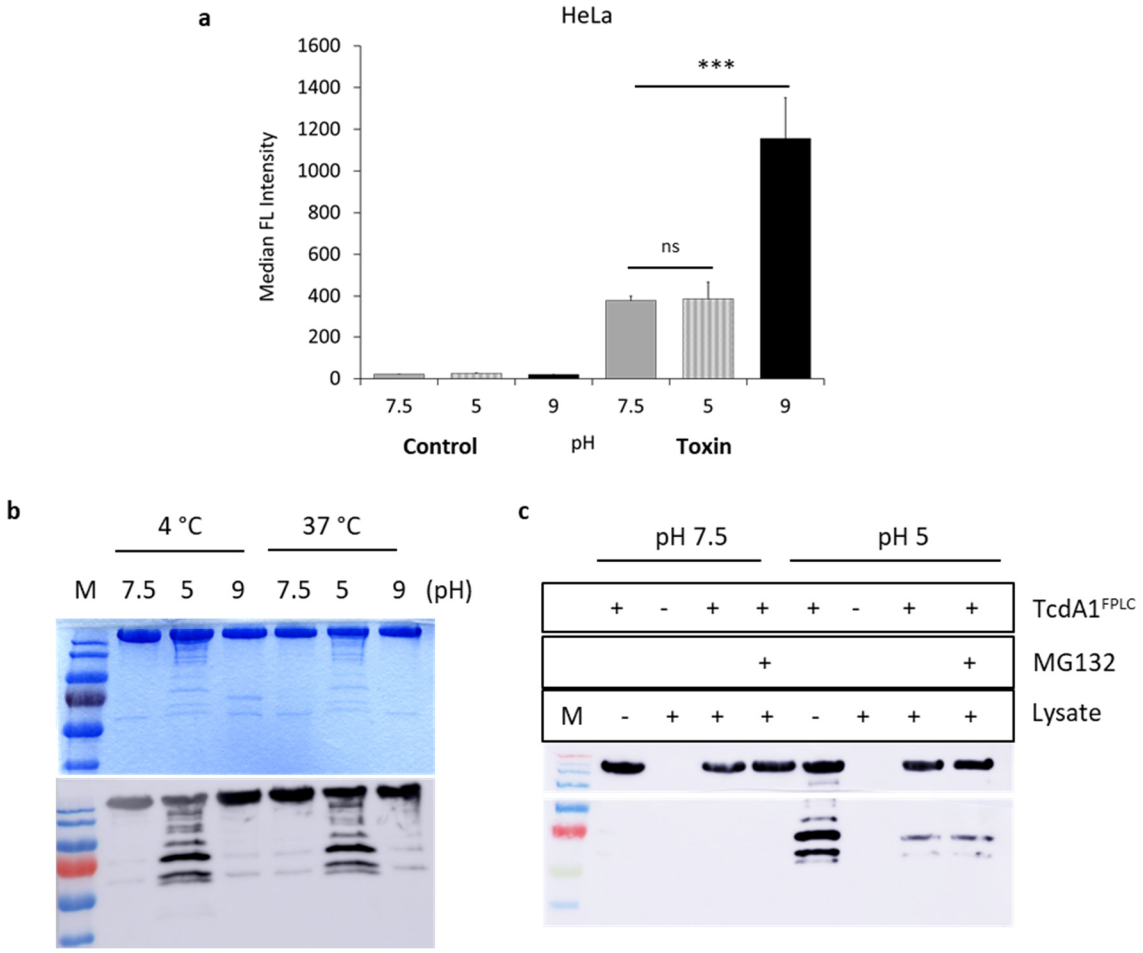


Figure 14: Cell binding of TcdA1 at different pH levels*. (**a**) Dylight-488-labeled TcdA1$^{FPLC/Collag}$ (1 µg/ml) was allowed to bind on either HeLa (left panel), or CHO-K1 cells (right panel), in PBS for 15 min at 4 °C, before washing and analysis by flow cytometry. The panels show median fluorescence intensity ± SEM of at least three independent experiments. Significance was tested by using the Student's t test (P < 0.05 (*), P < 0.001 (***), ns = not significant). (b) TcdA1^{FPLC} was incubated for 15 min at 4 °C or 37 °C in PBS adjusted to pH 7.5, 5 or 9. Afterwards the protein was separated on SDS-PAGE and then stained by Coomassie (top panel), or detected via Western blot with an antibody against the small 63 kDa fragment (bottom panel). (**c**) Instead of PBS, TcdA1^{FPLC} was incubated under the same conditions as (b) in Caco-2 cell lysate adjusted to either pH 7.5 or 5. The proteins were separated on SDS-PAGE and then detected via Western blot with an antibody against the small 63 kDa fragment.*

3.2.4. Cleaved toxin is no longer affected by bafilomycin A1 and dynasore

To assess if the cleaved toxin follows a similar intoxication mechanism as the uncleaved toxin, inhibitors affecting toxin uptake were used. It was previously shown that bafilomycin A1 inhibits cell intoxication by PTC3 and PTC5 (Lang *et al.*, 2010). Bafilomycin A1 is a macrolide antibiotic that blocks the vacuolar ATPase proton pump and thus inhibits endosomal acidification (Bowman *et al*., 1988). Dependence on endosomal acidification indicated that the PTC enters cells via the endocytic pathway. HeLa cells preincubated with 200 nM of

bafilomycin A1 for 1 h, were intoxicated with increasing concentrations of either $PTC3^{FPLC}$ or $PTC3^{FPLC/Collag}$, for 5 h. Surprisingly, intoxication by the cleaved toxin was not inhibited by bafilomycin A1 (Fig. 15a).

The effect of bafilomycin A1 was further observed with an electrical cell impedance sensing (ECIS) assay. Here, Caco-2 cells grown to form a complete monolayer, yielded high transepithelial electrical resistance values (TEER) when an electric current was passed through them. Intoxication of the cells would lead to interruption of cell junctions and hence a time-dependent reduction in subsequent resistance value. Therefore, fully formed monolayers, first preincubated with 200 nM of bafilomycin A1, were intoxicated with 0.07 nM and 1.4 nM of either cleaved or uncleaved toxin. The toxins were first allowed to bind for 1 h at 4 °C, before being transferred to 37 °C and resistance measurements continuously taken. Pre-binding ensured that intoxication commenced uniformly for all cells, thus limiting variations in the data. Similar to the HeLa cells, bafilomycin A1 had no effect on the cleaved toxin (Fig. 15b).

A similar effect was observed when the dynamin inhibitor, dynasore, which also inhibited intoxication of HeLa cells by the uncleaved toxin, was overcome by the cleaved one (Fig. 15c). Taken together, these results suggest that the cleaved toxin is active at the cell surface, and may intoxicate cells independent of endocytosis. Please note, the different toxin concentrations (0.07 nM and 1.4 nM) were used to account for the increased activity of the cleaved toxin. Therefore, even a 20x lower concentration of the cleaved toxin (0.07 nM), was able to overcome inhibition by bafilomycin A1, while 1.4 nM of the uncleaved toxin was not (Fig. 15a and b). This suggests that the altered toxin effects were due to cleavage, rather than the concentration-difference of the toxins.

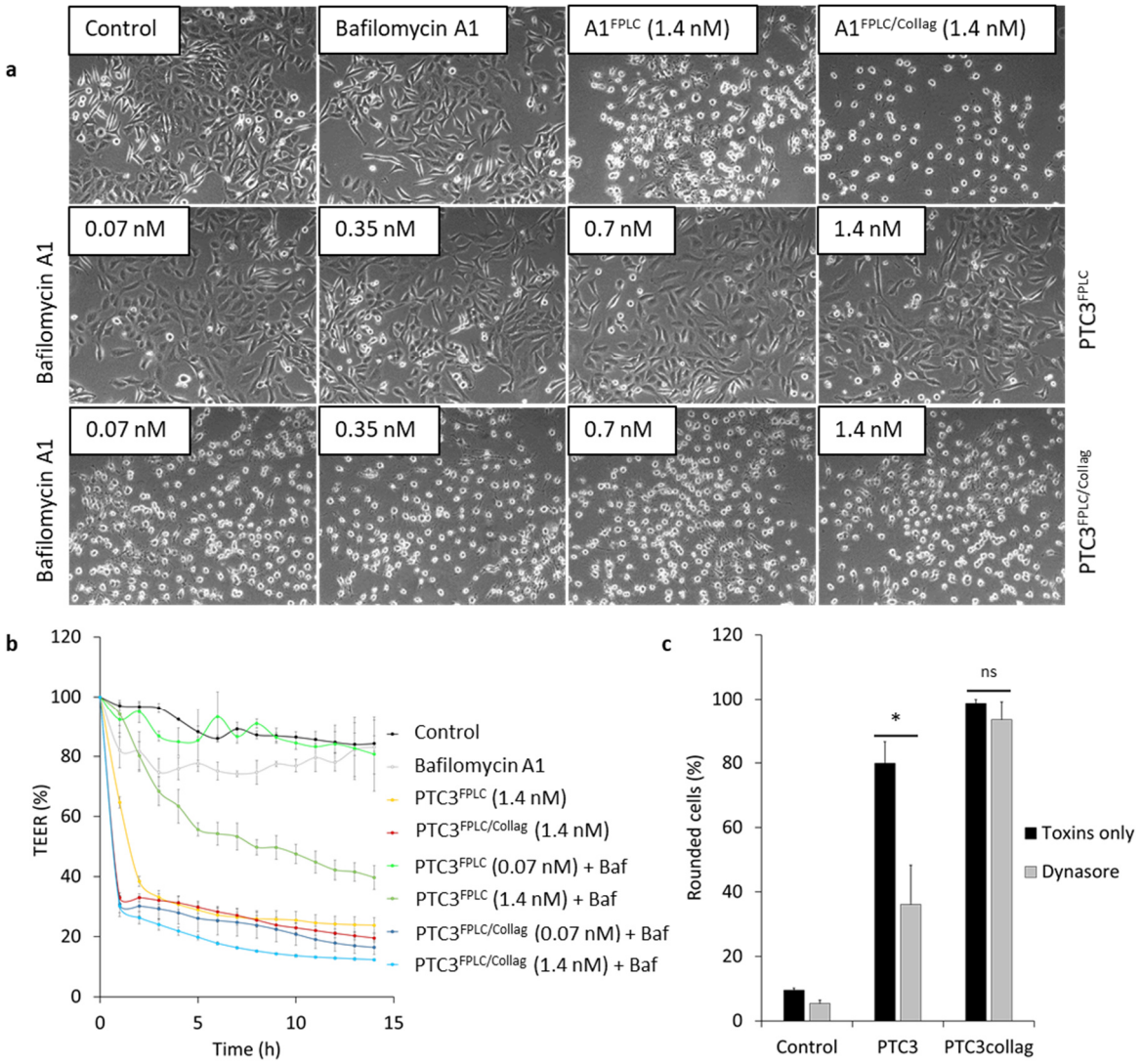


Figure 15: Cleaved toxin is resistant to bafilomycin A1 and dynasore*. (**a**) Intoxication of HeLa cells with increasing concentrations of PTC3^{FPLC} (middle panel) and PTC3$^{FPLC/Collag}$ (bottom panel) in the presence of bafilomycin A1 (200 nM). Cells were preincubated for 1 h with the inhibitor and then intoxicated for 5 h. (**b**) Electrical cell impedance sensing (ECIS) assay to measure the resistance of a Caco-2 cell monolayer after pretreatment with bafilomycin A1 (200 nM) (Baf) and intoxication with different concentrations of PTC3^{FPLC} and PTC3$^{FPLC/Collag}$ (0.07 nM and 1.4 nM). (**c**) Intoxication of HeLa cells in the presence, or absence, of dynasore (80 µM). Cells were preincubated for 1 h with the inhibitor and then intoxicated for 3 h. Pictures were then taken and the percentage of rounded cells quantified. Significance was tested by using the Student's t test (P < 0.05 (*), ns = not significant), n = 3.*

3.3. The *Photorhabdus* Tc toxin: an engineered peptide delivery nanomachine

3.3.1. The Tc toxin can deliver a foreign protein into target cells

The PTC holotoxin has multiple substrates for translocation into target cells. So far, two proteins, C3 and C5 hvrs, are known cargoes of the nanomachine (Lang *et al.*, 2010). It is therefore likely that other proteins could be similarly translocated. To study delivery of foreign

peptides through the PTC, the TccC3hvr was exchanged with the *C. botulinum* C3bot, a small (25 kDa) ADP-ribosyltransferase that targets Rho GTPases. The C3bot was cloned in place of the C3hvr downstream of the aspartyl-autoprotease cleavage site of the BC3 cocoon (Fig. 16a). This yielded a fusion complex of BC3-C3bot (271 kDa), consisting of a BC3-cocoon (246 kDa) encapsulating the C3bot (25 kDa). The fusion complex was expressed similar to the recombinant wildtype and visualized through an SDS-PAGE and Western blot (Fig. 16b). As expected, the recombinant fusion toxin was auto-proteolytically cleaved to release C3bot. To assess proper folding of C3bot, an *in vitro* time-dependent ADP-ribosylation assay of RhoA was performed, with radiolabeled NAD^+ as a co-substrate. Interestingly, the fusion complex modified RhoA, however, to a lower extent compared with C3bot alone (Fig. 16c). This indicated disruption of enzyme activity by the BC3 cocoon, which may shield the toxin from its substrate and/or co-substrate.

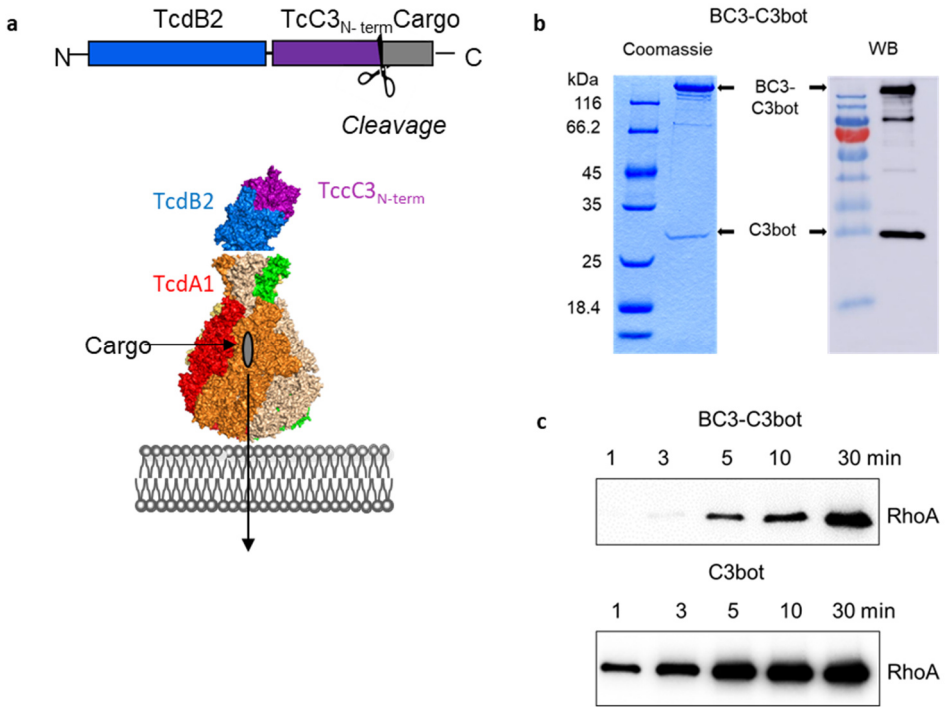


Figure 16: Exchange of C3hvr with foreign toxins and comparative activity of the fusion complex and enzyme alone. (***a***) *Experimental design of the chimeric BC3 toxin. The C3hvr was replaced with a selected cargo (grey box) downstream of the auto-proteolytic cleavage site. This cargo would then be packaged into the BC3 cocoon and translocated through TcdA1 into target cells (grey oval) (adapted from Meusch et al., 2014).* (***b***) *Purified BC3-C3bot fusion complex, separated by SDS–PAGE (Coomassie stained). For the Western blot, a C3bot-specific antibody was used* (Aktories *et al.*, 1989). (***c***) *Autoradiogram of in vitro ADP-ribosylation of RhoA by 3 nM of BC3-C3bot and 1.25 nM of recombinant wildtype C3bot.*

3.3.2. Influence of the BC3 cocoon on the enclosed enzyme

To study the influence of the cocoon on the enclosed enzyme, an *in vitro* ADP-ribosylation assay of actin comparing the recombinant wildtype BC3 toxin and recombinant C3hvr was performed. Interestingly, BC3 showed weaker activity compared to the C3hvr (Fig. 17a). Incubating both BC3 and C3hvr in increasing temperatures further elucidated the shielding effect of the cocoon. After incubation at the said temperatures, the samples were cooled back to 21°C and *in vitro* ADP-ribosylation was performed. Remarkably, increased activity of BC3 correlated with increase in temperature, suggesting destruction of the cocoon and subsequently, exposure of the enclosed hvr to the external environment (Fig. 17b). On the other hand, C3hvr alone showed no temperature-dependent increase in activity (Fig. 17c). Additionally, the hvr showed impressive stability even upon exposure to high temperatures. Similar to wildtype BC3 toxin and C3hvr, the fusion protein of BC containing the ADP-ribosyltransferase C3bot exhibited less enzyme activity than unfused C3bot alone.

To take it a step further, the question of whether an intact BC3 cocoon could shield the enclosed enzyme from the external environment was asked. For this, a BC3 fusion complex containing a modified fragment of the *Yersinia enterocolitica* effector YopT (30 KDa) was used. This fusion complex will be discussed further in the proceeding section. However, here, the fusion complex was fluorescently labelled with Dylight-488. The question was whether the closed cocoon would shield the enzyme from the dye. After labelling following the manufacturer's instructions, the protein was washed several times by centrifugation in Vivaspin 500 filter-tubes with a molecular weight cut-off of 10 KDa (Merck). Then, to confirm that there was no residual dye left in the sample, the *C. botulinum* C2I protein (50 kDa) was added to the sample and similarly incubated. Finally, the protein sample was resuspended in Laemmli buffer and removed in two rounds. The proteins were then separated on an SDS-PAGE gel and visualized, first by fluorescence and then by Coomassie staining. As shown in Fig. 17d, only the BC3 cocoon (lanes 2 and 3, left and middle panels) was stained. Both the YopT fragment (YopT-lys) and C2I (lane 4, Comassie staining, right panel) were either very minimally labelled, or not at all. This confirmed that the cocoon was indeed closed and could shield the enclosed enzyme from the external environment.

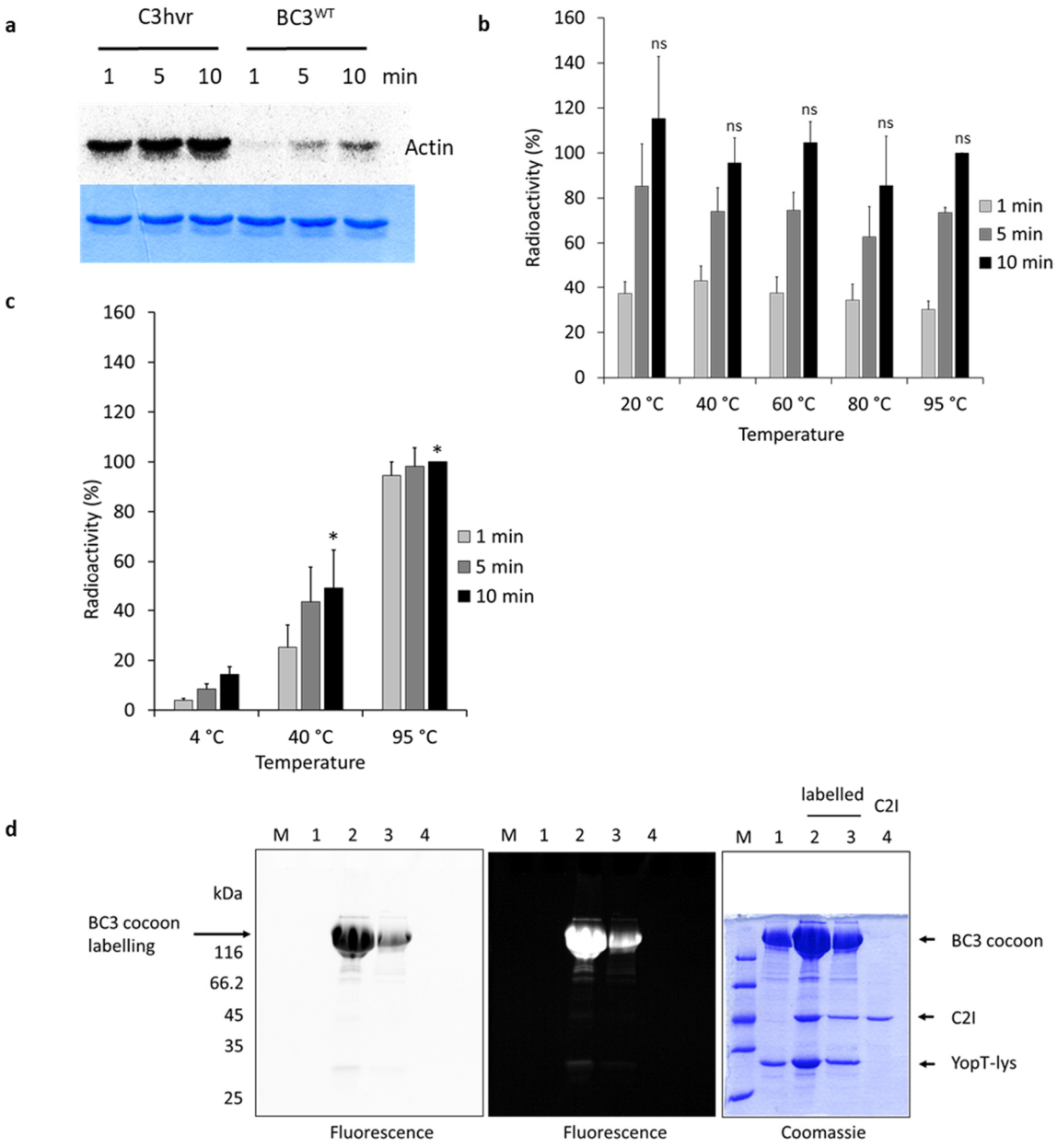


Figure 17: Influence of the BC3 cocoon on the enclosed enzyme. *(**a**) Autoradiogram of comparative in vitro ADP-ribosylation of actin by 3 nM of BC3 and C3hvr. (**b**, **c**) Quantification of temperature-dependent ADP-ribosylation of actin by 3 nM of BC3 (**b**) and C3hvr (**c**). The toxins were preincubated at the stated temperatures for 10 min, then cooled down to 21 °C for the assay. The 10 min time points for each temperature were statistically compared to the 10 min results of 4 °C (unheated). Significance was tested by using the Student's t test (P < 0.05 (*), ns = not significant), n = 3. (**d**) Fluorescent labelling of a BC3 fusion complex containing a fragment of the Y. enterocolitica effector YopT. After labelling, the toxin was washed severally using 10 KDa filter-tubes before the C. botulinum protein C2I was added to assess the presence of any residual dye. The proteins were then separated by SDS-PAGE and then visualized by fluorescence (left and middle panels) and Coomassie staining (right panel). M = molecular marker, lane 1 = unlabelled BC3-fusion complex, lanes 2 and 3 = labelled BC3-fusion complexes, lane 4 = C. botulinum C2I protein.*

3.3.3. Delivery of C3bot into target cells by the Tc toxin

C3bot does not possess any transport unit and is per se unable to enter cells at moderate concentrations (nM range) (Rohrbeck *et al.*, 2015; Rotsch *et al.*, 2012). Therefore, it was of great interest to clarify whether the BC3-C3bot fusion protein was delivered into cells. To this end, HeLa cells were incubated for 6 h with 10 nM of TcdA1^{aff} + BC3-C3bot (full toxin complex) and cells showing C3 morphology, that is, condensed cell bodies and arborisation-like effects (Fig. 18a) were quantified (Fig. 18b). As controls, TcdA1^{aff} alone, BC3-C3bot alone, recombinant C3bot alone, or untreated cells, were used. Only cells treated with TcdA1^{aff} + BC3-C3bot showed intoxication indicating *in vivo* RhoA inactivation (Fig. 18a, b). This suggested functional delivery of C3bot by the protein complex into HeLa cells. However, to confirm these results, a TEER assay using CaCo-2 cells was performed. Consistent with the HeLa cells, only cells treated with the full toxin complex (TcdA1^{aff} + BC3-C3bot) showed a time-dependent reduction in epithelial resistance (Fig. 18c), thus confirming that C3bot was injected by PTC as foreign cargo into mammalian cells.

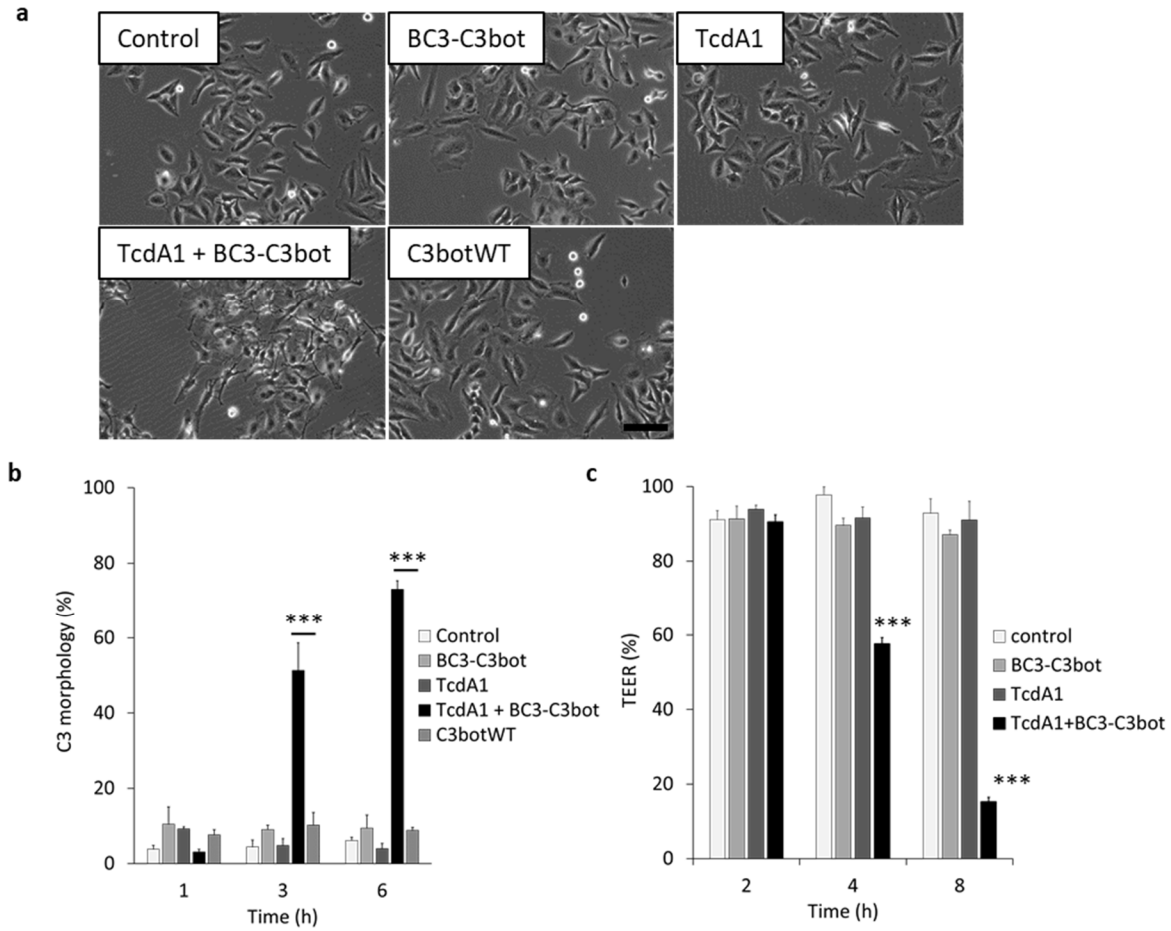


***Figure 18: Delivery of C3bot into target cells**. (**a**) Intoxication of HeLa cells with the BC3-C3bot fusion toxin. Cells were intoxicated for 6 h with 10 nM of TcdA1^{aff} + BC3-C3bot and 100 nM of C3bot (WT) at 37 °C. (**b**) Quantification of intoxication of HeLa cells treated as in (d). (**c**) TEER assay of CaCo-2 cells treated with 1.94 nM of TcdA1^{aff} and 5.1 nM of BC3-C3bot for 8 h. Unpaired, two-tailed t test (P < 0.001 (***), ±SEM). Scale bar: 100 µm, n = 3.*

3.3.4. Delivery of peptides through PTC depends on the size and charge of the cargo

To gain insight into the features of the cargo required for effective packing and transport, the properties of all naturally occurring peptides transported through the PTC were compared, although not all have been characterized. As shown in Table 12, all putative proteins injected by the PTC have a similar molecular weight (28 - 35 kDa). Additionally, they all (with one exception) have an isoelectric point above 8.5. C3bot showed similar characteristics with an even lower molecular weight (25 kDa) and a pI of 9.57. This could explain its successful delivery. Overall, these characteristics suggest that the size and the charge of the cargo are key for effective transport. However, so far, only ADP-ribosyltransferases have been successful. For further analyses, the *Y. enterocolitica* toxin YopT was chosen. It is a cysteine protease, which is injected by type- III secretion into mammalian cells to cleave and inactivate Rho GTPases.

Table 12: Properties of *Photorhabdus* proteins (in grey) injected by the PTC and other proteins selected for characterizing the complex's translocation capabilities

Gene/Protein name	Molecular weight (kDa)	pI	Reference
TccC3hvr_W14	32 KDa	9.68	Lang *et al*., 2010
TccC2_W14	28 KDa	8.7	
TccC4_W14	30 KDa	6.23	
TccC5_W14	30 KDa	8.67	Lang *et al*., 2010
TccC1_TT01	35 KDa	11.01	
TccC2_TT01	28 KDa	9.4	
TccC3_TT01	32 KDa	9.06	
TccC4_TT01	32 KDa	9.38	
TccC5_TT01	30 KDa	9.08	
TccC6_TT01	32 KDa	8.9	
TccC7_TT01	28 KDa	8.56	
C3bot	25 KDa	9.57	Aktories *et al.*, 1988
YopT$^{\Delta 1\text{-}74}$	30 KDa	6.2	Shao *et al.*, 2003

YopT$^{\Delta 1-74}$ – lys3	30 KDa	8.29	Shao *et al.*, 2003
YopT$^{\Delta 1-74}$ – lys6	30 KDa	8.91	Shao *et al.*, 2003
YopT$^{\Delta 1-30}$	33 KDa	9.04	Shao *et al.*, 2003
YopT full	36 KDa	8.53	Shao *et al.*, 2003
PaTox_GTD	37 KDa	8.95	Jank *et al.*, 2013
Toxin B_GTD	63 KDa	4.69	Just *et al.*, 1995
Metridia luciferase 7	16.5 kDa	8.66	Markova, Larionova and Vysotski, 2019

3.3.5. The Type III secretion effector YopT is injected through PTC into target cells

Different sizes of YopT with varying isoelectric points (Table 12) were fused to the truncated BC3 complex. Besides the full-length, smaller N-terminal truncations YopT$^{\Delta 1-30}$ and YopT$^{\Delta 1-74}$ which possessed full activity and stability, when expressed as recombinant proteins (Sorg *et al*., 2003), were cloned. Additionally, the acidic charge of YopT$^{\Delta 1-74}$ (pI = 6.2) was changed by the addition of three or six lysine residues at its C-terminus, resulting in pIs of 8.29 and 8.91, respectively. As expected, all clones were properly expressed and cleaved from the BC3 cocoon (Fig. 19a). To check the catalytic activity of the fusion proteins, a membrane-release assay of RhoA was performed. Purified HeLa cell membranes were incubated with or without BC3-YopT proteins and separated into pellet and supernatant by ultracentrifugation. Then, fractions were analysed by Western blot for the presence of RhoA. As explained above, active YopT should displace RhoA from cell membranes, because it cleaves the C-terminally isoprenylated cysteine, which is the membrane anchor of the GTPase. All constructs showed YopT activity (Fig. 19b). By contrast, heat-inactivated enzyme was not able to cleave and release RhoA.

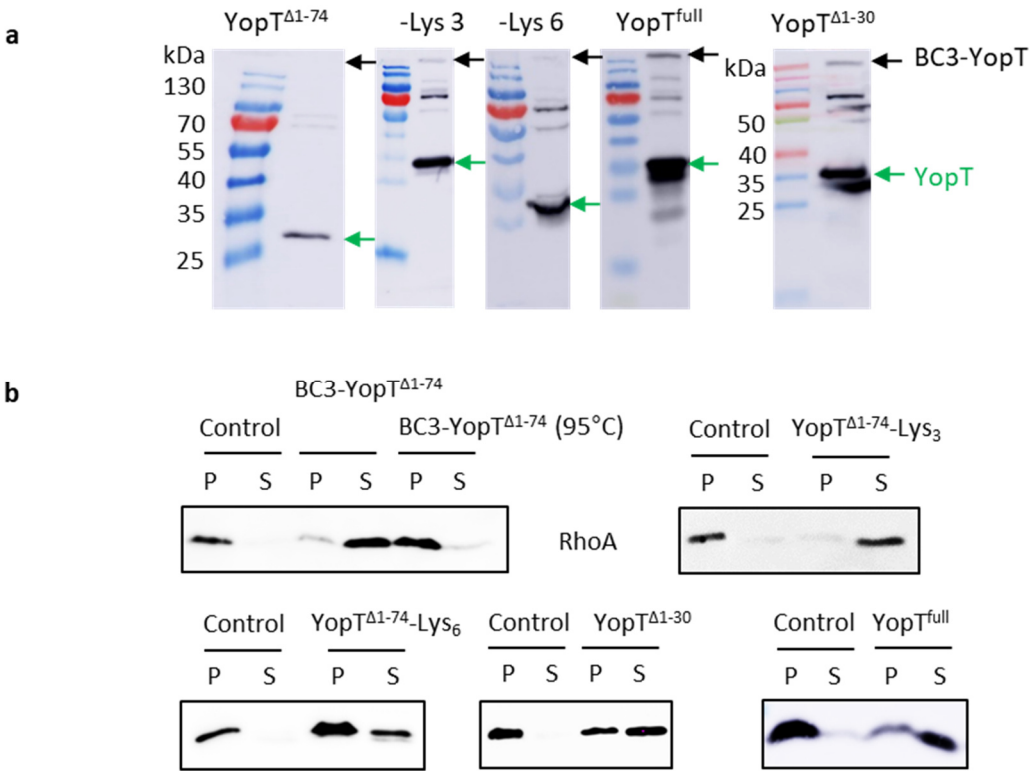


Figure 19: Purification and in vitro activity of BC3-YopT fusion toxins*. (**a**) Western blots of purified BC3-YopT chimeras showing YopT fragments (green arrows) cleaved from the BC3 cocoon (black arrows). The blots were performed using a YopT-specific antibody. (**b**) In vitro membrane release activity of RhoA from purified HeLa cell membranes by BC3-YopT fusion toxins. Pellet (membrane fraction); S, post-membrane supernatant (cytosolic fraction). N = 3*

Next, to analyse the functional delivery of YopT and its truncations, HeLa cells were incubated with the BC3-YopT constructs in the presence or absence of TcdA1, respectively. After overnight incubation, cells were fixed and stained for F-actin with rhodamine–phalloidin. As shown in Fig. 20a (top two panels), injection of YopT led to a reduction in stress fibre formation, indicating inactivation of RhoA-dependent signalling pathways. BC3-YopT$^{\Delta 1-74}$ showed little or no activity compared with the others, even in the presence of TcdA1, although it was able to cleave RhoA from purified membranes *in vitro*. However, addition of lysines to its C-terminal allowed its injection into cells and subsequent *in vivo* activity. Expectedly, TcdA1 alone showed no effects. To further analyse these effects, the cells were cointoxicated with cytotoxic necrotizing factor 1 (CNF1), a deamidase toxin from *E. coli*, that activates RhoA leading to increased stress fibre formation. It was previously shown that the YopT phenotype could not be reverted by CNF1 (Sorg *et al*., 2001). Therefore, as expected, the reduced stress fibre formation caused by YopT was more visible in the cells co-treated with CNF1 (Fig. 20a,

bottom two panels). Again, BC3-YopT$^{\Delta 1\text{-}74}$ and TcdA1 alone had no effect. These data indicated that YopT could be delivered into cells and besides the size of the protein, charge is also essential for functional delivery.

Further proof of YopT activity inside the cells was obtained through a membrane-release assay of HeLa cells preintoxicated with BC3-YopT^{full} and TcdA1, and co-treated with CNF1. Consistent with the previous results, there was a complete release of RhoA from membranes of YopT-treated cells (Fig. 20b). Additionally, less RhoA was released in cells treated with both YopT and CNF1. These data further confirmed successful delivery of YopT into the cells.

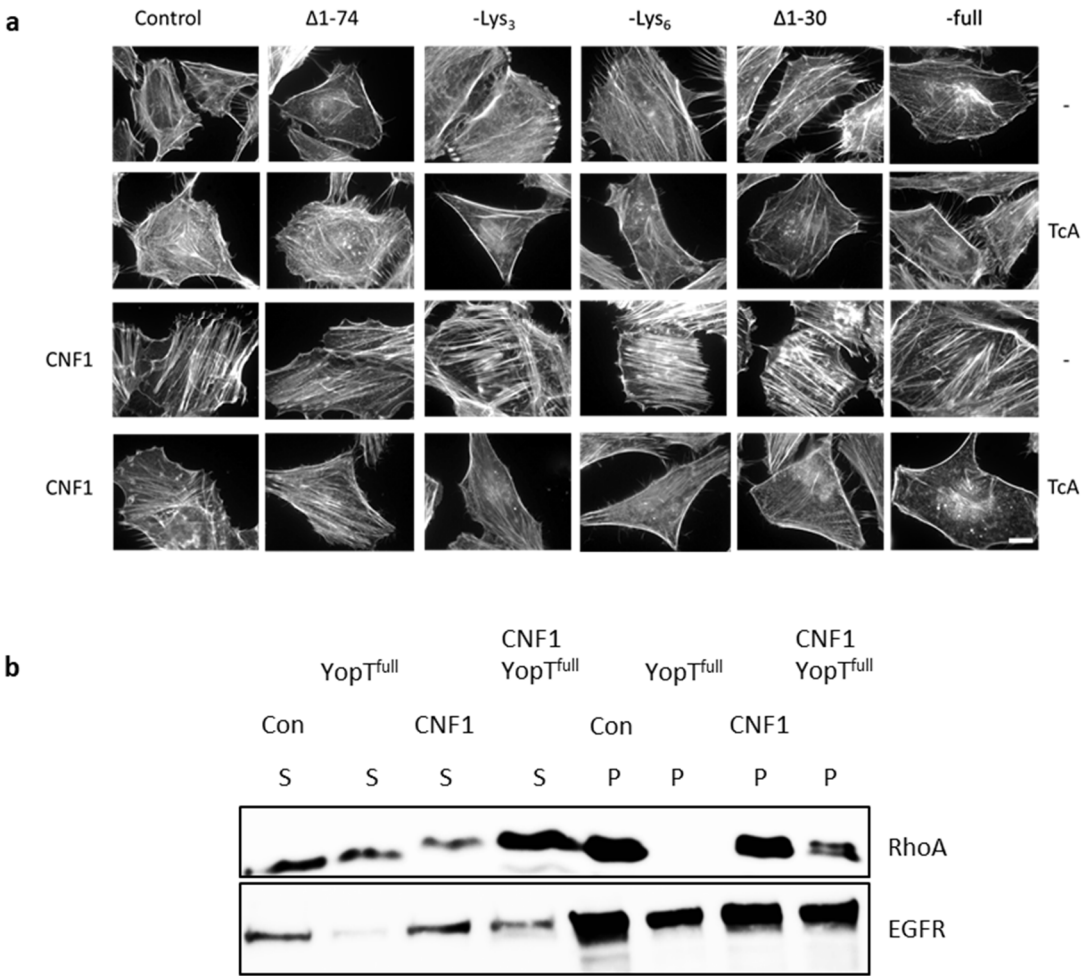


Figure 20: Delivery of YopT and YopT fragments into target cells. *(**a**) In vivo activity of BC3-YopT in HeLa cells intoxicated either overnight (top two panels) or for 4 h (bottom two panels) with TcdA1$^{FPLC/Collag}$ + BC3-YopTs (20 nM). Cells in the bottom two panels were also co-treated for 1 h with CNF1 (4 nM) before being fixed and stained for F-actin. (**b**) Biochemical analysis of in vivo cleavage and mislocalization of RhoA by BC3-YopT^{full} in HeLa cells intoxicated as in B above. Scale bar: 100 µm. P, Pellet (membrane fraction); S, post-membrane supernatant (cytosolic fraction). N = 3.*

In a next step, the temporal effect of YopT in cells was analysed through TEER assays of Caco-2 cells. As done previously, confluent CaCo-2 cells (TEER = 100%) were treated with increasing concentrations of each toxin chimera (1 - 30 nM). Then, transepithelial resistance was measured continuously for up to 20 h. PTC3 was used as a positive control and expectedly decreased the epithelial barrier within a few hours. On the other hand, consistent with the experiments performed with HeLa cells, all BC3-YopT chimeras (except BC3-YopT$^{\Delta 1\text{-}74}$) comparably increased the permeability of epithelial junctions in the presence of TcdA1 (Fig. 21a - e). A direct comparison of the YopT chimeras with a constant concentration of 30 nM, revealed a comparable effective concentration range of 15 – 20 nM (Fig. 21f). However, BC3-YopT$^{\Delta 1\text{-}30}$ was most effective, with an EC_{50} of 8.3 nM. This correlated with its highest pI compared to the other YopT fragments. YopT can therefore be injected into cells and leads to destruction of epithelial cell junctions.

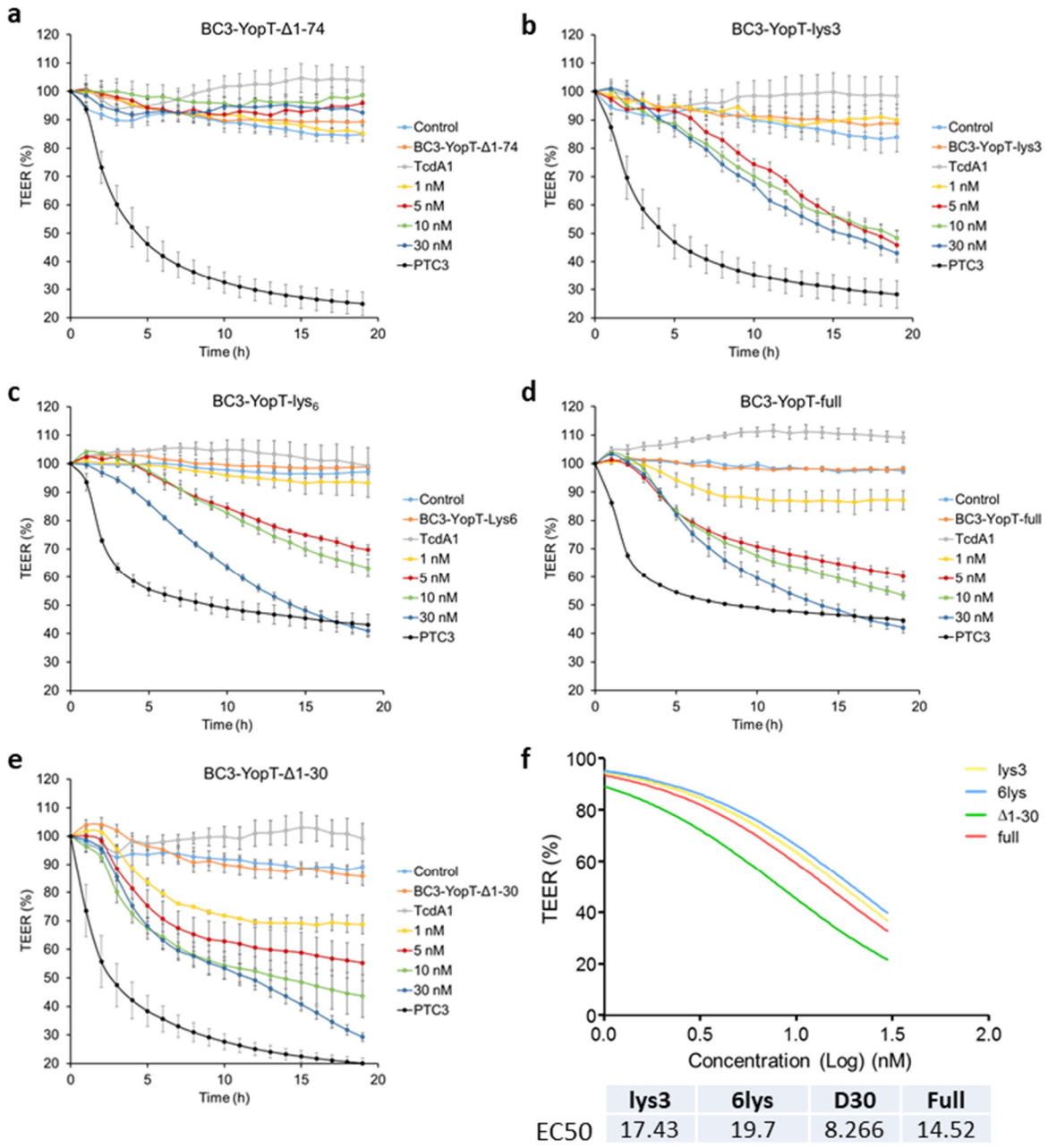


Figure 21: In vivo activity of BC3-YopT chimeras in CaCo-2 cells*. Confluent monolayers of CaCo-2 cells (TEER = 100%) were intoxicated with increasing concentrations of TcdA1^{aff} + BC3-YopT fusion toxins (1 – 30 nM) and transepithelial resistance was measured. The graphs show TEER as a percentage of the starting value. (**a**) BC3-YopTΔ1−74. (**b**) BC3-YopTΔ1−74-lys3. (**c**) BC3-YopTΔ1−74-lys6. (**d**) BC3-YopTΔ1−30. (**e**) BC3-YopTfull. (**f**) A direct comparison of all toxins (30 nM each) to obtain the effective concentration that causes 50% TEER (EC50). Recombinant wildtype PTC3^{aff} was used as a positive control. N = 3.*

3.3.6. BC3-chimeras of PaTox^{GTD} and Toxin B^{GTD} showed activity *in vitro* but were not delivered into cells

Other BC3 chimeras containing the glycosyltransferase domains (GTDs) of PaTox (37 kDa) and toxin B (63 kDa) from *P. luminescence* and *C. difficile* respectively, were also created. These

chimeras were also properly expressed and cleaved from their respective cocoons (Fig. 22a). Surprisingly, despite its large size, the Toxin B^{GTD} was expressed fused to BC3. PaTox, on the other hand, showed an unusual cleavage, which appeared to result in additional degradation. A persistent 25 kDa PaTox band was obtained instead of, or in addition to, the expected 37 kDa band (Fig. 22b). Nevertheless, catalytic activity of both enzymes was tested through *in vitro* glycosylation experiments of Rac1 using [^{14}C]-labelled UDP-glucose and UDP-GlcNAc as co-substrates for BC3-Toxin B and BC3-PaTox, respectively. As shown, in Fig. 22c and 22d, both toxins modified Rac1, however, their activity was much reduced compared to their wildtype counterparts. As previously observed, the BC3 cocoon could have played a role in reducing the enzyme activity. However, in the case of PaTox, the reduced activity could have also been due to the aforementioned degradation. The 25 kDa PaTox band was catalytically inactive. Both BC3-Toxin B and BC3-PaTox did not show any activity *in vivo* (data not shown). Factors that could be responsible for the absence of *in vivo* activity include the large size of Toxin B^{GTD} as well as its low pI (4.69). In the case of PaTox, while its size and pI were in the favourable range (Table 12), the degradation highly reduced the amount of active enzyme, hence masking its activity *in vivo* (if any).

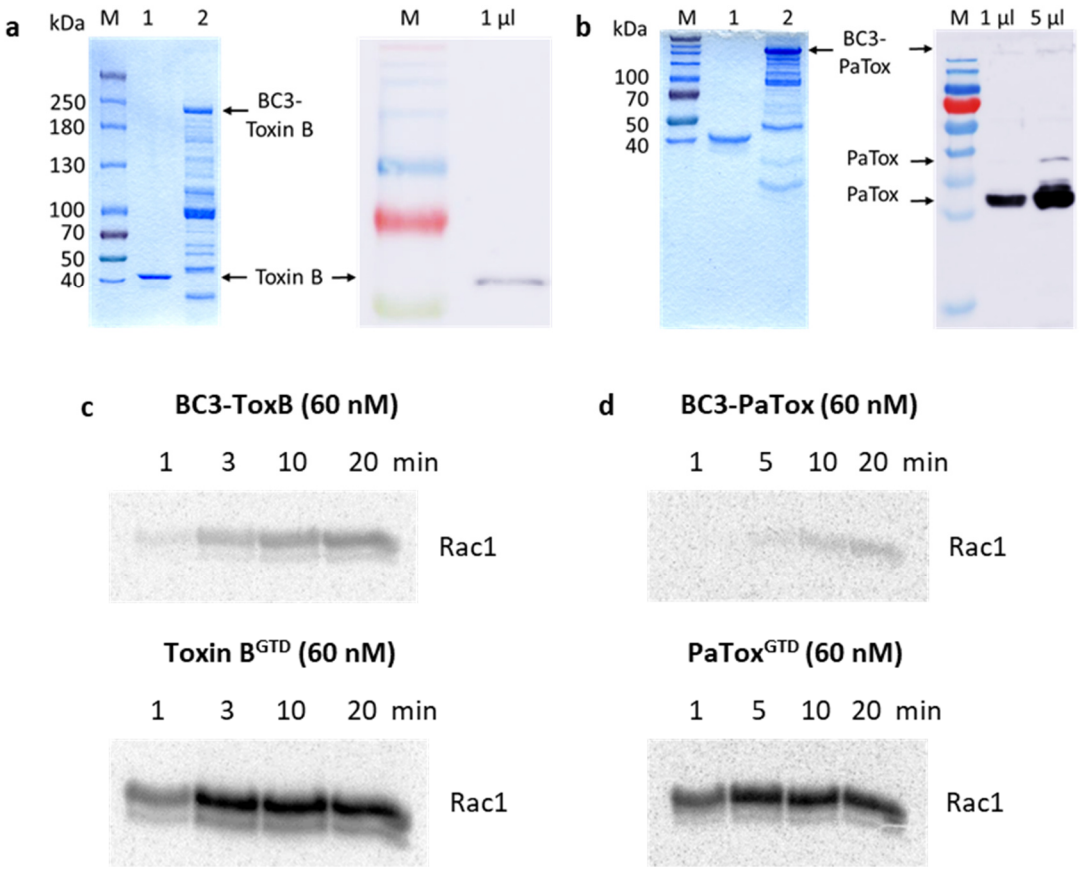


***Figure 22: Activity of BC3-Toxin B and BC3-PaTox chimeras**. (**a, b**) SDS-PAGE gels and Western blots of purified chimeras showing fragments cleaved from the BC3 cocoon. The blots were performed using toxin-specific antibodies (Lane 1 = βγ actin, Lane 2 = toxin, M = molecular mass). (**c, d**) Autoradiograms of in vitro glycosylation of Rac1 by 60 nM of BC3-Toxin B (**c**) and BC3-PaTox (**d**) respectively.*

3.3.7. Quantification of the injection efficiency of the PTC3 nanomachine

Having proven that the PTC can inject heterologous peptide cargoes into target cells, the question of efficiency of delivery was addressed. Considering that only a few molecules of toxin are required for modification of substrate and cellular effects, the highly active toxins were not apt for the exact quantification of cargo injected into each cell. Based on the structure of the PTC, each toxin complex injects a single enzyme into the cell (Meusch *et al*., 2014). However, how many PTCs, at a given concentration, actually bind to a single cell and functionally release the payload, remains unknown. To address this, a quantifiable cargo was required. Therefore, a small secreted luciferase (MLuc7) produced by the marine copepod *Metridia longa* was chosen (Markova *et al*., 2004). Mluc7 met the criteria for a suitable payload with a size of 16 kDa without the signal peptide, and a pI above 8 (see Table 11). Additionally, despite having no sequence or structural similarities to better-known *Renilla* or firefly luciferases, Mluc7, like *Renilla* luciferase, also uses commercially available cell-penetrating coelenterazine as a substrate for producing blue light.

First, *in vitro* bioluminescence activity of BC3-Mluc7 was tested in different buffers (Fig. 23a). BC3-Mluc7 was suspended in each buffer and incubated for 2 min at 37 °C. Then, 30 µl of coelenterazine (Stop&Glo® solution, Promega) was added and bioluminescence was measured. Compared to passive lysis buffer (PLB) (Promega), luminescence was significantly reduced in all buffers except collagenase buffer. However, it was most reduced in trypsin/EDTA and collagenase where BC3-Mluc7 was most likely degraded by protease activity. Additionally, medium, which has proteins like albumin that can bind the enzyme (Zhao *et al*., 2004), and EDTA, which chelates divalent cations (Rodionova and Petushkov, 2006), also significantly reduced luminescence.

Next, to analyse luciferase activity in detail, increasing concentrations of BC3-Mluc7 were dissolved in either HeLa cell lysate or PLB. Subsequently, the samples were aliquoted onto a white plate and transferred to an Infinite M200 microplate reader (Tecan) for analysis. Using the in-built injector of the reader, 30 µl of Stop&Glo® was added to the samples and bioluminescence was measured (Fig. 23b). BC3-Mluc7 showed comparable concentration-

dependent bioluminescence activity in both conditions. Simple linear regression analysis was then performed to determine the number of toxin molecules producing one light unit (LU) in lysate (Fig. 23c). This analysis revealed that 3.24×10^8 molecules were required to produce one LU.

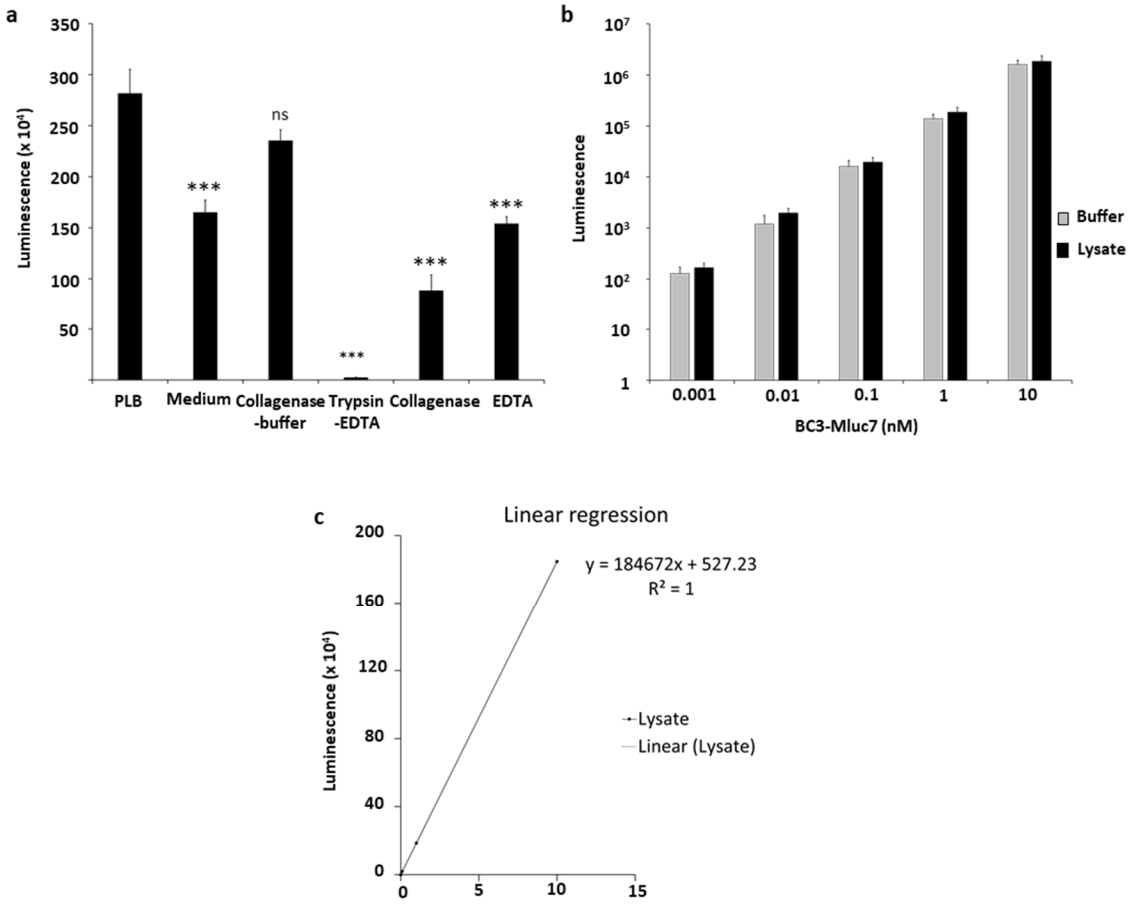


***Figure 23: In vitro activity of BC3-Mluc7 chimera**. (**a**) In vitro activity of BC3-Mluc7 in different buffers. BC3-Mluc7 (10 nM) was incubated for 2 min at 37 °C in each buffer before luminescence was measured. (**b**) In vitro activity of BC3-MLuc7 in passive lysis buffer (Buffer) or HeLa cell lysate. (**c**) Simple linear regression analysis of the assay performed in (b) to determine the number of protein molecules producing one light unit. Unpaired, two-tailed t test (P < 0.001 (***), ±SEM). N = 3.*

Next, to assess the *in vivo* activity of Mluc7, HeLa cells were intoxicated with 10 nM of $TcdA1^{FPLC/Collag}$ + BC3-Mluc7. As a control, 10 nM of BC3-Mluc7 alone was used. Note, as the most active form of the toxin, $TcdA1^{FPLC/Collag}$, was used for all intoxication experiments (see section 3.2). After an incubation of 1 h at 37 °C, the cells were washed, lysed and bioluminescence was measured upon injection of substrate. As shown in Fig. 24a, a higher luminescence value was obtained, when cells were incubated with the full toxin, compared to the cocoon alone. This indicated that the luciferase-containing full complex was taken up by

the cells as expected. However, for determination of injection efficiency, a distinction between uptake (endocytosis) and injection was required.

To discriminate between uptake and injection, cells pretreated with bafilomycin A1 (to block acidification of endosomes) were incubated at 4 °C with either BC3-Mluc7 alone, or with the full toxin complex (PTC3-Mluc7), or with increasing concentrations of each. Bafilomycin A1 and the constant 4 °C were meant to limit uptake via the endosome, thereby restricting toxin effects to only those molecules bound at the cell surface. This setup would, therefore, enable quantification of only injection efficiency, rather than uptake. To induce injection, the medium was changed to acidic medium (pH 5) for 2 min, or left at neutral pH 7.5. Next, the cells were washed and treated with trypsin/EDTA to degrade extracellular protein and to, simultaneously, detach them from the plate. Afterwards, the cells were harvested, lysed, and the lysates used for detecting luciferase activity. As was previously observed, the highest luminescence was obtained when the full toxin was used (Fig. 24b, c, d). However, a much higher luminescence was measured in acidic pH than at pH 7.5 (Fig. 24b), indicating favourable injection of the enzyme at this pH. Interestingly, compared to endocytotic uptake, injection at the cell surface produced much higher luminescence, especially at pH 5 (compare Fig. 24a and b).

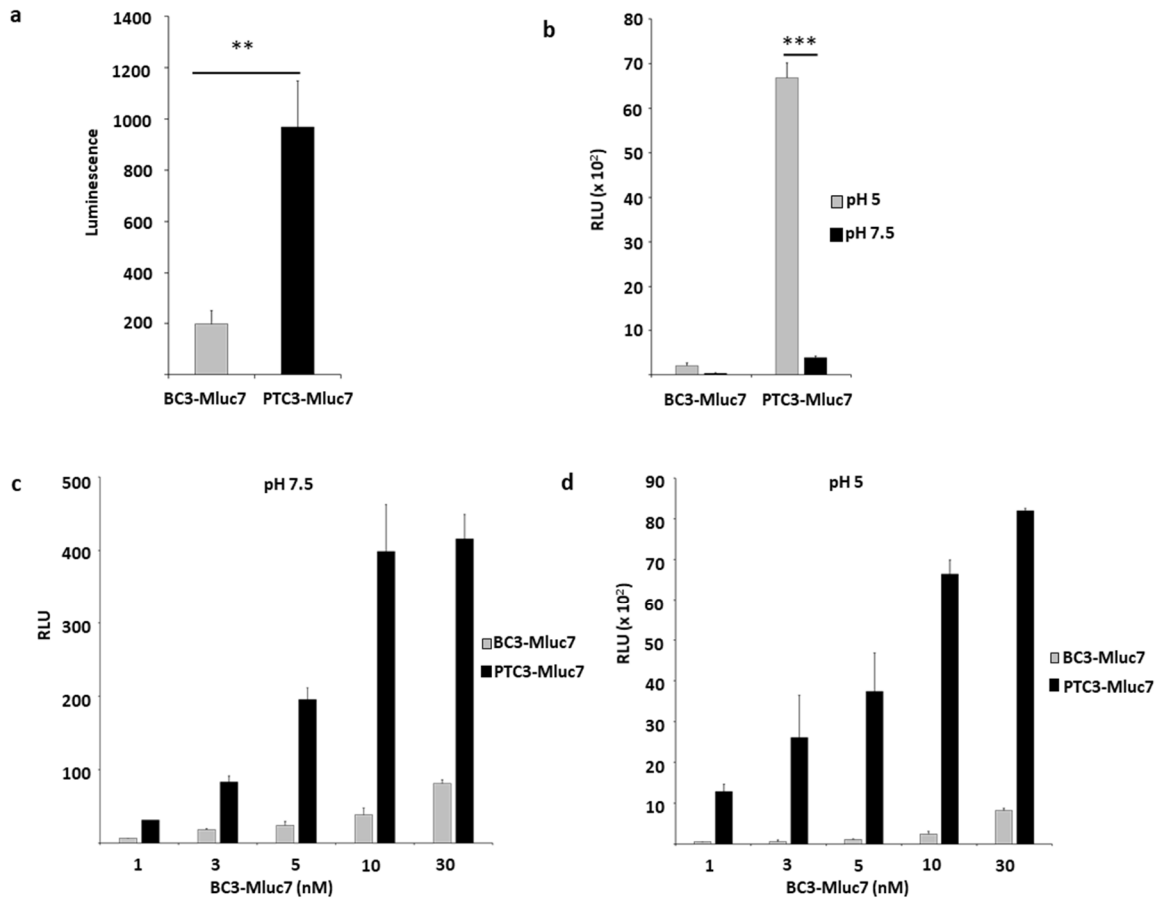


***Figure 24: In vivo activity of BC3-Mluc7 chimera**. (**a**) In vivo activity of MLuc7 after treatment of HeLa cells with 10 nM of TcdA1$^{FPLC/Collag}$ + BC3-MLuc7 for 1 h at 37°C (endocytosis). (**b**) pH-dependent delivery of MLuc7 across the plasma membrane of HeLa cells. (**c, d**) Plasma membrane injection of MLuc7 at pH 7.5 (**c**) and pH 5 (**d**). Unpaired, two-tailed t test (P < 0.001 (**), P < 0.001 (***), ±SEM). N = 3.*

To further prove the distinction between endocytosis- and plasma membrane–based toxin deliveries, a similar experiment using the microtubule disrupting inhibitor, nocodazole was conducted. As shown in Fig. 25a, intoxication of HeLa cells via endocytosis-based toxin delivery was inhibited by nocodazole but not plasma-membrane-based injection of luciferase by PTC3-MLuc7 fusion complex (Fig. 25b). It is worth to note that, plasma membrane-based injection closely mimics the type-III secretion system where effectors are also injected via the plasma membrane.

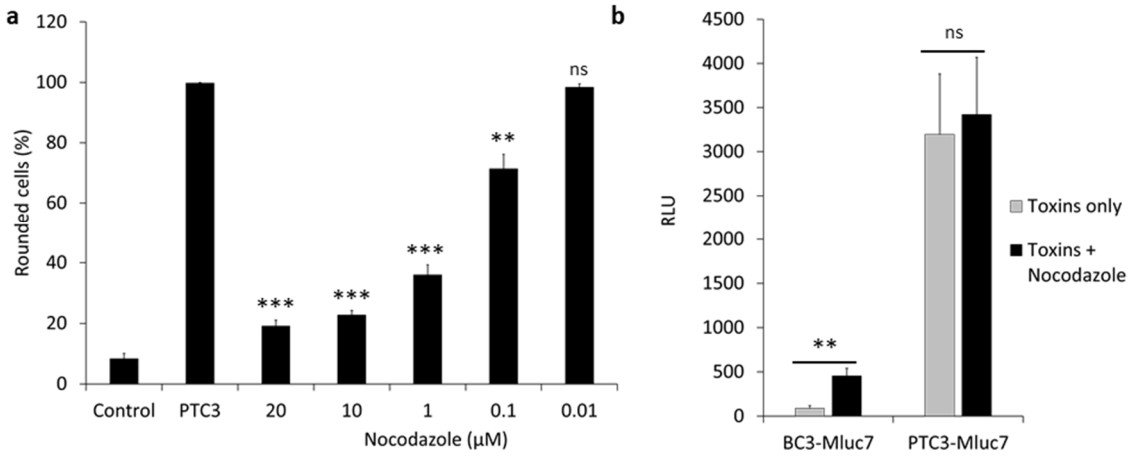


***Figure 25: Influence of nocodazole on endocytosis- and plasma membrane-based intoxication**. (**a**) Intoxication of HeLa cells with 1.4 nM of TcdA1$^{FPLC/Collag}$ + 3.5 nM of BC3 in the presence of increasing concentrations of nocodazole. (**b**) Plasma membrane injection of MLuc7 across the cell membrane at pH 5 in the presence or absence of 20 µM of nocodazole. Unpaired, two-tailed t test (P < 0.001 (**), P < 0.001 (***), ns = not significant, ±SEM.*

Having obtained data for injection of Mluc7, the efficiency of PTC could be estimated. The first component (**1**) required for the estimation was the number of molecules producing one light unit (LU) in cell lysate (3.24×10^8 molecules) (Fig. 23b, c). The next components were standard curves detecting luminescence produced by injecting Mluc7 into cells from the plasma membrane at pH 7.5 (Fig. 24c) (**2**) and pH 5 (Fig. 24d) (**3**). From these curves, the number of light units produced by 1 nM of full toxin at pH 7.5 and 5 were 30.5 LUs (10^{10} molecules) and 961.5 LUs (3.1×10^{11} molecules) respectively. Finally, the average number of cells utilized in each treatment (**4**) was calculated as 10^6 cells. Therefore, the number of molecules injected into each cell at pH 7.5 was about 10^4 molecules, while pH 5 was 3.1×10^5 molecules.

To further prove intracellular localization of the injected luciferase, CaCo2 cells were grown overnight on glass bottom plates. Then, they were incubated for 1 h at 4 °C with either buffer only, TcdA1$^{FPLC/Collag}$ alone (10 nM), BC3-MLuc7 alone (10 nM), or PTC3-MLuc7 (10 nM TcdA1$^{FPLC/Collag}$ + BC3-MLuc7). Subsequently, extracellular protein was inactivated by treatment with trypsin/EDTA and the cells were washed. Next, in the microscope, cell-permeable coelenterazine (Biotium) was added and luminescence detected for 30 minutes by live-cell microscopy. Bioluminescence images (between 450 and 700 nm) were acquired with a GaAsP detector set to maximum gain. The GaAsP detector (gallium arsenide phosphide) enables higher quantum yields and thereby higher sensitivity compared with conventional

photomultiplier tubes. Consistent with previous results, cells treated with the full toxin showed the highest bioluminescence, indicating successful delivery of the luciferase (Fig. 26).

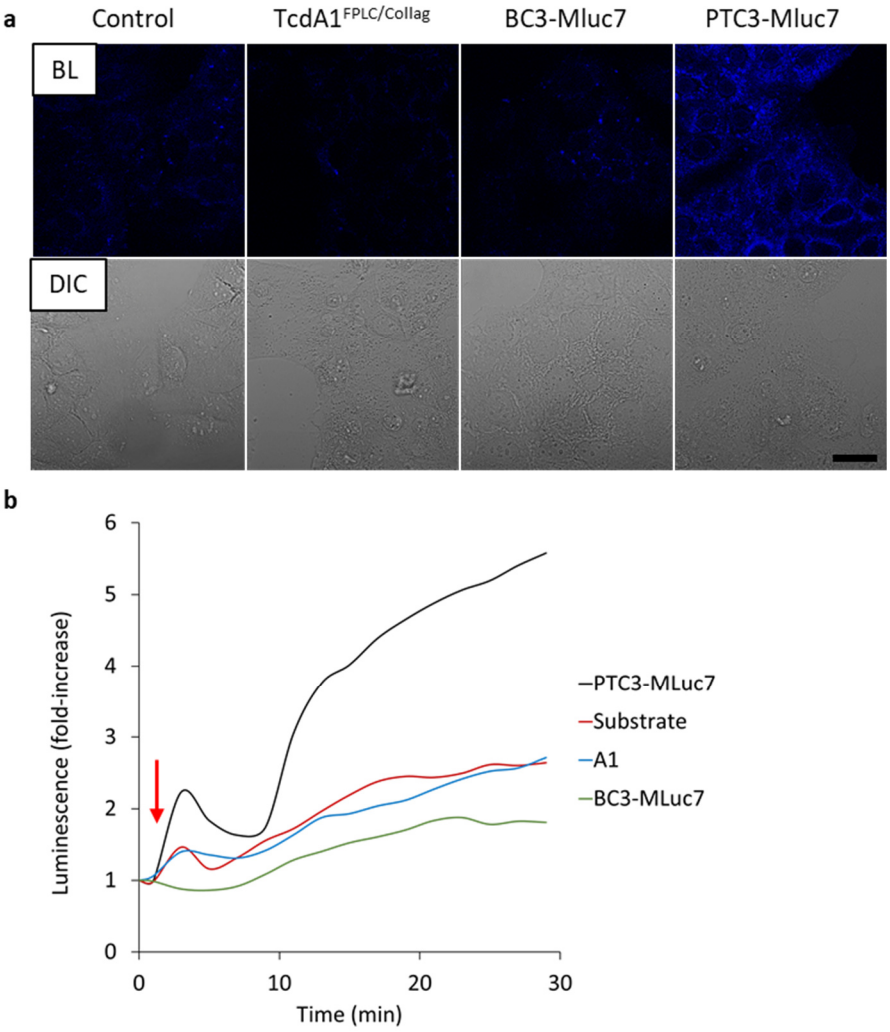


Figure 26: In vivo luminescence activity of injected BC3-MLuc7 fusion toxin*. (**a**) Live-cell microscopy images of bioluminescence after 10 min. Membrane delivery of toxin (10 nM) into HeLa cells was performed for 1 h at 4°C. Then, the cells were transferred to a live-cell imaging microscope, where 2 µg/ml of coelenterazine H (Biotium) was added and visualization started immediately. Figure represents one of two independent experiments. Scale bar: 15 µm. BL, bioluminescence; DIC, differential interference contrast. (**b**) Quantification of the live-cell imaging setup in (a) over a 30 min period. The graph represents one of three independent experiments.*

3.4. The *Photorhabdus* TcdA1 binds to N-linked glycans on the cell surface

3.4.1. TcdA1 binds to fucosylated Lewis X and Y N-glycans

The receptor(s) for the *Photorhabdus* Tc complex remain unknown. However, considering that a number of bacterial toxins, including YenTcA, a member of the Tc toxin family, bind to cell surface glycans, there is a likelihood that the PTC could also interact with glycans. Its variety of putative receptor binding domains and wide target-host range, are also indicators of

interaction with glycans. Therefore, to investigate the possible glycan binding of TcdA1, a glycan microarray screen was performed with the help of the Consortium for Functional Glycomics (CFG) (www.functionalglycomics.org).

The screening procedure was performed by the CFG. Briefly, two concentrations (5 and 50 µg) of $TcdA1^{FPLC}$ and $TcdA1^{FPLC/Collag}$ were used. $TcdA1^{FPLC/Collag}$ was directly labelled with Dylight-488, while $TcdA1^{FPLC}$ was left unlabelled and visualized using a TcdA1-specific primary antibody and a Horseradish peroxidase-tagged anti-mouse secondary antibody. The proteins were diluted in binding buffer and allowed to bind to glycan-printed slides for 1 h, at room temperature (RT) in a dark humidified chamber. The slides were then washed, fluorescently screened, and images of the slides read in a Perkin Elmer Microscanarray XL4000 scanner. Six replicates of each screen were performed. After scanning, the amount of protein bound to each glycan spot was quantified by fluorescence, tabulated and averaged.

The screen data was further analysed using the online Glycan Array Dashboard (GLAD) software (Mehta and Cummings, 2019). First, a force graph showing interaction of the protein samples with the top glycan hits was generated using default parameters (Fig. 27a). All top hits with an average relative fluorescence unit (RFU) above 1000 were included in the analysis. The hits were then grouped according to their interaction with each sample. As shown in Table 13, ten hits met these criteria and were grouped according to the protein sample(s) they interacted with (Fig. 27a). The overall top hits were N-glycans, with a majority of them having terminal galactose and fucosylated Lewis X or Y structures. (Fig. 27b-e). See Appendix 2 for the list of all the glycans used in the glycan screen.

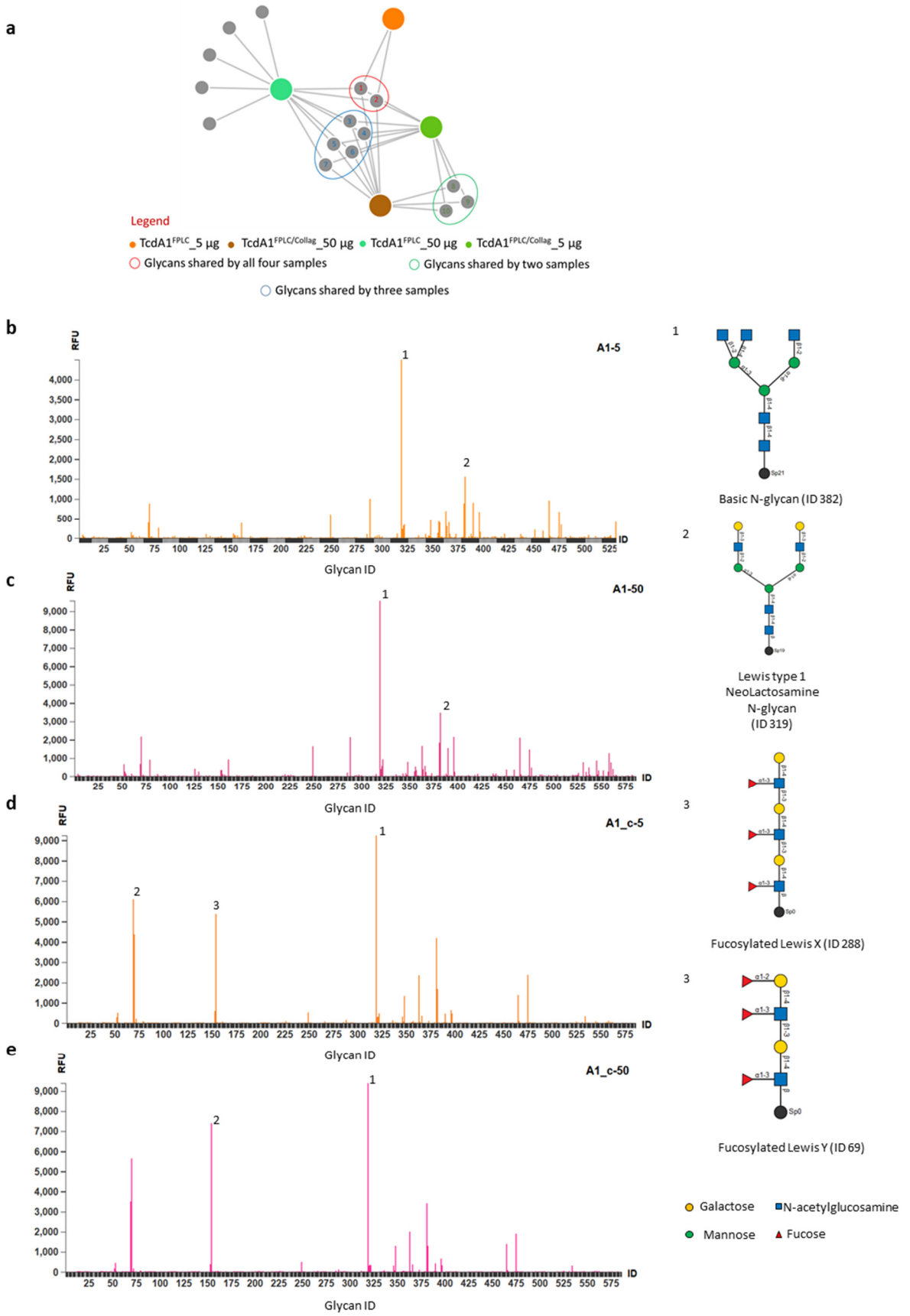


Figure 27: Results of glycan microarray screens of TcdA1. *(**a**) Force graph of the top hits interacting with the four TcdA1 samples. All hits with an average relative fluorescence unit above 1000 were linked to each TcdA1 sample to reveal shared interactions. The hits marked in red (red oval) were shared by*

*all four samples, those in green (green oval) were shared by three samples and those in blue (blue oval) were shared by two samples). (**b** – **e**) Graph of all hits for 5 µg TcdA1^{FPLC} (**b**), 50 µg TcdA1^{FPLC} (**c**), 5 µg TcdA1$^{FPLC/Collag}$ (**d**) and 50 µg TcdA1$^{FPLC/Collag}$ (**e**). 1 = Overall top hit (Basic N-glycan), 2 = Second top hit (Lewis type 1 Neolactosamine N-glycan), 3 = Third top hit (Lewis X/ Y N-glycan).*

Table 13: Summary of the top hits of a glycan microarray screen shared by all samples of TcdA1^{FPLC} and TcdA1$^{FPLC/Collag}$ (the hits are colour-coded according to the groups in Fig. 27a)

Number	Glycan ID	IUPAC name	Common name
1	382	GlcNAcb1-2Mana1-6(GlcNAcb1-4(GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	N-Glycan Basic
2	319	Galb1-3GlcNAcb1-2Mana1-6(Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp19	Neo Lactosamine, N-Glycan Basic, Lewis Type 1
3	363	Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-4(Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	Lactosamine, N-Glycan Basic, Lewis Type 2
4	70	Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	Fucosylated, Lewis X, Lewis Y
5	475	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	Sialylated, Fucosylated, N-Glycan Basic, Lewis Type 2
6	465	Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-6(Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	Fucosylated, N-Glycan Basic, Lewis X, Lewis Y
7	381	Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-2)Mana1-6(Galb1-4GlcNAcb1-4(Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	Lactosamine, N-Glycan Basic, Lewis Type 2
8	154	Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	Fucosylated, Lewis X
9	348	Galb1-3GlcNAcb1-2Mana1-6(Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp22	
10	69	Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	Fucosylated, Lewis X, Lewis Y

3.4.2. N-glycan deficient cells are resistant to intoxication by PTC3

To gain insight into N-glycan binding, CHO-Lec1 cells, which do not produce complex and hybrid N-glycans were used. These cells have a mutated MgatI gene, which codes for the N-acetylglucosaminyltransferase I (GlcNAc-TI). Due to the mutation of MgatI, these cells have an inactive GlcNAc-TI and hence lack N-glycans (Puthalakath *et al*., 1996). However, they have short oligomannose trees and produce normal O-linked glycans (North *et al*., 2010). Cells were intoxicated with $PTC3^{aff}$ and as a positive control, with C2 toxin, which was shown to interact with N-glycans (Eckhardt *et al.*, 2000). As a negative control, Toxin B, which has yet unknown N-glycan-binding partners was used. As shown in Fig. 28a and b, Lec1 cells were resistant to PTC3 and C2 toxin, but not to Toxin B, indicating a dependence on N-glycans for PTC3 binding. A consistent result was obtained with CHO-RK14 cells, which have a similar phenotype as the Lec1 cells (data not shown).

Resistance was confirmed through a post-intoxication ADP-ribosylation assay. Here, CHO-K1 wildtype and Lec1 mutant cells were either intoxicated for 3 h with $PTC3^{aff}$, or left untreated. Then, the cells were lysed and lysates treated with 0.7 µM BC3 and 50 µM biotin-NAD^+ in a buffer containing 50 mM Hepes and 50 µM $MgCl_2$, pH 7.4. All samples were incubated at 21 °C for 20 min before the reaction was stopped by addition of Laemmli buffer. The samples were then separated by SDS-PAGE and analysed via Western blot using a horseradish peroxidase-avidin D antibody (Vector labs). As shown in Fig. 28c, only the CHO-K1 cells were intoxicated (top panel) and all lysates except the pre-intoxicated CHO-K1 cells, were modified by BC3 (bottom panel). The lack of modification in this lane confirmed the initial intoxication by PTC3, as all actin was already ADP-ribosylated and could not be modified any further by TccC3hvr. As a control, recombinant βγ-actin was used in the presence or absence of biotin-NAD^+. Only actin treated with both toxin (C3hvr) and biotin-NAD^+ showed a signal and not one treated with biotin-NAD^+ only (Con) (Fig. 28c, bottom panel). Unmodified actin appeared to be degraded when incubated at similar conditions (actin control).

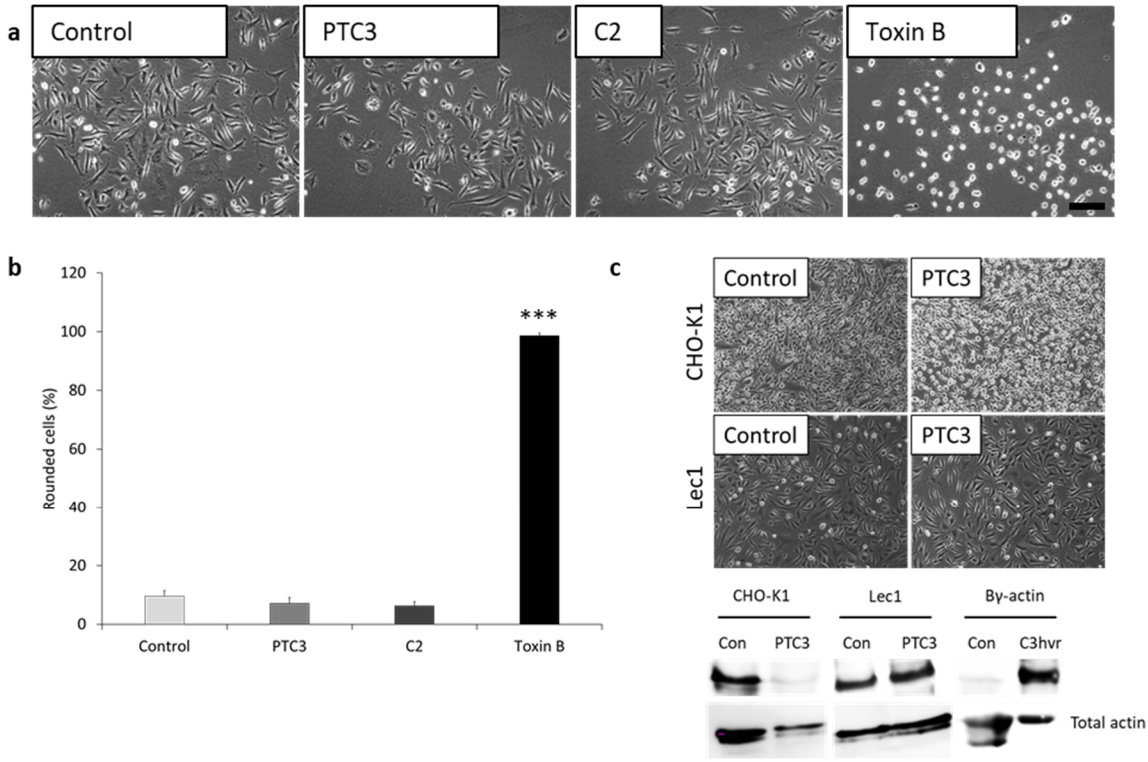


Figure 28: Resistance of CHO-Lec1 cells to intoxication by PTC3*. (**a**) Intoxication of Lec1 cells with PTC3 (1.4 nM TcdA1^{aff} + 3.5 nM BC3), Toxin B (4 pM) and C2 toxin (7.5 nM C2IIa + 6 nM C2I). Pictures were taken after 5 h of incubation at 37 °C. (**b**) Quantification of cells intoxicated in (a). (**c**) Post-intoxication ADP-ribosylation of CHO-K1 and Lec1 cell lysates from cells pre-intoxicated for 3 h with PTC3^{aff} (top panel). Lysates were treated for 20 min at 21 °C with BC3 (0.7 µM) and biotin-NAD+ (50 µM), then separated by SDS-PAGE and visualized via Western blot using horseradish peroxidase-avidin D antibody (bottom panel). Unpaired, two-tailed t test (P < 0.001 (***), ns = not significant, ±SEM). N=3.*

To further study the resistance of Lec1 cells to intoxication by the PTC3, binding of the toxin to the cells was analysed by flow cytometry. Increasing concentrations of Dylight-488-labeled TcdA1$^{FPLC/Collag}$ were incubated with the cells for 15 min at 4 °C, before washing and analysis. Compared to wildtype CHO-K1 cells, Lec1 cells showed only minimal binding (Fig. 29a). Taken a step further, binding studies performed at pH 5 and 9, showed no binding at pH 9 and only little binding at pH 5 on Lec1 cells (Fig. 29b). This was contrary to results obtained with HeLa cells, where higher binding of the cleaved toxin was observed at pH 9 (see Fig. 14a). Furthermore, exogenous addition of various monosaccharides (100 mM) did not restore sensitivity of Lec1 cells to PTC3 (Fig. 29c).

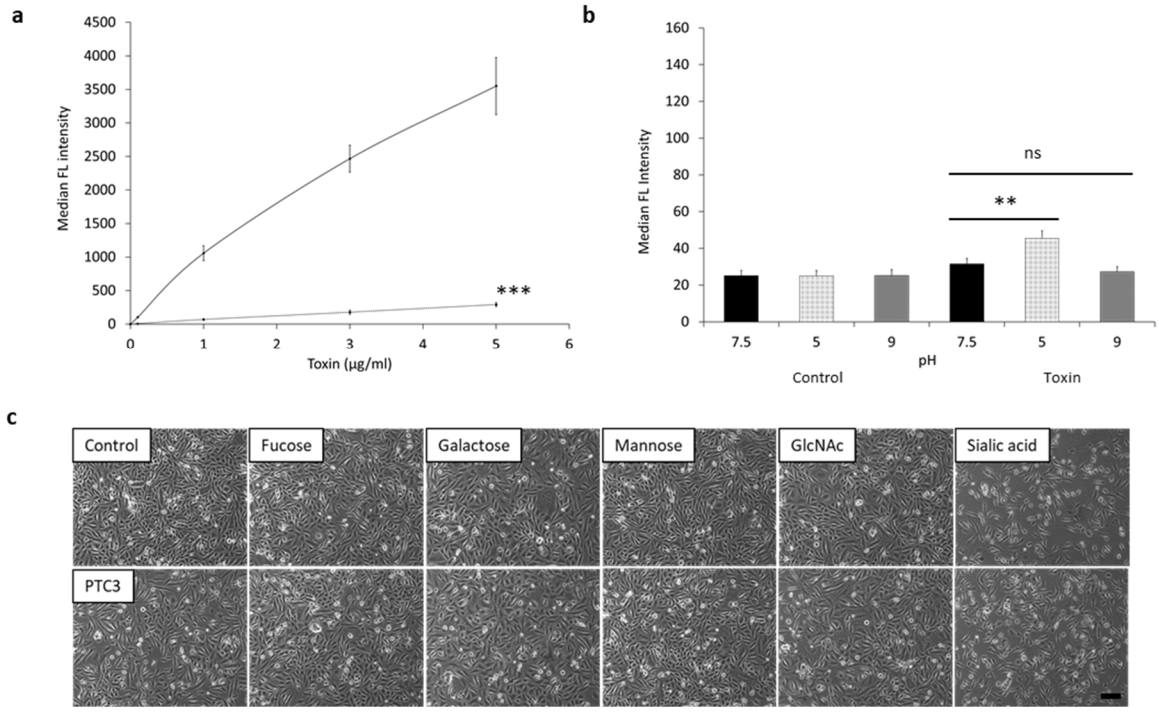


Figure 29: Binding of TcdA1 to CHO-Lec1 cells and influence of sugars on intoxication. *(**a**) Flow cytometry-based binding studies of Dylight-488-TcdA1$^{FPLC/Collag}$ on CHO-K1 and Lec1 cells. Toxin was allowed to bind for 15 min at 4 °C and then the cells were washed and analysed. (**b**) Flow cytometry-based binding studies of Dylight-488-TcdA1$^{FPLC/Collag}$ on Lec1 cells at pH 7.5, 5 and 9. Toxin was allowed to bind for 15 min at 4 °C in each pH buffer and then the cells were washed and analysed. (**c**) Intoxication of Lec1 cells with PTC3 (1.4 nM TcdA1^{FPLC} + 3.5 nM BC3) in the presence of 100 mM of sugars. TcdA1$^{FPLC/Collag}$ was first premixed with the sugars, incubated for 5 min at RT and then added to the cells. Unpaired, two-tailed t test P < 0.01 (**), P < 0.001 (***), ns = not significant, n=3, ±SEM.*

3.4.3. Exogenous addition of Lewis X trisaccharides reduced intoxication kinetics

To confirm the direct involvement of fucosylated Lewis X (Le^X) antigens on cell intoxication by PTC3, fucosylated Le^X trisaccharides (Fig. 30a) were synthesised (Biosynth, Basel, Switzerland). Then, to test their influence in cell intoxication, TcdA1^{aff} (1.4 nM) was premixed with increasing concentrations of the trisaccharide and then used to intoxicate HeLa cells, together with BC3 (3.5 nM). As shown in Fig. 30b and d, the Le^X trisaccharide delayed cell intoxication in a concentration-dependent manner, with 1 mM leading to 50 % inhibition. On the other hand, Toxin B, used as a control, was not affected by addition of the sugar (Fig. 30c and e). Furthermore, exogenous addition of a mixture of the sugars constituting the Le^X trisaccharide (Galactose, Fucose and N-acetylglucosamine (GlcNAc)) did not affect intoxication either (Fig. 30f). These data therefore confirmed the involvement of Le^X trisaccharides in interaction of PTC3 with the cell surface. They also confirmed that the sugars have to be combined into an N-glycan structure to have an effect, and not simply mixed as monosaccharides.

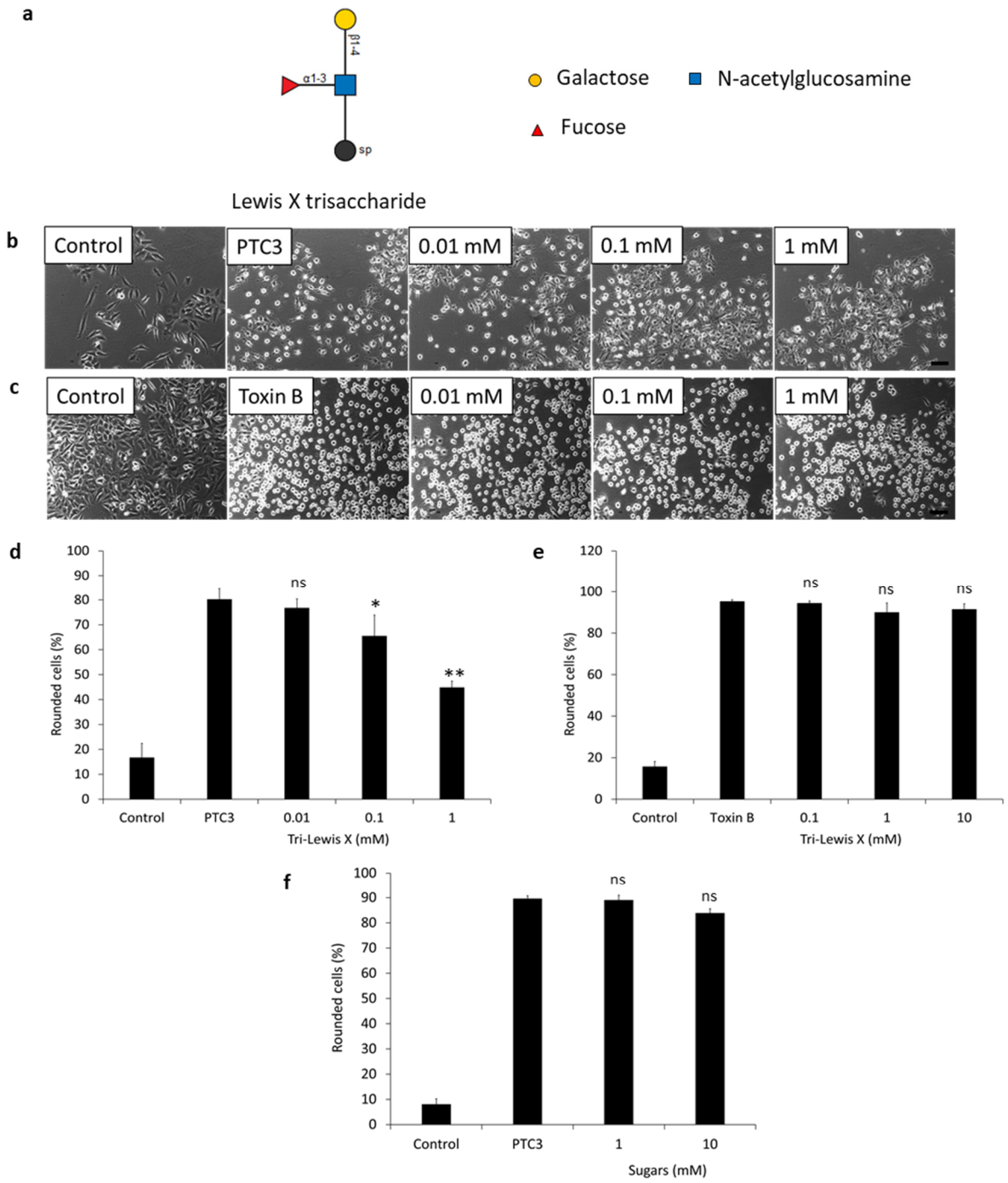


Figure 30: Effects of exogenous addition of Lewis X trisaccharides to cell intoxication by the PTC*. (**a**) Structure of the artificially synthesised fucosylated Lewis X trisaccharide. (**b**) Intoxication of HeLa cells for 4 h with PTC3 consisting of TcdA1^{off} (1.4 nM) premixed with the indicated concentrations of sugars, and BC3 (3.5 nM). (**c**) Intoxication of HeLa cells for 4 h with Toxin B (4 pM) premixed with the indicated concentrations of sugars. (**d**) Quantification of the number of rounded cells in cells intoxicated in (b). (**e**) Quantification of the number of rounded cells in cells intoxicated in (c). (**f**) Intoxication of HeLa cells for 4 h with PTC3 consisting of TcdA1^{off} (1.4 nM) premixed with monosaccharides constituting the Lewis X trisaccharide in (a). Unpaired, two-tailed t test P < 0.05 (*), P < 0.01 (**), ns = not significant, n=3, ±SEM. The scale bar is 80 µm.*

3.4.4. TcdA1 may interact with glycosaminoglycans (GAGs)

To test whether proteoglycans may be involved in TcdA1 binding, a mutant HeLa cell line deficient in heparan sulfate proteoglycans (HSPGs) was used. This cell line was a knock-out mutant of exostosin-2 (EXT2) (Tao *et al*., 2019), a glycosyltransferase required for the elongation of the heparan sulfate (HS) chain in the synthesis of HSPGs (Kreuger and Kjellén, 2012). Therefore, the knock-out mutant had reduced cell surface heparan sulfate levels compared with the wild-type (Tao *et al*., 2019). After treatment for 5 h with PTC3^{aff}, the mutant cells showed a slightly higher resistance to intoxication compared to wildtype HeLa cells (Fig. 31a). This correlated with slightly reduced, but statistically insignificant, binding of Dylight-488-labeled TcdA1$^{FPLC/Collag}$ (Fig. 31b). However, a Chinese hamster ovary (CHO) mutant (pgsA-745) lacking xylosyltranferase-I activity, did not show a difference in sensitivity compared to the wildtype (Fig. 31c). Xylosyltranferase-I is the chain-initiating enzyme in the biosynthesis of proteoglycans. Specifically, it initiates the biosynthesis of the glycosaminoglycan (GAG) linkage tetrasaccharide in proteoglycans (Götting *et al*., 2000; Kuhn *et al*., 2003). Therefore, CHO-pgsA-745 cells do not synthesise chondroitin or heparan sulfate (Esko *et al*., 1985).

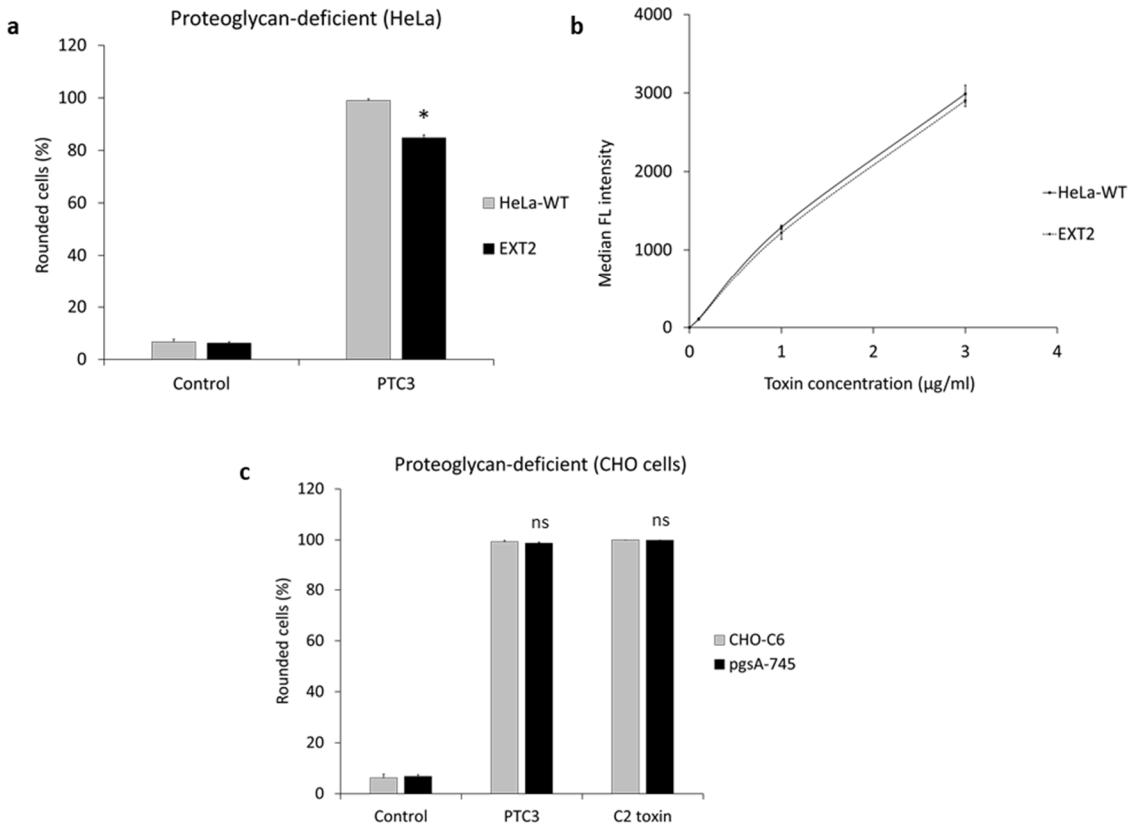


***Figure 31: Intoxication of proteoglycan-deficient cells by PTC3**. (**a**) Intoxication of HSPG-deficient HeLa cell mutants, EXT2, with PTC3 (1.4 nM TcdA1^{aff} + 3.5 nM BC3). Cells were intoxicated for 5 h at 37 °C before pictures were taken and the number of intoxicated cells was determined. (**b**) Flow cytometry-based binding analysis of Dylight-488-labeled TcdA1$^{FPLC/Collag}$ (0.7 nM) on HSPG-deficient HeLa cell mutant. Toxin was allowed to bind for 15 min at 4 °C before the cells were washed and analysed. (**c**) Intoxication of HSPG-deficient CHO mutant, pgsA-745, with PTC3 (1.4 nM TcdA1^{aff} + 3.5 nM BC3). Cells were intoxicated for 5 h at 37 °C before pictures were taken and the number of intoxicated cells was determined. Unpaired, two-tailed t test (P < 0.05 (*), ns = not significant, ±SEM). N=3.*

Glycosaminoglycans (GAGs) are linear, anionic polysaccharides that provide an ideal landing pad for proteins, macromolecules, bacteria, viruses and parasites (Kamhi *et al.*, 2013; Tao *et al.*, 2019). To test their involvement in binding, TcdA1^{aff} was premixed with heparin (HP), heparan sulfate (HS), the semi-synthetic GAG analogue dextran sulfate (DexS), and chondroitin sulfate (CS), before incubation for 2 – 5 min at RT. Then, together with BC3, it was used to intoxicate HeLa cells. Consistent with data from the HS-deficient cells, both HP and HS reduced cell intoxication in a concentration-dependent manner (Fig. 32a and b). DexS also marginally reduced intoxication (Fig 32c), but CS had no effect (Fig. 32d). Lower concentrations of HP and HS were required to inhibit intoxication than DexS. These results further suggested the influence of GAGs on the binding of TcdA1 to cells. Additionally, their variability in inhibiting intoxication suggested the influence of charge, rather than the sulfate group per se. However, taken together, these results indicate that GAGs may have a marginal influence on intoxication by PTC3, compared to N-glycans.

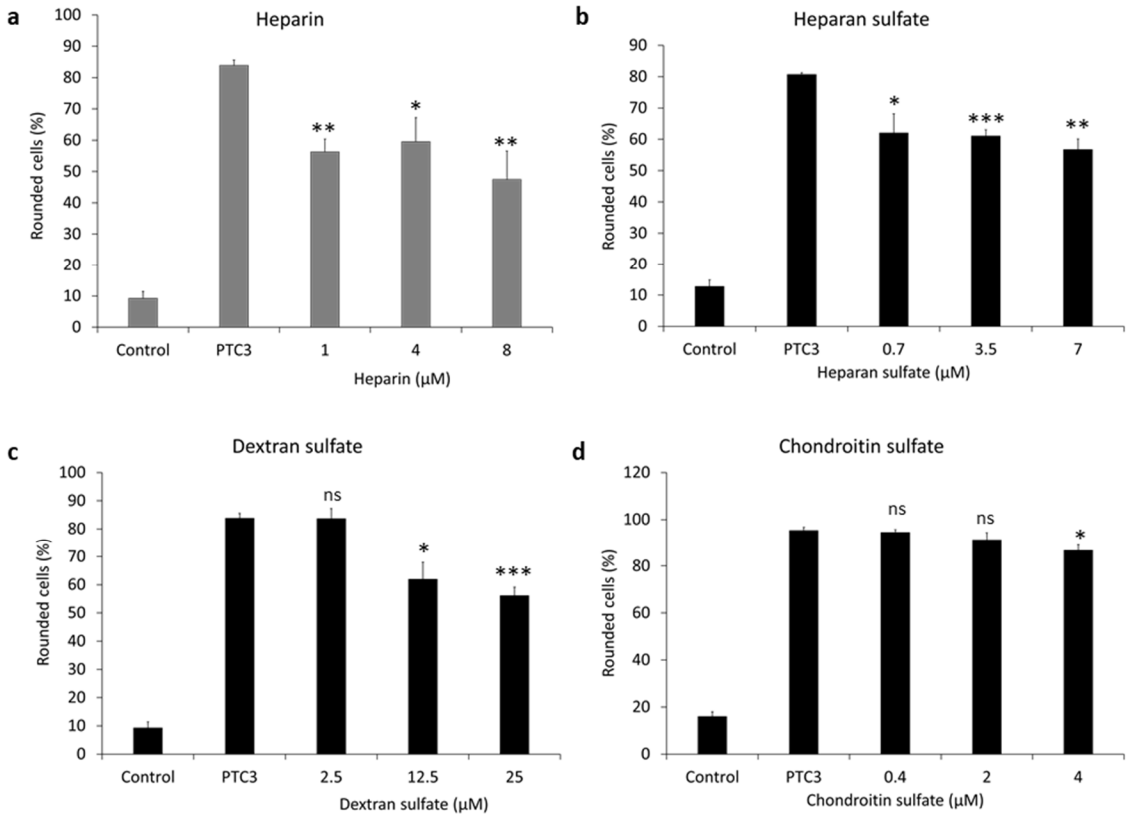


Figure 32: Effects of glycosaminoglycans on cell intoxication by PTC3*. (**a-d**) Intoxication of HeLa cells with TcdA1^{aff} (1.4 nM) premixed with increasing concentrations of heparin (**a**), heparan sulfate (**b**), dextran sulfate (**c**) and chondroitin sulfate (**d**). Unpaired, two-tailed t test (P < 0.05 (*), P < 0.01 (**), P < 0.001 (***), ns = not significant, ±SEM). N=3.*

3.4.5. Sialic acid-, galactose-, fucose- and glycolipid-deficient cells were sensitive to PTC3

To narrow down to the influence of individual sugars that constitute N-glycan chains, different monosaccharide-deficient mutant cell lines were employed. Specifically, CHO Lec2 and Lec8 cells were used to study the influence of sialic acid and galactose, respectively (North *et al*., 2010). Fucose-deficient leukocyte-adhesion deficiency II (LADII) cells (Wild *et al*., 2002) were also studied. Additionally, a mouse B16 melanoma cell line, GM95, which lacks sialyllactosylceramide, and hence reduced glycolipids, was used together with its wildtype counterpart, MEB4 (Ichikawa *et al*., 1994). All cells were intoxicated with PTC3 (1.4 nM TcdA1^{aff} + 3.5 nM BC3) for 5 h and the percentage of rounded cells was quantified. As shown in Fig. 33, none of these cell lines showed any resistance to intoxication by PTC3. Similarly, neither Lec2, Lec8, nor GM95 showed decreased toxin binding compared to the wildtypes (Fig. 33c and f), GM95 even showed a slight increase in binding compared to MEB4 (Fig. 33f).

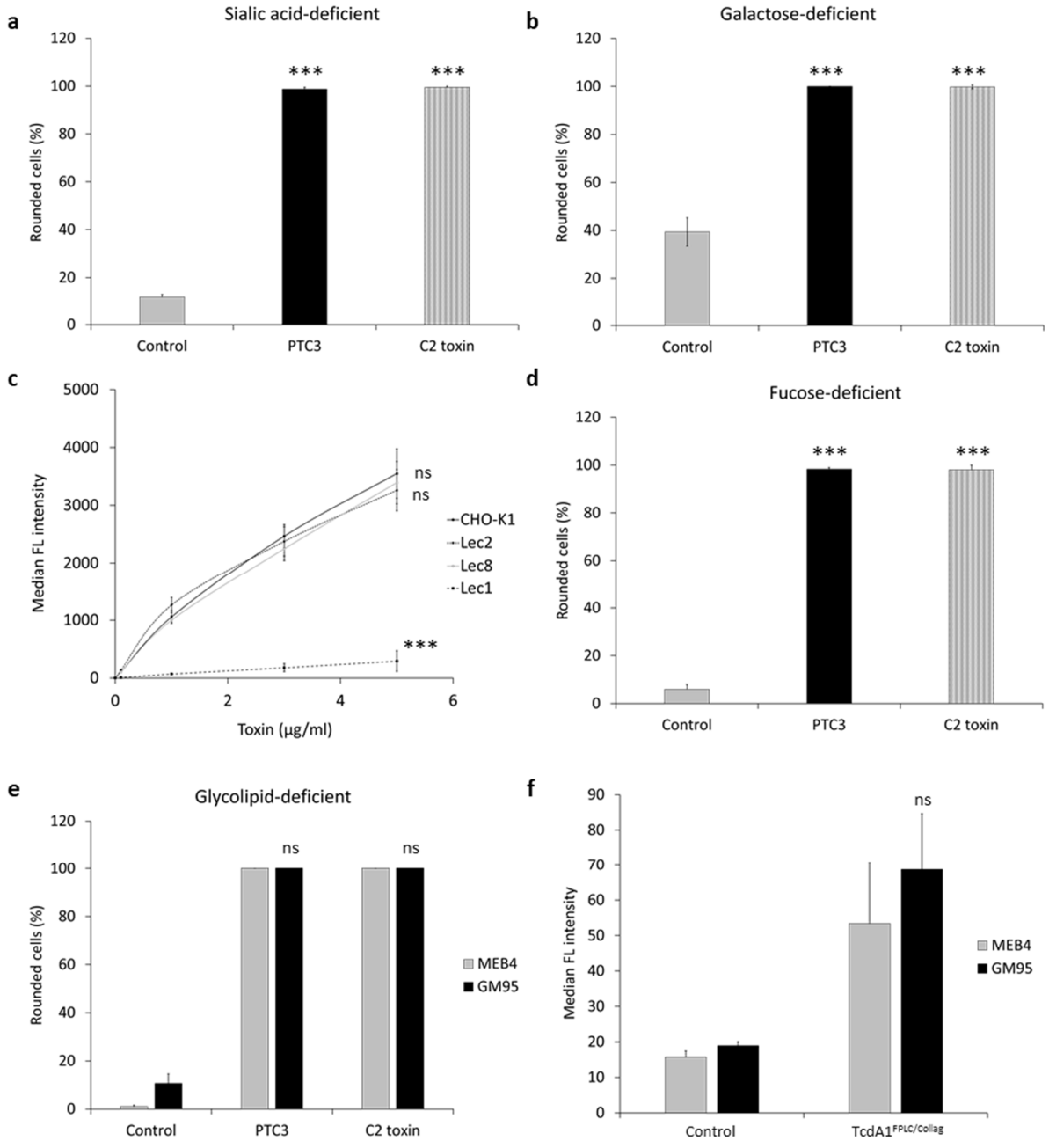


Figure 33: Intoxication and binding of PTC3 on monosaccharide- and glycolipid-deficient cells*. PTC3 (1.4 nM TcdA1^{qff} + 3.5 nM BC3)-treated cells, Lec2 (**a**), Lec8 (**b**) LADII (**d**), GM95 and MEB4 (**e**), were incubated for 5 h at 37 °C and then the proportion of rounded cells was quantified. Flow cytometry-based binding analyses of Lec2 and Lec8 (**c**), as well as MEB4 and GM95 cells (**f**) were performed by allowing TcdA1$^{FPLC/Collag}$ to bind for 15 min at 4 °C before washing and analysis. Unpaired, two-tailed t test P < 0.001 (***), ns = not significant, n=3, ±SEM.*

3.4.6. Exogenous addition of sialic acid, but not other monosaccharides, inhibited cell intoxication

Complementary to the aforementioned glycan-deficient cell lines, TcdA1 was pre-exposed to various monosaccharides before binding and/or intoxication studies. The toxin was premixed

in a sugar solution and incubated for 2 – 5 min at RT before cell intoxication. As shown in Fig. 34a, neither D-glucose, L-fucose, D-xylose, D-galactose nor D-mannose, at 10 and 100 mM concentrations, had an effect on intoxication of HeLa cells by PTC3. However, sialic acid exhibited a strong inhibition at a low concentration of 0.1 mM (Fig. 34b, c). This inhibition was not cell-type-specific as it was also observed in CHO-K1 cells (Fig. 34d). Additionally, sialic acid blocked binding of dylight-488-labeled TcdA1$^{FPLC/Collag}$ on HeLa cells (Fig. 34e). Interestingly, while 0.1 mM only reduced binding one-fold, it completely inhibited intoxication (compare Fig. 34c and e). This raised a discrepancy between binding and intoxication.

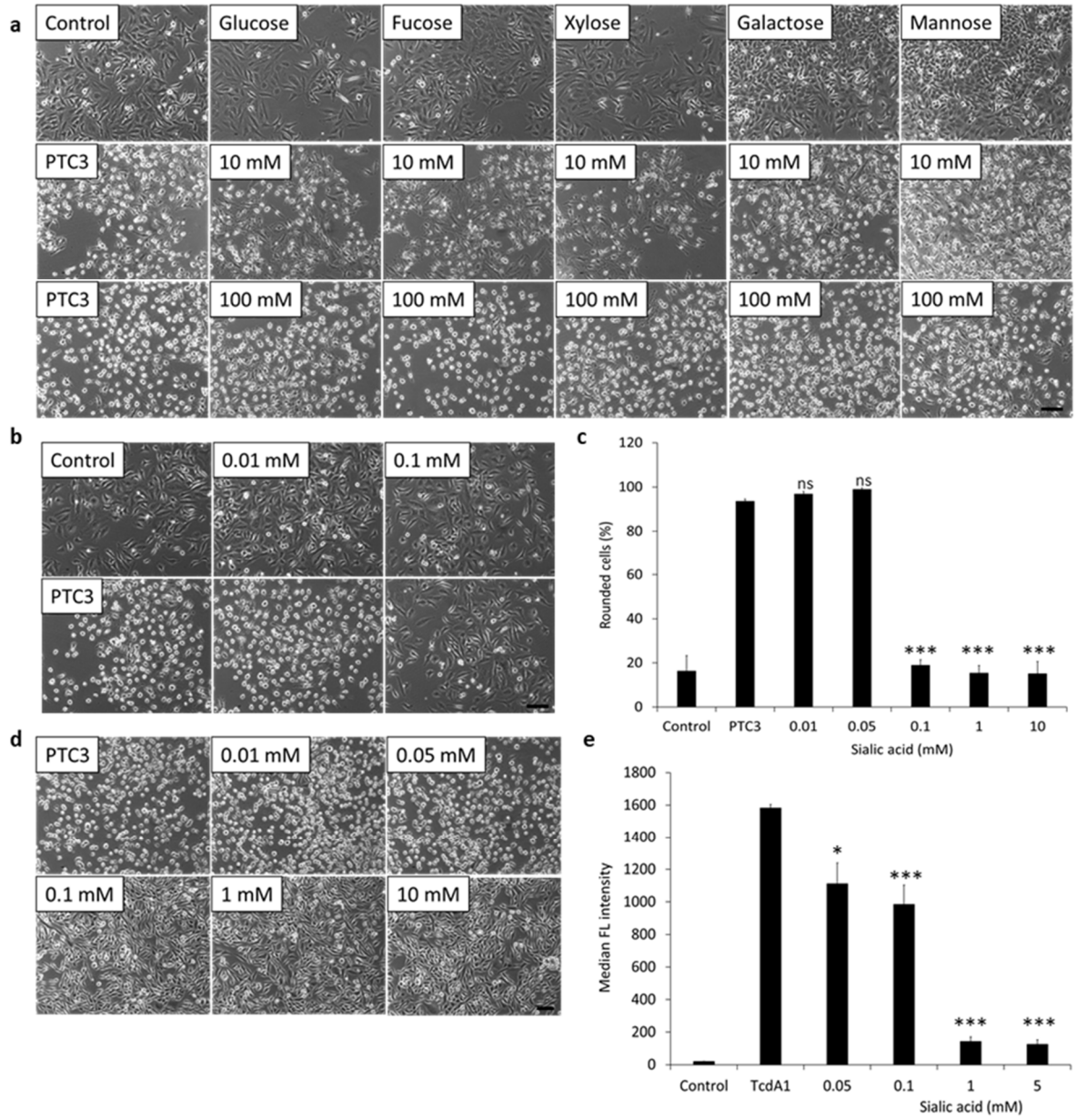


Figure 34: Effects of exogenous addition of sugars to cell intoxication by the PTC*. (**a**) Intoxication of HeLa cells with PTC3 consisting of TcdA1^{qff} (1.4 nM) premixed with the indicated concentrations of sugars, and BC3 (3.5 nM). The pictures were taken after 5 h of intoxication. The top panel are control*

*cells treated with the sugars only, whereas the two bottom panels are cells treated with toxin premixed with the corresponding sugar. (**b**) Intoxication of HeLa cells with TcdA1^{qff} (1.4 nM) premixed with the indicated concentrations of sialic acid and BC3 (3.5 nM). The top panel are control cells treated with the sialic acid only, whereas the bottom panel are cells treated with toxin + sialic acid. Pictures taken after 5 h. (**c**) Quantification of rounded cells in HeLa cells intoxicated in (b). (**d**) Intoxication of CHO-K1 cells with PTC3 as described in (b). (**e**) Flow cytometry analysis of the binding of Dylight-488-labeled TcdA1$^{FPLC/Collag}$ premixed with the indicated concentrations of sialic acid. Unpaired, two-tailed t test (P < 0.05 (*), P < 0.01 (**), P < 0.001 (***), ns = not significant, ±SEM). N=3. The scale bar is 80 µm.*

To clarify the effect at 0.1 mM of sialic acid, BC3-Mluc7 was used in order to provide a more sensitive readout. In this regard, pH 5-dependent plasma-membrane-injection of Mluc7 was performed, as described above (sub-section 3.3.7). First, TcdA1 was pre-mixed with different concentrations of sialic acid. Then, together with BC3-Mluc7, it was allowed to bind to the surface of HeLa cells at 4 °C. Finally, after washing, pore-formation was induced by pH-change to pH 5, to deliver the luciferase into cells. As shown in Fig 35a, a minimal luminescence, significantly above the background (BC3-Mluc7 only), was observed. This confirmed that indeed TcdA1 was able to bind in the presence of 0.1 mM of sialic acid, and could inject the luciferase into cells, thus supporting the flow cytometry data (Fig. 34e). However, by observing cell rounding, this small difference could not be distinguished because very few cells were intoxicated. The results at the other concentrations of sialic acid correlated with both the intoxication and flow cytometry data (compare Fig. 35a with Fig. 34c and e). Sialic acid did not have any effect on BC3-Mluc7 alone (Fig. 35b) and did not affect holotoxin formation (Fig. 35c).

To analyse the effect of sialic acid further, HeLa cells were treated with toxins pre-mixed either with sialic acid alone, or sialic acid and other sugars (GlcNAc and fucose). In one set-up, TcdA1 was premixed with equal concentrations of sialic acid and GlcNAc. Alternatively, TcdA1 was premixed with GlcNAc, fucose and sialic acid before intoxication. As shown in Fig. 35d, the effect of sialic acid was abrogated by the addition of other sugars. This indicated a probable competition for binding sites on the toxin. However, comparably, a sialic acid analog, Zanamivir® (Cayman chemical), used in similar concentrations as sialic acid, showed a much weaker effect (Fig. 35e). With Zanamivir, 100x more sugar was required to achieve a minimal inhibition of intoxication.

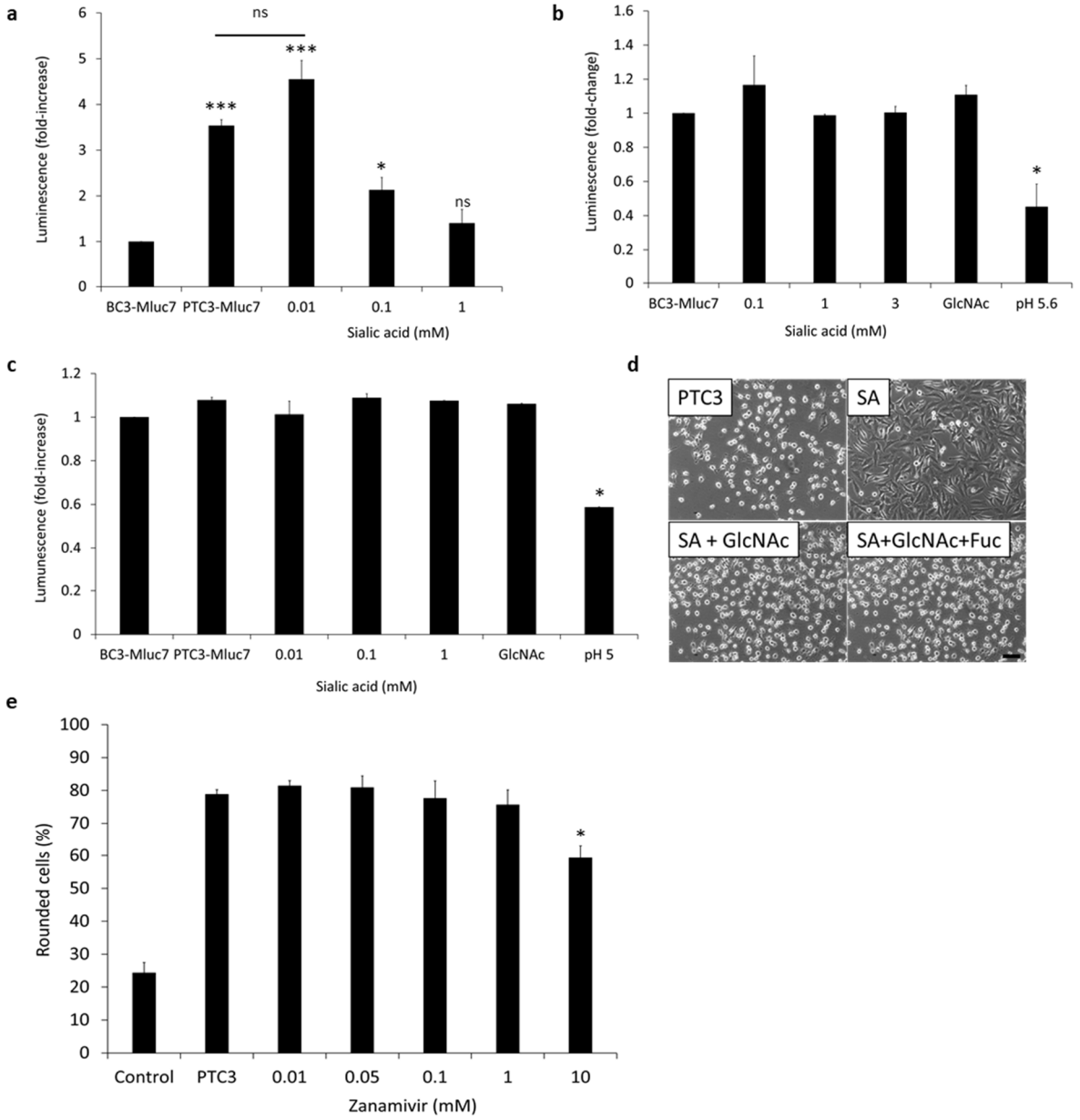


Figure 35: Effects of sialic acid on the PTC*. (**a**) Luciferase assay to assess binding and translocation by TcdA1. Toxins premixed with the indicated concentrations of sialic acid were allowed to bind on cells for 30 min at 4 °C. Then the pH of the surrounding medium was reduced to pH 5 to induce injection. Subsequently, the cells were washed and lysed before analysis of luminescence. (**b**) Bioluminescence analysis of BC3-Mluc7 (1 nM) premixed with the indicated concentrations of sialic acid, as well as, GlcNAc (3 mM) and a pH 5.6 buffer (PLB and 20 mM MES, pH 5). (**c**) Bioluminescence analysis of PTC3-Mluc7 (1 nM) premixed with the indicated concentrations of sialic acid, as well as, GlcNAc (3 mM) and a pH 5.6 buffer. (**d**) Intoxication of HeLa cells with TcdA1^{off} (1.4 nM) premixed with 0.1 mM of the indicated sugars and BC3 (3.5 nM). Pictures taken after 6 h. (**e**) Intoxication of HeLa cells with TcdA1^{off} (1.4 nM) premixed with the indicated concentrations of Zanamivir® and BC3 (3.5 nM). Unpaired, two-tailed t test (P < 0.05 (*), P < 0.01 (**), P < 0.001 (***), ns = not significant, ±SEM). N=3. The scale bar is 80 µm.*

3.4.7. TcdA1 may bind to a protein (co)receptor

In addition to glycan-binding, a possible interaction with cell-surface proteins was also studied. Detached HeLa cells were treated with 1 mg/ml of either proteinase K, or pronase, for 30 min at 37 °C and then bathed in medium with 10 % FCS to stop the reaction. Later, the cells were washed and the toxin was allowed to bind. The whole process was conducted in the presence of cycloheximide, an inhibitor of protein synthesis (Chow *et al*., 1995), to prevent the regeneration of cell surface proteins. After 15 min of toxin binding, the cells were washed and then analysed by flow cytometry. Epidermal growth factor (EGF) was used as a positive control as its receptor is ubiquitously expressed and upregulated in many cancer cells (Wee and Wang, 2017).

Cleavage of cell surface proteins minimally reduced TcdA1 binding to HeLa cells (Fig. 36a), while EGF binding was significantly reduced (Fig. 36b). To further analyse this effect, the same experiment was performed with glycolipid-deficient GM95 cells and their wildtype counterpart, MEB4. With fewer surface glycolipids, it was expected that the removal of the (presumably) exposed glycoproteins would show clearer results. However, as shown in Fig. 36c, there was, similarly, a minimal reduction in toxin binding. However, contrary to the HeLa cells, EGF did not show any change of binding to GM95 cells (Fig. 36d). This could be due to cell-type-dependent differential expression of the EGF receptor (Carpenter and Cohen, 1979). It is possible that the strong involvement of cell surface glycans in TcdA1-binding could have masked the effect of removing cell surface proteins, since many other glycans were likely still present of the cell surface, for example, glycolipids.

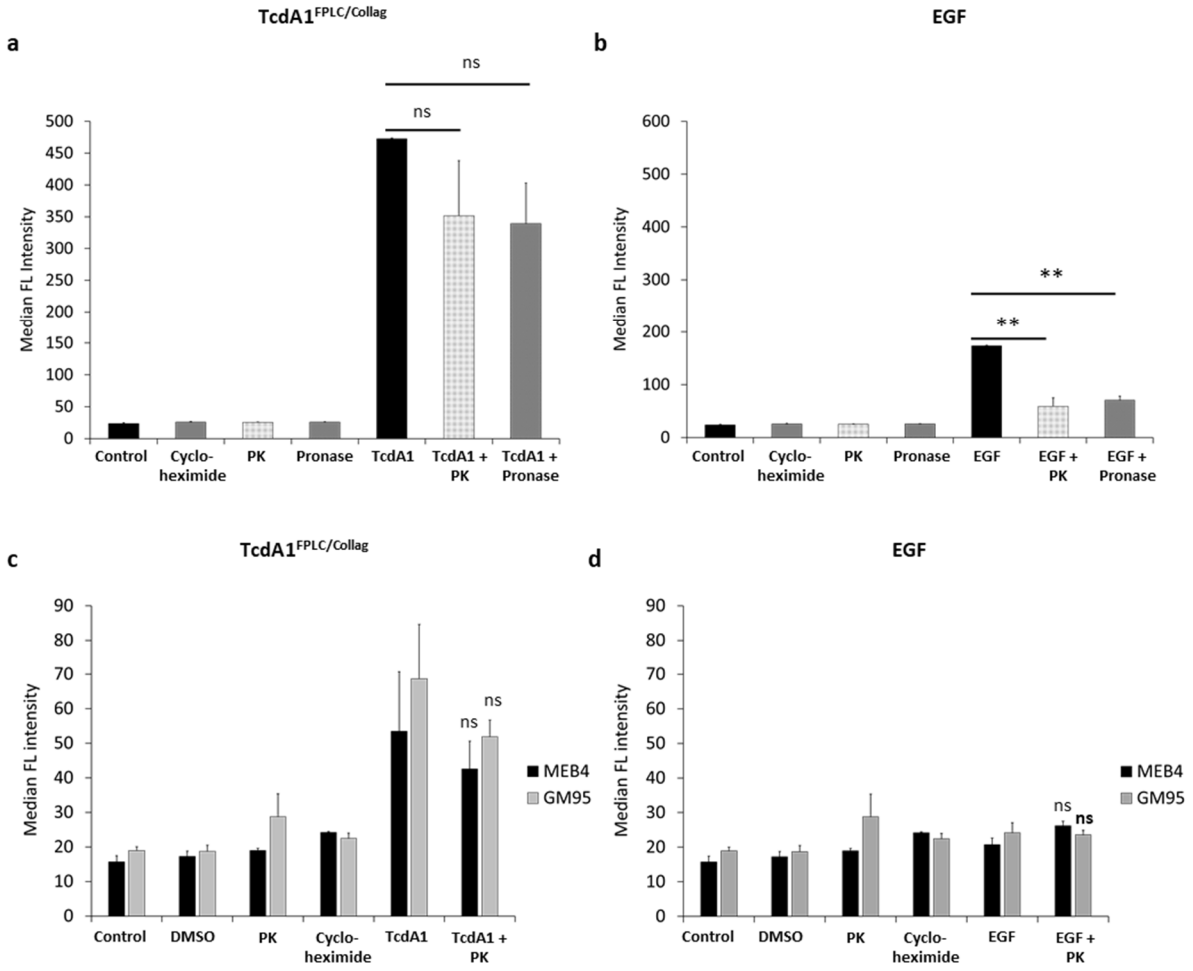


Figure 36: Cleavage of cell surface proteins on HeLa cells with proteinase K (PK) and Pronase*. Cells were treated for 30 min at 37 °C with the proteases (1mg/ml) in the presence of cycloheximide (4 µM). The reaction was then stopped by addition of medium containing 10 % FCS. Afterwards, they were washed and Dylight-488-TcdA1$^{FPLC/Collag}$ (**a**, **c**), or Alexa-488-EGF (**b**, **d**) was bound for 15 min at 4 °C. Finally, after washing, the cells were analysed by flow cytometry. Unpaired, two-tailed t test P < 0.05 (*), P < 0.01 (**), ns = not significant, n=3, ±SEM.*

4. Discussion

4.1. Activation of the TcA protein by metalloproteases leads to increased activity

The *Photorhabdus* toxin complex (PTC), also known as the Tc toxin, is a prominent member of the ABC-type Tc toxin family (ffrench-Constant and Waterfield, 2005). It is highly efficient in intoxication and modification of host cellular factors (Hinchliffe *et al*., 2010; Lang *et al*., 2011). It is formed through a combination of three proteins TcA, TcB and TcC. The three first combine into two separate complexes, which then come together to form the full active holotoxin (Meusch *et al*., 2014). One is the TcB-TcC complex, which forms a hollow cocoon-like structure that encloses an active enzyme, the TcC C-terminal hypervariable region (hvr). Two TcC hvrs have been characterized, C3hvr and C5hvr, which are ADP-ribosyltransferases targeting actin and small Rho GTPases respectively (Lang *et al*., 2011).

The second complex forming the holotoxin is the TcA complex, which is responsible for binding to target cells and translocation of the active enzyme. The TcA complex therefore, may influence not only host specificity, but also the efficiency of delivery of the toxic enzyme into target cells. Recent work by Stefan Ost (Ost and Ng'ang'a *et al.*, 2019), showed that some protease inhibitors including MG132, which is also known as a proteasome inhibitor, blocked the cytopathic effects caused by PTC. However, MG132 did not affect toxin complex formation, cell binding and uptake into early endosomes, or ADP-ribosylation of actin and Rho GTPases. Therefore, suggesting that MG132 affected a specific step of the intoxication process, which might be the translocation of the toxic hvr into the cytosol. Additionally, MG132-related cathepsin and calpain inhibitors like E64d, CA-074 Me, calpeptin and CalpIII also inhibited intoxication by PTC, thus implicating cathepsin-like enzymes in the protecting effects of MG132. Taken together, these data suggested that the PTC might be subjected to some form of protease-dependent activation in order to achieve full activity.

Indeed, as was previously mentioned, a number of bacterial toxins undergo cleavage by host proteases in order to achieve full activity. However, considering that the toxins used in this study consisted of recombinant proteins expressed in *E. coli*, it is also possible that it could be activated by a bacterial protease, perhaps from *P. luminescence*. The bacterium produces proteases that can cleave a 281 kDa Tc toxin protein into two fragments, 208 kDa and 63 kDa (Guo *et al*., 1999). However, this report did not reveal the identity and activity of the said Tc toxin. Coincidentally, our TcdA1 (283 kDa) was cleaved in a similar manner. Other studies have

also mentioned cleavage of TcdA1 (Blackburn *et al*., 1998; Bowen *et al*., 1998; Liu *et al*., 2003), however, here, the Tc toxins were either found cleaved in the bacterial supernatant, or, the information was based on patent publications. In every instance, the TcAs were reported to be cleaved at both the N- and C-termini (Fig. 37). A 10 kDa fragment is cleaved at the N-terminus (the first 88 amino acids), and a 63 kDa fragment from the C-terminus, where cleavage was suggested to occur between amino acids 1937 and 1938. An attempt to purify the toxin without the 10 kDa N-terminal fragment was unsuccessful, because the shortened toxin was unstable.

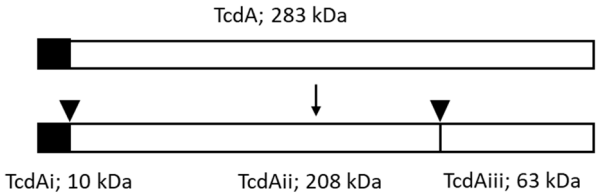


Figure 37: Putative cleavage sites of TcdA*. The first site is found at the N-terminal end, after amino acid 88, which releases TcdAi. The remaining large peptide is cleaved internally to produce the large (TcdAii) and small (TcdAiii) subunits found in the toxin complex. Molecular sizes are calculated from the amino acid sequence of Photorhabdus TcdA1. Filled box, N-terminal TcdAi segment; inverted filled triangles, protease cleavage sites. Figure adapted from* Liu *et al.*, 2003*.*

Treatment of TcdA1 with the *P. luminescence* protease, PrtA1 provided some key insights into the effect of processing. First, the cleaved toxin, which was additionally separated from the protease by gel filtration, showed enhanced toxic activity which was unaffected by MG132. Secondly, the gel-filtrated, but uncleaved toxin, exhibited much lower toxicity, which was further inhibited by MG132. Thirdly, upon gel filtration, the cleaved fragment eluted in one peak with the rest of the protein, suggesting that it remains associated with the toxin and probably acts in concert with it during intoxication. Fourth, when the gel-filtrated, uncleaved toxin, was then cleaved and filtrated again, it exhibited similarly enhanced activity. Fifth, TcdA1 requires processing to achieve full activity. It remains unknown exactly where PrtA1 naturally acts on the toxin (if at all), but it may be either in the bacteria, or after secretion of the toxins (Bowen *et al*., 2003; Silva *et al*., 2002). However, considering that the uncleaved toxin is blocked by MG132, it is likely that the toxin is processed after secretion, that is, by a mammalian protease, which can be blocked by the inhibitor.

The mammalian proteases, trypsin, chymotrypsin and collagenase all caused the release of a 63 kDa fragment (Fig. 12a). However, only treatment by collagenase, a matrix metalloprotease

like PrtA1, resulted in increased toxin activity. MG132 had no effect on cleavage or activity of the PrtA1/collagenase-treated toxin. Additionally, competitive binding between collagenase- and PrtA1-cleaved toxins exhibited similar binding profiles, thus suggesting that cleavage by both enzymes yields similar, but not identical, products. The canonical trypsin cleavage site comes C-terminal to an Arg or Lys residue (Keil, 1971; Olsen *et al*., 2004). However, it has also been shown to cleave non-canonically at sites before a Pro (Perutka and Šebela, 2018; Rodriguez *et al*., 2008). Chymotrypsin preferentially cleaves at Phe, Tyr, Trp, and Leu sites (Schaefer *et al*., 2005). Collagenases, on the other hand, typically cleave native type I, II, and III collagens at distinct hyper-reactive sites along repeating Gly-X-Y motifs (X and Y are different amino acids) (French *et al*., 1992). Various studies using collagen and collagen-like sequences have revealed that, clostridial collagenases (like the one used in this study) prefer to cleave at unique Gly-Leu/Ile bonds (Miller *et al*., 1976; Yu *et al*., 2012). However, they generally cleave at Yaa-Gly bonds in the Gly-X-Y repeats (aa = aromatic amino acid) (Mookhtiar *et al*., 1985; Van Wart, 2013; Van Wart and Steinbrink, 1985). They have also been shown to cleave at Gly-Pro-X and Gly-X-Hyp (hydroxyproline) motifs, which were previously thought to be inaccessible to the enzyme due to the tightly bound conformation of collagen at these points (Eckhard *et al*., 2014). It would therefore be important to analyse the TcdA1 sequence around the cleaved fragments, to assess the presence and form of these, or similar, cleavage sites. However, the key question remains the mechanism underlying toxin activation.

To understand the mechanism of toxin activation, one has to consider the effects thereof; (1) the cleaved toxin exhibited significantly increased binding onto target cells. This contributed to highly increased intoxication kinetics. Notably, the increase in binding was higher in insect cells (4-fold) (Fig. 13c), which are the native targets, than in HeLa cells (2-fold) (Fig. 13b). MG132 had no effect on binding and intoxication. Therefore, processing might increase affinity to the cell surface receptors, and this could lead to increased endocytosis, translocation and ultimately, intoxication. However, (2) the processed toxin was no longer affected by the inhibitors, bafilomycin A1 and dynasore (Fig. 15). Both of these inhibitors impact heavily the endocytotic process, which, as was previously shown (Lang *et al.*, 2010; Gatsogiannis *et al.*, 2013; Meusch *et al.*, 2014), is required for the function of the uncleaved toxin. Therefore, these results suggest that the processed toxin is already active at the cell surface and might not depend on endocytosis and the subsequent pH change, to translocate the enzyme into the cells. However, as was observed with pH 5-dependent injection at the

plasma membrane by PTC3-Mluc7 (Fig. 24), pH still plays a role in injection. This is concurrent with data that showed increased binding of the processed toxin at high pH. Notably, the uncleaved toxin also showed the highest pore-forming activity at pH 11 (Gatsogiannis *et al*., 2013). Therefore, pH definitely has a strong impact on toxin function, even though the precise mechanisms of function may differ between the cleaved and uncleaved TcdA1s. Overall, the cleavage-dependent effects on binding (Fig. 13 and 15) and translocation (Fig. 24 and 25) could explain the fast intoxication kinetics of the processed toxin (Fig. 11 and 12).

Some structural considerations might also explain the mechanism of activation. Intriguingly, the cleaved 63 kDa fragment encompasses the small lobe of the α-helical shell of TcdA1 (amino acids 1762-1972), which was described by Meusch *et al.* (2014). This lobe forms the moving domain of the α-helical shell as it contains a hinge (amino acids 93-105, 1034-1036 and 1903-1930) that is essential for the opening of the shell during injection, that is, the switch from pre-pore to pore state. The proposed cleavage site (between amino acids 1937 and 1938), which was unresolved in the crystal structure (PDB ID 1VW1); is just after this hinge (Fig. 38). Cleavage might therefore provide this domain with increased freedom of movement. Furthermore, this small lobe is directly linked to the entropic spring (amino acids 1973–2015) that is involved in the syringe-like injection of the pore structure into membranes (Gatsogiannis *et al*., 2013, 2016). Therefore, cleavage-related structural changes in this region might strongly influence injection of the channel. Steered molecular dynamics studies suggested that the injection process is exergonic and could happen spontaneously (Meusch *et al*., 2014). In this regard, the cleaved toxin might translocate much faster and more efficiently than the uncleaved toxin upon receptor binding. This process may be (partially) independent of endocytosis and pH change. It would therefore, be of interest to analyse in detail the consequences of proteolytic processing, on the structure-function relationship of TcdA1.

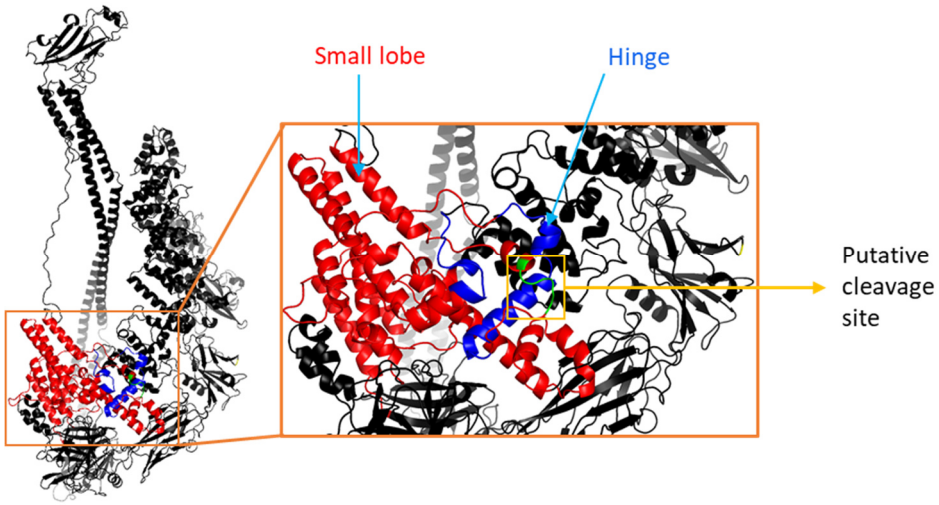


Figure 38: Structure of TcdA1 (PDB ID 1VW1A) showing the small lobe (red), hinge (blue) and putative cleavage site between amino acids 1937 and 1938 (green)*. Figure adapted from Meusch et al., 2014.*

4.2. The PTC3: a molecular delivery tool for foreign peptides

Pore-forming bacterial toxins have been used to deliver recombinant proteins, small membrane impermeable molecules (e.g., nucleotides and ions), as well as antibodies across cell membranes (Aktories and Schmidt, 2015; Barbier and Gillet, 2015). Particularly, the *B. anthracis* protective antigen (PA) and the *C. botulinum* C2 toxin have proven useful in delivering peptides across the cell membrane (Barth *et al.*, 2002; Collier and Young, 2003). However, some toxic bacterial proteins rely on their producing bacteria for delivery through the type-III secretion system (Cornelis, 2000) and indeed some of these bacteria have been genetically manipulated as delivery vectors for vaccination. However, this strategy requires the live bacteria to come into contact with their target cells (Derouazi *et al.*, 2010; Epaulard *et al.*, 2006; Nishikawa, 2006).

The *Photorhabdus* toxin complex is a standalone delivery nanomachine that has the capacity to deliver a variety of peptides into target cells (Table 11), even though most of its cargoes are yet to be characterized. Two enzymes delivered by the complex, the ADP-ribosyltransferases C3hvr and C5hvr, were characterized (Lang *et al.*, 2011). This presented the opportunity to test the possibility of transporting a foreign cargo through this system. The first candidate was C3bot, a small single-domain ADP-ribosyltransferase that modifies and inactivates Rho GTPases (Aktories *et al.*, 1989). C3bot encompasses only a catalytic domain (Aktories *et al.*, 1988). Therefore, it was a suitable test substrate for the PTC. C3bot activity was not only preserved upon fusion to the BC3 cocoon, it was also active *in vivo* after treatment of cells for 6 h with PTC3-C3bot. C3-like proteins were previously delivered into cells through the C2 and

iota toxins (Barth *et al.*, 1998; Marvaud *et al.*, 2002). In both cases, *in vivo* activity was observed in 2-4 h, suggesting a more efficient delivery. Nevertheless, compared to these toxins, the PTC might present an advantage in its enclosed structure, which might shield the cargo from degradation by proteases. However, this has not been proven.

An interesting observation was the reduced *in vitro* activity of the BC3-fusion proteins compared to their wildtype counterparts. This was further elucidated using wildtype BC3, which also showed lower *in vitro* activity compared to C3hvr. The BC3 cocoon was elaborately described by Meusch *et al.* (2014) as an enclosed cage-like structure. Various experiments performed in this study corroborated this. (1) The cocoon appeared to be opened by increasing the temperature to expose the cargo and, (2) the BC3 cocoon shielded the contained cargo from Dylight-488-labelling. This also correlates with previous studies that showed that the cocoon blocks TcdA1 pores in lipid bilayers (Lang *et al*., 2013). It is likely therefore, that the cocoon hinders the enclosed enzyme from interacting with its substrate(s). The observed *in vitro* activity could then be from misfolded BC3 complexes, or partial temperature-dependent unfolding, or exposed enzymes that are, in a yet unknown fashion, associated with the cocoon.

Comparison of C3bot with the *Photorhabdus* hvrs indicated that the size and charge of the cargo might be key to successful delivery. Therefore, to study the possibility of delivering a non-ADP-ribosyltransferase, and simultaneously assess the influence of charge, the *Y. enterocolitica* type-III secretion effector YopT was fused to BC3. YopT is a cysteine protease that localizes to the plasma membrane. Here, it releases mainly RhoA by cleaving its C-terminal isoprenylated tail that links it to the membrane (Shao *et al*., 2003). Additionally, cleaved-RhoA is released from its association with GDI, leading to its proteasomal degradation and subsequently, disruption of stress fibres (Aepfelbacher and Wolters, 2016; Zumbihl *et al*., 1999). In addition to the full protein, two N-terminally truncated fragments, YopT$^{\Delta 30}$ and YopT$^{\Delta 74}$, were also fused to BC3. Furthermore, to assess the influence of charge, YopT$^{\Delta 74}$ which had the lowest pI (6.2), was modified by addition of either three (YopT$^{\Delta 74}$-lys3) or six (YopT$^{\Delta 74}$-lys6) lysines to its C-terminus, thus raising its pI to 8.29 and 8.91 respectively. All these chimeras of YopT showed activity *in vitro* by releasing RhoA from purified HeLa cell membranes, compared to control samples and a heat-inactivated BC3-YopT $^{\Delta 74}$.

The crucial question of delivery of YopT into cells was then answered. HeLa cells were intoxicated and their F-actin stained to observe the inhibition of stress fibre formation. Contrary to previous studies where YopT was either microinjected, or expressed in target cells to yield very rapid cell rounding, in this study, complete cell rounding was not observed. Additionally, disruption of stress fibres and the opening of cell junctions was much slower, especially when only a few molecules were injected. This is possibly because, in our case, YopT was delivered through the endosomes. As was previously shown, the spatial localization of YopT may influence its intracellular substrate specificity, even though it does not show any preference to a particular Rho GTPase *in vitro* (Sorg *et al*., 2001). From the endosomes, YopT is likely more exposed to Rac, Cdc42 and other GTPases, which also show endosomal and Golgi localization, rather than RhoA (Michaelson *et al*., 2001). In *Yersinia*-infected macrophages, RhoA seems to be the preferred substrate of YopT (Aepfelbacher *et al*., 2003). Nevertheless, disruption of stress fibres was observed even in the presence of the Rho GTPase activating toxin CNF1. It was previously shown that CNF1 activity could be overcome by YopT, but not vice versa (Sorg *et al*., 2001).

Interestingly, the most active YopT *in vitro* ($YopT^{\Delta 74}$) showed almost no activity in HeLa and Caco-2 cells. Two possible factors could explain this: (1) $YopT^{\Delta 74}$ was incompletely unfolded, which is a requirement for loading into PTC3 (Meusch *et al*., 2014; Roderer and Raunser, 2019), or, (2) the acidic pI of $YopT^{\Delta 74}$ disfavoured translocation via the TcdA1 pore, which is cation-selective (Lang *et al*., 2013). The latter was confirmed by (a) the successful delivery of $YopT^{\Delta 74}$-lys3 and $YopT^{\Delta 74}$-lys6 into cells, as well as (b) the fact that $YopT^{\Delta 30}$, which had the highest pI (9.04) also had the lowest EC_{50} (8.3). However, a possible exception to this rule might be the *Photorhabdus* C4hvr (Table 11) which also has a low pI (6.23). It would be interesting to study its translocation through the PTC. Additionally, it is worth to note that $YopT^{\Delta 74}$ was delivered by PTC (upon increase of PI) despite lacking the N-terminal amino acids which are involved in the binding of its chaperone SycT (Büttner *et al*., 2005). Interaction with SycT is typically required for efficient translocation through the type-III secretion system (Trülzsch *et al*., 2004).

Finally, having obtained a proof of concept on delivery of foreign peptides through the PTC, an attempt was made to estimate its efficiency by calculating how many toxin complexes bind to a single cell and inject their load. To this end, a small marine luciferase MLuc7 from *M. longa* was fused to BC3. Mluc7 oxidises coelenterazine to produce light and like *Renilla*

luciferase, it does not require ATP as a cofactor. The enzyme is released by *M. longa* "as a bright luminous liquid "bomb" in sea water", in order to blind dark-adapted potential predators and thereby provide cover for the escaping copepod (Markova *et al*., 2019). The luciferase reaction displays flash-type kinetics, with an initial flash followed by quick time-dependent decay of the emitted light. This correlates with the live cell imaging data (Fig. 20b), which showed an initial peak upon addition of the substrate, followed by a continuous signal in cells treated with PTC3-Mluc7. The continuous signal was likely a result of the continuous interaction of intracellular coelenterazine and the injected luciferase.

Reducing conditions in the interior of living cells are not optimal for oxidative luciferase activity. Therefore, the estimated toxin efficiency might be lower compared to an *in vitro* set-up. However, cell lysate per se did not show a major effect on luciferase activity compared to buffer only. Calculation of the number of molecules injected per cell yielded 10^4 molecules at pH 7.5 and 3.1×10^5 at pH 5. This indicated that about 0.5 % of the original toxin concentration was injected at pH 5, while 0.016 % was injected at pH 7.5. Notably, marginal activity was always observed with BC3-Mluc7 alone, suggesting that it may bind unspecifically to the cells. This could not be observed with any of the other BC3 constructs including the wildtype. Therefore, BC3-Mluc7 presents an unprecedented opportunity to gain novel insights into the PTC, in regards to toxin activation, receptor binding and pore formation.

4.3. TcdA1 binds to cell surface N-glycans

Tc toxins are widespread and have shown a tropism for a variety of targets ranging from insect, to mammalian and human cells (ffrench-Constant and Waterfield, 2005; Hinchliffe *et al*., 2010; Sheets and Aktories, 2017). However, their receptors and specific mechanism of uptake into the target cells remains unknown. Additionally, as shown in this study, the complexity of their interaction with target cells increases when one considers the differences between the processed and unprocessed TcAs. These differences are most likely linked to their receptor(s) and can only be fully understood by identifying them. A very strong hint for potential cell surface interactors of TcA was provided by YenTcA, the TcA from the insect pathogen *Yersinia entomophaga* (Landsberg *et al*., 2011). It possesses two chitinase domains in complex with the TcA, which may facilitate toxin entry through the insect midgut via degradation of the peritrophic membrane (Busby *et al*., 2012). It was also shown that YenTcA interacts with glycan structures, which could potentially facilitate cellular binding. Both the TcA and chitinase domains of the toxin interacted with glycans (Piper *et al*., 2019). Similarly, cholera toxin

interacts with GM1 ganglioside and fucosylated Le^X glycans (Cervin *et al*., 2018), *C. difficile* toxin A binds to HSPGs (Tao *et al*., 2019), botulinum neurotoxin A binds to the N-glycosylated synaptic vesicle glycoprotein 2 (SV2) (Yao *et al*., 2016) and *C. botulinum* C2 toxin interacts with N-glycans (Eckhardt *et al.*, 2000).

Like YenTcA, the glycan screen performed in this study also revealed interactions with a range of N-glycan structures including those incorporating fucose (Fuc), galactose (Gal), mannose (Man) and N- aceytylglucosamine (GlcNAc) motifs. Complex and hybrid N-glycan structures have been identified in insects (Aoki *et al*., 2007; Vandenborre *et al*., 2011; Walski *et al*., 2017), which are the primary targets of these toxins, and also in vertebrates (North *et al*., 2010; Schiller *et al*., 2012), to which the toxin has showed activity. The resistance of N-glycan-deficient cells (Lec1) to TcdA1 binding and intoxication confirmed this result (Fig. 28). Even the most efficient version of the toxin, that is, cleaved-toxin at pH 9, did not bind to these cells (Fig. 29), testament to the high dependence on N-glycans for binding. The ADP-ribosylating C3hvr enzyme was able to modify actin in un-intoxicated CHO-K1 control cells and (un-)intoxicated Lec1 cells, but not in previously intoxicated K1 cells. This confirmed that the resistance to intoxication was due to lack of TcdA1 binding and not translocation or enzyme activity per se.

Predominant in the top hits of interacting saccharides were α-1,3-fucosylated Lewis antigens, expecially lewis X/Y ($Le^{X/Y}$) tri- and tetrasaccharides. Lewis antigens are mostly found on human red blood cells (Marcus and Cass, 1969) and are targets for human pathogens like cholera toxin (Cervin *et al*., 2018; Heim *et al*., 2019; Wands *et al*., 2015) and norovirus (Kubota *et al*., 2012). Similarly, Lewis-like structures are present in insects (Stanton *et al*., 2017), however, unlike in vertebrates, α-1,3-fucosylated structures are less abundant than α-1,6- and difucosylated- (α-1,6 and α-1,3) structures (Rendić *et al*., 2006, 2010). This, therefore, suggests that these structures might not be universal targets of TcdA1 on insects, but only on some, like Lepidopterans, which do possess α-1,3-fucosylated Lewis-like structures on glycoproteins (Stanton *et al*., 2017). Notably, Le^X-trisaccharide-dependent inhibition of intoxication was rather weak, in the micromolar to millimolar range, indicating medium-to-low affinity. This suggests that they may be co-receptors, acting co-ordinately or synergistically with other receptors to facilitate binding. The large size of the toxin supports this argument, as it would probably require multiple interactions to gain and maintain stability on the surface for injection. Additionally, whereas fucose and galactose seem important due to their presence

on all the top hits in the screen, they did not have an effect as monosaccharides, neither on fucose- and galactose-deficient cells, nor when added exogenously. They however, reduced toxin activity as part of the Le^X-trisaccharide. Indeed, a similar result was observed for all monosaccharides except sialic acid, suggesting that they probably need to be incorporated into glycan structures to have an effect.

Glycosaminoglycans (GAGs) may also be involved in the binding of TcdA1 to cells. They are a large family of polyanionic glycan structures consisting of hyaluronan, heparin (HP), heparan- (HS), chondroitin- (CS), keratan- (KS), and dermatan- (DS) sulfates (Lindahl *et al*., 2015). Of these, HP and HS are the most complex and have highly variable repeating disaccharide units with HP being more highly sulfated than HS. However, HS chains are typically more heterogeneous than those of HP and are richer in GlcNAc and D-glucuronic acid (GlcA). Furthermore, HP is produced by a limited number of cells, mainly mast and bipotential glial progenitor cells, while HS is made by virtually all types of cells (Lindahl *et al*., 2015). Moreover, HP is O-linked to Serglycin, a proteoglycan core protein on hematopoietic and endothelial cells, while HS is O-linked to a number of different core proteins (Kamhi *et al*., 2013). In this study, HS-deficient HeLa cells showed a marginal resistance to the toxin. Similarly, addition of the sulfated sugars minimally protected cells from intoxication. However, both HP and HS were more efficient in inhibiting intoxication than DexS, while CS showed no inhibition.

Interestingly, YenTcA also showed HS-binding on both TcA and one of the chitinases (Chi2) (Piper *et al*., 2019). Considering the structural similarity of the YenTcA α-helical shell to *P. luminescence* TcdA1, this result suggests that HS, and perhaps other GAGs, may be true interactors. It would therefore be interesting to perform structural studies of the TcdA1-HS interaction. Binding to TcdA1 may be via electrostatic interactions with positive surfaces on the toxin as has been observed with some viruses and bacteria (Kamhi *et al*., 2013). However, like the Le^X antigens, the marginal effects observed indicate that GAGs are also not high-affinity targets. Notably, Lec1 cells showed a 2.4-fold increase in GAG expression when compared to wildtype cells (Fujitani *et al*., 2013), yet they were resistant to TcdA1. Overall, GAGs form long negatively charged polymers on the cell surface, extending 20–150 nm from the plasma membrane, longer than Lewis antigens (Cohen, 2015; Lindahl *et al*., 2015). Therefore, they could act as primary low-affinity receptors that provide a landing pad for the toxin, followed by binding to N-glycans, particularly to Lewis antigens as co-receptors, then perhaps, subsequently to a protein receptor. These fluid interactions may resemble HSPG-

binding viruses, which have been shown to be able to migrate along the cell surface in a phenomenon called 'viral surfing' (Oh *et al*., 2010).

Surprisingly, exogenous addition of sialic acid (SA) to TcdA1 resulted in very strong inhibition of binding and intoxication. However, SA-deficient Lec2 cells were sensitive to PTC3. As with the Fuc-, Gal- and glycolipid-deficient cells, it is possible that any effects resulting from the missing monosaccharides could have been masked by the multivalency of TcdA1. Various studies have indicated that weak (micromolar) individual binding-site interactions are often enhanced by multivalency, resulting in high (nanomolar) affinity binding (Ascencio *et al*., 1993). This may also apply for the synergistic binding to GAGs and Lewis antigens. Importantly, glycan structures on Lec1 cells are much reduced, leaving only $Man_5GlcNAC_2$ core structures and sialylated O-glycans (North *et al*., 2010). Therefore, binding to N-glycans may require, at least, the next level of GlcNAc addition, which is catalysed by the missing Mgat1 gene. Note, after the core, the resultant next level of biantennal structures ($GlcNAC_{1\text{-}4}Man_5GlcNAC_2$) from subsequent N-glycan processing by Mgat1, Golgi class II mannosidase (α-Man-IIa) and Mgat2, represents the structures on Lec8 cells (Kawar *et al*., 2005), which were sensitive to PTC3. These Lec8 structures terminate with GlcNAc (Fig. 39b), while those on Lec1 terminate with mannose (Fig. 39a), indicating the involvement of GlcNAc.

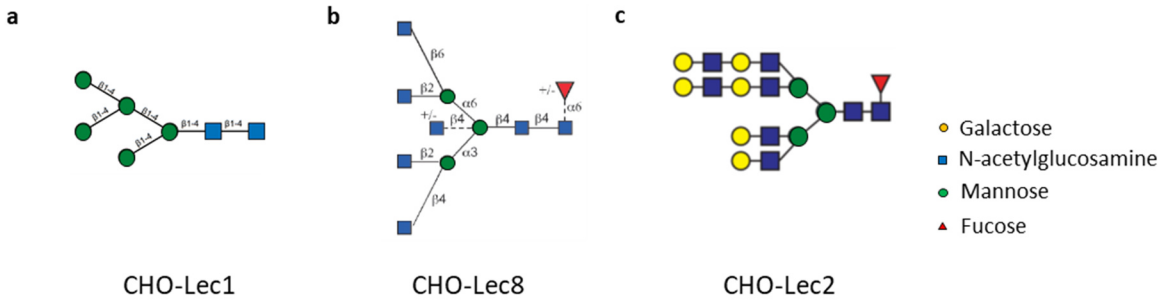


***Figure 39: Examples of N-glycan structures on mutant CHO cells**. (**a**) Lec1. (**b**) Lec8. (**c**) Lec2. Figure Adapted from Kawar et al., 2005 and North et al., 2010.*

Since SAs are mostly terminal in vertebrate glycan structures (Varki, 2007, 2009), the remnant complex N-glycan structures on Lec2 cells still comprise a variety of sugars (North *et al*., 2010) that are enough to facilitate intoxication due to multivalency (Fig. 39c). Interestingly, the glycan structures on Lec2 cells resemble the TcdA1-binding structures obtained from the glycan microarray (compare structures on Fig. 27 and Fig. 39c). The same can be said for the glycolipid-deficient cells, which lack ceramide glucosyltransferase activity and hence no GM3 glycolipids, but their glycoproteins are unaffected, therefore the cells are sensitive (Ichikawa

et al., 1994). The multivalency of TcdA1 could also explain the marginal reduction of toxin binding after removal of cell surface proteins by proteinase-K and pronase, barring inefficiency of the enzymes.

At least five reasons could account for the inhibitory properties of SA on TcdA1; (1) the sugar binds to the receptor-binding domains of the protein and thus inhibits binding to glycan structures on the cell surface. (2) SA binds to the protein via the neuraminidase-like domain and inhibits the opening of the gate for pore-formation. (3) SA binds to the protein resulting in a different structural conformation that disfavours binding. (4) SA binds to the protein and inhibits proper formation of the TcdA1 pentamer, resulting in non-functional aggregates. (5) SA binds to the cell surface, thereby changing the surface architecture, or modulating physiological effects in the cells that result in an unfavourable cell surface architecture. However, all these possibilities require more in-depth studies, including structural data to elucidate the specific toxin-SA interaction, which remains an open question. Nevertheless, SA did not have an effect on the BC3 cocoon, or formation of the PTC3 holotoxin, implying TcdA1-specific effects. Additionally, Zanamivir showed only a marginal effect, suggesting that inhibition was specific to SA. Surprisingly, inhibition of intoxication by SA at low concentration (0.1 mM) was overcome when another sugar, e.g. GlcNAc or Fuc was added. This result indicated competitive binding; however, addition of similar concentrations of GlcNAc or Fuc alone (Fig. 34a), or in combination (Fig. 30f), did not inhibit binding; suggesting that they themselves do not bind to TcdA1 alone and therefore could not compete with SA for binding sites on TcdA1. This leaves, charge-dependent binding of SA to TcdA1, reminiscent of the effect of GAGs. SA is negatively charged and has even been used as a mimic for some GAG functions (Kamhi *et al*., 2013; Nyström *et al*., 2010). However, SA has a slightly higher acidity (pKa = 2.6) (Vimr *et al*., 2004)) than HP (pKa ≈ 3.0) (Wang *et al*., 1991), which could explain its stronger inhibitory effects (compare Fig. 32 and Fig. 34). However, more work is required to characterize these effects further.

5. Suggestions for further research

The data obtained in this study also raises new questions and presents potential areas for further study, which would yield more insights into the *Photorhabdus* toxin complex and other Tc toxins. A few of the key questions to be addressed are presented below.

1. To further analyse the activation of TcdA1 to determine the following:
    - The exact cleavage site and involvement of the cleaved fragment in toxin activity
    - Comparison of TcdA1 cleavage with cleavage in other Tc toxins

These questions could be addressed by in depth sequence analysis, coupled with mutational and structure studies. Mutation of putative cleavage sites could aid their characterization while structure studies (X-ray crystallography and Cryo-electron microscopy (Cryo-EM)) will give proper insight into the consequences of cleavage on the protein complex.

2. To study translocation of other cargoes through the PTC in order to determine other factors that influence success. Particularly, TccC4hvr, which has a low pI. Here, biochemical and cell-biological studies including intoxication assays with/without inhibitors would be important. Additionally, structural studies of the BC cocoon would help elucidate physical features of the cargo that may affect its delivery.

3. To further analyse glycan-binding of TcdA1 to elucidate the following;
    - The exact binding affinities of Le^X and GAGs
    - The specific glycan-binding sites on TcdA1
    - To compare receptor binding of TcdA1 with other Tc toxins

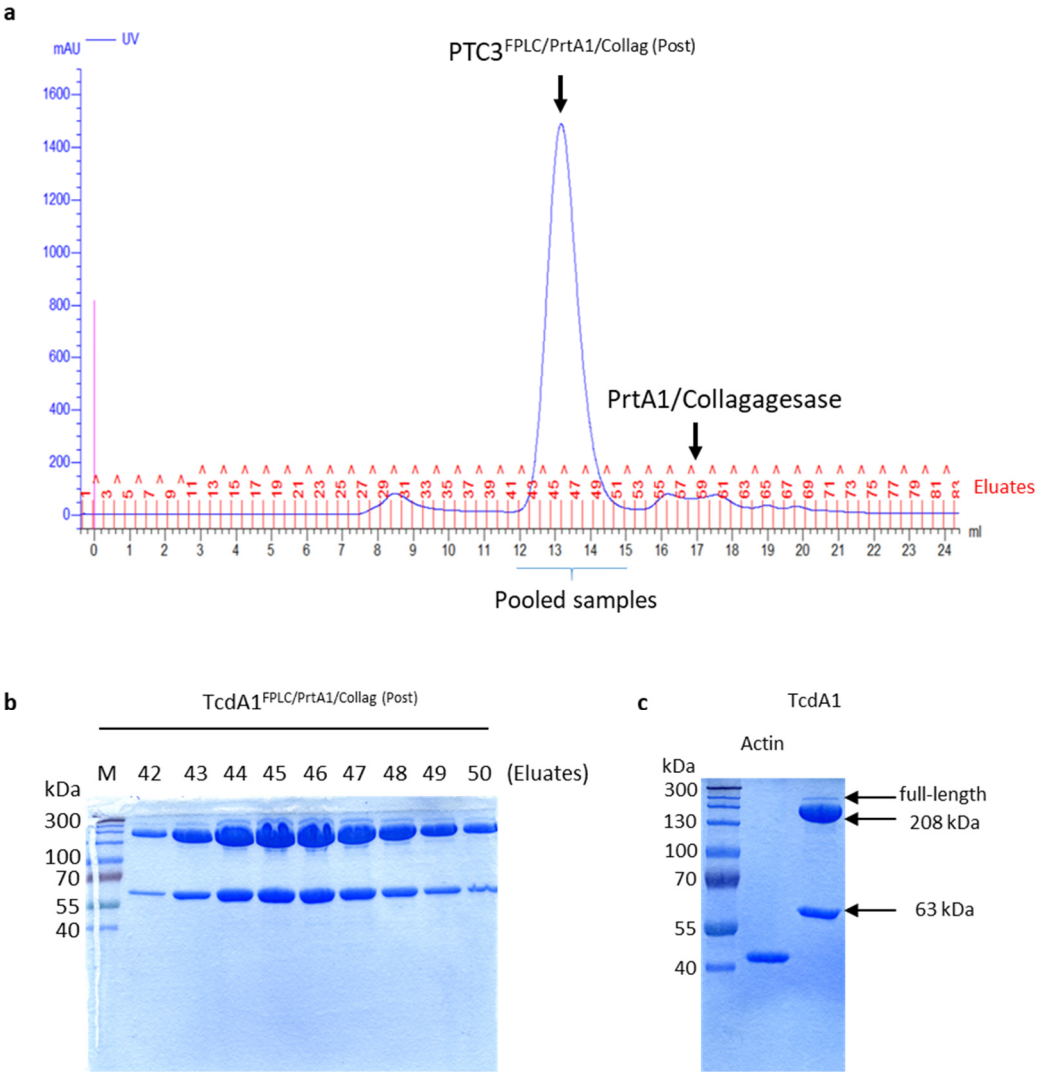
Bio-layer interferometry (BLI) assays could be performed to elucidate the binding affinities of the sugars. Subsequently Cryo-EM studies could be used to resolve the binding sites, study the binding of the cleaved toxin and compare binding by other Tc toxins.

In conclusion, this study showed that TcdA1 can be activated by treatment with metalloproteases. Secondly, activation has some effects including increased cell binding and intoxication, effects that are not inhibited by MG132 as well as inhibitors affecting toxin uptake like bafilomycin A1 and dynasore. Thirdly, the PTC injection machinery can be used to deliver foreign peptides into target cells, as long as they meet the described requirements

(size and charge). Additionally, the luciferase complex could be useful in further characterizing and quantifying toxin effects, as it provides a relatively sensitive and efficient read-out. Finally, TcdA1 interacts with $Le^{X/Y}$ antigens and GAGs, which probably work cooperatively to facilitate TcdA1 binding onto target cells. This binding may be influenced by activation of TcdA1, or vice versa.

6. Appendix

Appendix 1: **Purification of cleaved TcdA1 by fast protein liquid chromatography (FPLC)**. The enzyme-treated TcdA1 and control TcdA1 were further purified by gel filtration to separate the toxin from the enzymes as described in experimental procedures. (**a**) Histogram plot of purified TcdA1 showing the TcdA1 and protease peaks. (**b**) Eluates within the TcdA1 peak that were pooled afterwards. (**c**) Final purified protein.



Appendix 2: List of all the glycans used in the glycan screen (**CFG**, www.functionalglycomics.org). Also shown are their hit numbers according to Fig. 27a and Table 13.

Chart ID	Sample (conc.) Secondary (conc.) Barcode# Slide # Request # Date Initials	Top 10 hits (Table 13)
1	Gala-Sp8	
2	Glca-Sp8	
3	Mana-Sp8	
4	GalNAca-Sp8	
5	GalNAca-Sp15	
6	Fuca-Sp8	
7	Fuca-Sp9	
8	Rhaa-Sp8	
9	Neu5Aca-Sp8	
10	Neu5Aca-Sp11	
11	Neu5Acb-Sp8	
12	Galb-Sp8	
13	Glcb-Sp8	
14	Manb-Sp8	
15	GalNAcb-Sp8	
16	GlcNAcb-Sp0	
17	GlcNAcb-Sp8	
18	GlcN(Gc)b-Sp8	
19	Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-3)GalNAca-Sp8	
20	Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-3)GalNAc-Sp14	
21	GlcNAcb1-6(GlcNAcb1-4)(GlcNAcb1-3)GlcNAc-Sp8	
22	6S(3S)Galb1-4(6S)GlcNAcb-Sp0	
23	6S(3S)Galb1-4GlcNAcb-Sp0	
24	(3S)Galb1-4(Fuca1-3)(6S)Glc-Sp0	
25	(3S)Galb1-4Glcb-Sp8	
26	(3S)Galb1-4(6S)Glcb-Sp0	
27	(3S)Galb1-4(6S)Glcb-Sp8	
28	(3S)Galb1-3(Fuca1-4)GlcNAcb-Sp8	
29	(3S)Galb1-3GalNAca-Sp8	
30	(3S)Galb1-3GlcNAcb-Sp0	
31	(3S)Galb1-3GlcNAcb-Sp8	
32	(3S)Galb1-4(Fuca1-3)GlcNAc-Sp0	
33	(3S)Galb1-4(Fuca1-3)GlcNAc-Sp8	
34	(3S)Galb1-4(6S)GlcNAcb-Sp0	
35	(3S)Galb1-4(6S)GlcNAcb-Sp8	
36	(3S)Galb1-4GlcNAcb-Sp0	
37	(3S)Galb1-4GlcNAcb-Sp8	
38	(3S)Galb-Sp8	
39	(6S)(4S)Galb1-4GlcNAcb-Sp0	
40	(4S)Galb1-4GlcNAcb-Sp8	
41	(6P)Mana-Sp8	
42	(6S)Galb1-4Glcb-Sp0	
43	(6S)Galb1-4Glcb-Sp8	
44	(6S)Galb1-4GlcNAcb-Sp8	
45	(6S)Galb1-4(6S)Glcb-Sp8	
46	Neu5Aca2-3(6S)Galb1-4GlcNAcb-Sp8	
47	(6S)GlcNAcb-Sp8	

48	Neu5,9Ac$_2$a-Sp8	
49	Neu5,9Ac2a2-6Galb1-4GlcNAcb-Sp8	
50	Mana1-6(Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
51	Mana1-6(Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp13	
52	GlcNAcb1-2Mana1-6(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
53	GlcNAcb1-2Mana1-6(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp13	
54	Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
55	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
56	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Man-a1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	
57	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp24	
58	Fuca1-2Galb1-3GalNAcb1-3Gala-Sp9	
59	Fuca1-2Galb1-3GalNAcb1-3Gala1-4Galb1-4Glcb-Sp9	
60	Fuca1-2Galb1-3(Fuca1-4)GlcNAcb-Sp8	
61	Fuca1-2Galb1-3GalNAca-Sp8	
62	Fuca1-2Galb1-3GalNAca-Sp14	
63	Fuca1-2Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glcb-Sp0	
64	Fuca1-2Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glcb-Sp9	
65	Fuca1-2Galb1-3GlcNAcb1-3Galb1-4Glcb-Sp8	
66	Fuca1-2Galb1-3GlcNAcb1-3Galb1-4Glcb-Sp10	
67	Fuca1-2Galb1-3GlcNAcb-Sp0	
68	Fuca1-2Galb1-3GlcNAcb-Sp8	
69	Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	10
70	Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	4
71	Fuca1-2Galb1-4(Fuca1-3)GlcNAcb-Sp0	
72	Fuca1-2Galb1-4(Fuca1-3)GlcNAcb-Sp8	
73	Fuca1-2Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
74	Fuca1-2Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
75	Fuca1-2Galb1-4GlcNAcb-Sp0	
76	Fuca1-2Galb1-4GlcNAcb-Sp8	
77	Fuca1-2Galb1-4Glcb-Sp0	
78	Fuca1-2Galb-Sp8	
79	Fuca1-3GlcNAcb-Sp8	
80	Fuca1-4GlcNAcb-Sp8	
81	Fucb1-3GlcNAcb-Sp8	
82	GalNAca1-3(Fuca1-2)Galb1-3GlcNAcb-Sp0	
83	GalNAca1-3(Fuca1-2)Galb1-4(Fuca1-3)GlcNAcb-Sp0	
84	(3S)Galb1-4(Fuca1-3)Glcb-Sp0	
85	GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb-Sp0	
86	GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb-Sp8	
87	GalNAca1-3(Fuca1-2)Galb1-4Glcb-Sp0	
88	GlcNAcb1-3Galb1-3GalNAca-Sp8	
89	GalNAca1-3(Fuca1-2)Galb-Sp8	
90	GalNAca1-3(Fuca1-2)Galb-Sp18	
91	GalNAca1-3GalNAcb-Sp8	
92	GalNAca1-3Galb-Sp8	
93	GalNAca1-4(Fuca1-2)Galb1-4GlcNAcb-Sp8	
94	GalNAcb1-3GalNAca-Sp8	
95	GalNAcb1-3(Fuca1-2)Galb-Sp8	
96	GalNAcb1-3Gala1-4Galb1-4GlcNAcb-Sp0	
97	GalNAcb1-4(Fuca1-3)GlcNAcb-Sp0	

98	GalNAcb1-4GlcNAcb-Sp0	
99	GalNAcb1-4GlcNAcb-Sp8	
100	Gala1-2Galb-Sp8	
101	Gala1-3(Fuca1-2)Galb1-3GlcNAcb-Sp0	
102	Gala1-3(Fuca1-2)Galb1-3GlcNAcb-Sp8	
103	Gala1-3(Fuca1-2)Galb1-4(Fuca1-3)GlcNAcb-Sp0	
104	Gala1-3(Fuca1-2)Galb1-4(Fuca1-3)GlcNAcb-Sp8	
105	Gala1-3(Fuca1-2)Galb1-4GlcNAc-Sp0	
106	Gala1-3(Fuca1-2)Galb1-4Glcb-Sp0	
107	Gala1-3(Fuca1-2)Galb-Sp8	
108	Gala1-3(Fuca1-2)Galb-Sp18	
109	Gala1-4(Gala1-3)Galb1-4GlcNAcb-Sp8	
110	Gala1-3GalNAca-Sp8	
111	Gala1-3GalNAca-Sp16	
112	Gala1-3GalNAcb-Sp8	
113	Gala1-3Galb1-4(Fuca1-3)GlcNAcb-Sp8	
114	Gala1-3Galb1-3GlcNAcb-Sp0	
115	Gala1-3Galb1-4GlcNAcb-Sp8	
116	Gala1-3Galb1-4Glcb-Sp0	
117	Gala1-3Galb1-4Glc-Sp10	
118	Gala1-3Galb-Sp8	
119	Gala1-4(Fuca1-2)Galb1-4GlcNAcb-Sp8	
120	Gala1-4Galb1-4GlcNAcb-Sp0	
121	Gala1-4Galb1-4GlcNAcb-Sp8	
122	Gala1-4Galb1-4Glcb-Sp0	
123	Gala1-4GlcNAcb-Sp8	
124	Gala1-6Glcb-Sp8	
125	Galb1-2Galb-Sp8	
126	Galb1-3(Fuca1-4)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	
127	Galb1-3GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	
128	Galb1-3(Fuca1-4)GlcNAc-Sp0	
129	Galb1-3(Fuca1-4)GlcNAc-Sp8	
130	Fuca1-4(Galb1-3)GlcNAcb-Sp8	
131	Galb1-4GlcNAcb1-6GalNAca-Sp8	
132	Galb1-4GlcNAcb1-6GalNAc-Sp14	
133	GlcNAcb1-6(Galb1-3)GalNAca-Sp8	
134	GlcNAcb1-6(Galb1-3)GalNAca-Sp14	
135	Neu5Aca2-6(Galb1-3)GalNAca-Sp8	
136	Neu5Aca2-6(Galb1-3)GalNAca-Sp14	
137	Neu5Acb2-6(Galb1-3)GalNAca-Sp8	
138	Neu5Aca2-6(Galb1-3)GlcNAcb1-4Galb1-4Glcb-Sp10	
139	Galb1-3GalNAca-Sp8	
140	Galb1-3GalNAca-Sp14	
141	Galb1-3GalNAca-Sp16	
142	Galb1-3GalNAcb-Sp8	
143	Galb1-3GalNAcb1-3Gala1-4Galb1-4Glcb-Sp0	
144	Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glcb-Sp0	
145	Galb1-3GalNAcb1-4Galb1-4Glcb-Sp8	
146	Galb1-3Galb-Sp8	
147	Galb1-3GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
148	Galb1-3GlcNAcb1-3Galb1-4Glcb-Sp10	
149	Galb1-3GlcNAcb-Sp0	
150	Galb1-3GlcNAcb-Sp8	
151	Galb1-4(Fuca1-3)GlcNAcb-Sp0	
152	Galb1-4(Fuca1-3)GlcNAcb-Sp8	
153	Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	

154	Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	8
155	Galb1-4(6S)Glcb-Sp0	
156	Galb1-4(6S)Glcb-Sp8	
157	Galb1-4GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb-Sp8	
158	Galb1-4GalNAcb1-3(Fuca1-2)Galb1-4GlcNAcb-Sp8	
159	Galb1-4GlcNAcb1-3GalNAca-Sp8	
160	Galb1-4GlcNAcb1-3GalNAc-Sp14	
161	Galb1-4GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	
162	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
163	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
164	Galb1-4GlcNAcb1-3Galb1-4Glcb-Sp0	
165	Galb1-4GlcNAcb1-3Galb1-4Glcb-Sp8	
166	Galb1-4GlcNAcb1-6(Galb1-3)GalNAca-Sp8	
167	Galb1-4GlcNAcb1-6(Galb1-3)GalNAc-Sp14	
168	Galb1-4GlcNAcb-Sp0	
169	Galb1-4GlcNAcb-Sp8	
170	Galb1-4GlcNAcb-Sp23	
171	Galb1-4Glcb-Sp0	
172	Galb1-4Glcb-Sp8	
173	GlcNAca1-3Galb1-4GlcNAcb-Sp8	
174	GlcNAca1-6Galb1-4GlcNAcb-Sp8	
175	GlcNAcb1-2Galb1-3GalNAca-Sp8	
176	GlcNAcb1-6(GlcNAcb1-3)GalNAca-Sp8	
177	GlcNAcb1-6(GlcNAcb1-3)GalNAca-Sp14	
178	GlcNAcb1-6(GlcNAcb1-3)Galb1-4GlcNAcb-Sp8	
179	GlcNAcb1-3GalNAca-Sp8	
180	GlcNAcb1-3GalNAca-Sp14	
181	GlcNAcb1-3Galb-Sp8	
182	GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
183	GlcNAcb1-3Galb1-4GlcNAcb-Sp8	
184	GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
185	GlcNAcb1-3Galb1-4Glcb-Sp0	
186	GlcNAcb1-4-MDPLys	
187	GlcNAcb1-6(GlcNAcb1-4)GalNAca-Sp8	
188	GlcNAcb1-4Galb1-4GlcNAcb-Sp8	
189	GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAcb1-Sp8	
190	GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAcb1-Sp8	
191	GlcNAcb1-4GlcNAcb1-4GlcNAcb-Sp8	
192	GlcNAcb1-6GalNAca-Sp8	
193	GlcNAcb1-6GalNAca-Sp14	
194	GlcNAcb1-6Galb1-4GlcNAcb-Sp8	
195	Glca1-4Glcb-Sp8	
196	Glca1-4Glca-Sp8	
197	Glca1-6Glca1-6Glcb-Sp8	
198	Glcb1-4Glcb-Sp8	
199	Glcb1-6Glcb-Sp8	
200	G-ol-Sp8	
201	GlcAa-Sp8	
202	GlcAb-Sp8	
203	GlcAb1-3Galb-Sp8	
204	GlcAb1-6Galb-Sp8	
205	KDNa2-3Galb1-3GlcNAcb-Sp0	
206	KDNa2-3Galb1-4GlcNAcb-Sp0	
207	Mana1-2Mana1-2Mana1-3Mana-Sp9	
208	Mana1-2Mana1-6(Mana1-2Mana1-3)Mana-Sp9	

209	Mana1-2Mana1-3Mana-Sp9	
210	Mana1-2Mana1-6(Mana1-2Mana1-3)Mana1-6(Mana1-2Mana1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
211	Mana1-6(Mana1-3)Mana-Sp9	
212	Mana1-2Mana1-2Mana1-6(Mana1-3)Mana-Sp9	
213	Mana1-6(Mana1-3)Mana1-6(Mana1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
214	Mana1-6(Mana1-3)Mana1-6(Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
215	Manb1-4GlcNAcb-Sp0	
216	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	
217	(3S)Galb1-4(Fuca1-3)(6S)GlcNAcb-Sp8	
218	Fuca1-2(6S)Galb1-4GlcNAcb-Sp0	
219	Fuca1-2Galb1-4(6S)GlcNAcb-Sp8	
220	Fuca1-2(6S)Galb1-4(6S)Glcb-Sp0	
221	Neu5Aca2-3Galb1-3GalNAca-Sp8	
222	Neu5Aca2-3Galb1-3GalNAca-Sp14	
223	GalNAcb1-4(Neu5Aca2-8Neu5Aca2-8Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp0	
224	GalNAcb1-4(Neu5Aca2-8Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp0	
225	Neu5Aca2-8Neu5Aca2-8Neu5Aca2-3Galb1-4Glcb-Sp0	
226	GalNAcb1-4(Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp0	
227	Neu5Aca2-8Neu5Aca2-8Neu5Aca-Sp8	
228	GalNAcb1-4(Neu5Aca2-3)Galb1-4GlcNAcb-Sp0	
229	GalNAcb1-4(Neu5Aca2-3)Galb1-4GlcNAcb-Sp8	
230	GalNAcb1-4(Neu5Aca2-3)Galb1-4Glcb-Sp0	
231	Neu5Aca2-3Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glcb-Sp0	
232	Neu5Aca2-6(Neu5Aca2-3)GalNAca-Sp8	
233	Neu5Aca2-3GalNAca-Sp8	
234	Neu5Aca2-3GalNAcb1-4GlcNAcb-Sp0	
235	Neu5Aca2-3Galb1-3(6S)GlcNAc-Sp8	
236	Neu5Aca2-3Galb1-3(Fuca1-4)GlcNAcb-Sp8	
237	Neu5Aca2-3Galb1-3(Fuca1-4)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	
238	Neu5Aca2-3Galb1-4(Neu5Aca2-3Galb1-3)GlcNAcb-Sp8	
239	Neu5Aca2-3Galb1-3(6S)GalNAca-Sp8	
240	Neu5Aca2-6(Neu5Aca2-3Galb1-3)GalNAca-Sp8	
241	Neu5Aca2-6(Neu5Aca2-3Galb1-3)GalNAca-Sp14	
242	Neu5Aca2-3Galb-Sp8	
243	Neu5Aca2-3Galb1-3GalNAcb1-3Gala1-4Galb1-4Glcb-Sp0	
244	Neu5Aca2-3Galb1-3GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
245	Fuca1-2(6S)Galb1-4Glcb-Sp0	
246	Neu5Aca2-3Galb1-3GlcNAcb-Sp0	
247	Neu5Aca2-3Galb1-4(6S)GlcNAcb-Sp8	
248	Neu5Aca2-3Galb1-4(Fuca1-3)(6S)GlcNAcb-Sp8	
249	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	
250	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	
251	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb-Sp8	
252	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb-Sp8	
253	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4GlcNAcb-Sp8	
254	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
255	Neu5Aca2-3Galb1-4GlcNAcb-Sp0	
256	Neu5Aca2-3Galb1-4GlcNAcb-Sp8	
257	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
258	Fuca1-2Galb1-4(6S)Glcb-Sp0	
259	Neu5Aca2-3Galb1-4Glcb-Sp0	
260	Neu5Aca2-3Galb1-4Glcb-Sp8	
261	Neu5Aca2-6GalNAca-Sp8	
262	Neu5Aca2-6GalNAcb1-4GlcNAcb-Sp0	

263	Neu5Aca2-6Galb1-4(6S)GlcNAcb-Sp8	
264	Neu5Aca2-6Galb1-4GlcNAcb-Sp0	
265	Neu5Aca2-6Galb1-4GlcNAcb-Sp8	
266	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	
267	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
268	Neu5Aca2-6Galb1-4Glcb-Sp0	
269	Neu5Aca2-6Galb1-4Glcb-Sp8	
270	Neu5Aca2-6Galb-Sp8	
271	Neu5Aca2-8Neu5Aca-Sp8	
272	Neu5Aca2-8Neu5Aca2-3Galb1-4Glcb-Sp0	
273	Galb1-3(Fuca1-4)GlcNAcb1-3Galb1-3(Fuca1-4)GlcNAcb-Sp0	
274	Neu5Acb2-6GalNAca-Sp8	
275	Neu5Acb2-6Galb1-4GlcNAcb-Sp8	
276	Neu5Gca2-3Galb1-3(Fuca1-4)GlcNAcb-Sp0	
277	Neu5Gca2-3Galb1-3GlcNAcb-Sp0	
278	Neu5Gca2-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	
279	Neu5Gca2-3Galb1-4GlcNAcb-Sp0	
280	Neu5Gca2-3Galb1-4Glcb-Sp0	
281	Neu5Gca2-6GalNAca-Sp0	
282	Neu5Gca2-6Galb1-4GlcNAcb-Sp0	
283	Neu5Gca-Sp8	
284	Neu5Aca2-3Galb1-4GlcNAcb1-6(Galb1-3)GalNAca-Sp14	
285	Galb1-3GlcNAcb1-3Galb1-3GlcNAcb-Sp0	
286	Galb1-4(Fuca1-3)(6S)GlcNAcb-Sp0	
287	Galb1-4(Fuca1-3)(6S)Glcb-Sp0	
288	Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-3(Fuca1-4)GlcNAcb-Sp0	
289	Galb1-4GlcNAcb1-3Galb1-3GlcNAcb-Sp0	
290	Neu5Aca2-3Galb1-3GlcNAcb1-3Galb1-3GlcNAcb-Sp0	
291	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-3GlcNAcb-Sp0	
292	4S(3S)Galb1-4GlcNAcb-Sp0	
293	(6S)Galb1-4(6S)GlcNAcb-Sp0	
294	(6P)Glcb-Sp10	
295	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-6(Galb1-3)GalNAca-Sp14	
296	Galb1-3Galb1-4GlcNAcb-Sp8	
297	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
298	Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-3)Galb1-4GlcNAc-Sp0	
299	GlcNAcb1-6(Galb1-4GlcNAcb1-3)Galb1-4GlcNAc-Sp0	
300	Galb1-4GlcNAca1-6Galb1-4GlcNAcb-Sp0	
301	Galb1-4GlcNAcb1-6Galb1-4GlcNAcb-Sp0	
302	GalNAcb1-3Galb-Sp8	
303	GlcAb1-3GlcNAcb-Sp8	
304	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
305	GlcNAcb1-3Man-Sp10	
306	GlcNAcb1-4GlcNAcb-Sp10	
307	GlcNAcb1-4GlcNAcb-Sp12	
308	MurNAcb1-4GlcNAcb-Sp10	
309	Mana1-6Manb-Sp10	
310	Mana1-6(Mana1-3)Mana1-6(Mana1-3)Manb-Sp10	
311	Mana1-2Mana1-6(Mana1-3)Mana1-6(Mana1-2Mana1-2Mana1-3)Mana-Sp9	
312	Mana1-2Mana1-6(Mana1-2Mana1-3)Mana1-6(Mana1-2Mana1-2Mana1-3)Mana-Sp9	
313	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-3)GalNAca-Sp14	

314	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
315	Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
316	Neu5Aca2-8Neu5Acb-Sp17	
317	Neu5Aca2-8Neu5Aca2-8Neu5Acb-Sp8	
318	Neu5Gcb2-6Galb1-4GlcNAc-Sp8	
319	Galb1-3GlcNAcb1-2Mana1-6(Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp19	2
320	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
321	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
322	Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-6(Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp20	
323	Neu5,9Ac2a2-3Galb1-3GlcNAcb-Sp0	
324	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-3GlcNAcb-Sp0	
325	Neu5Aca2-3Galb1-3(Fuca1-4)GlcNAcb1-3Galb1-3(Fuca1-4)GlcNAcb-Sp0	
326	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
327	Gala1-4Galb1-4GlcNAcb1-3Galb1-4Glcb-Sp0	
328	GalNAcb1-3Gala1-4Galb1-4GlcNAcb1-3Galb1-4Glcb-Sp0	
329	GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
330	GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
331	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-6(Neu5Aca2-3Galb1-3)GalNAc-Sp14	
332	GlcNAca1-4Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
333	GlcNAca1-4Galb1-4GlcNAcb-Sp0	
334	GlcNAca1-4Galb1-3GlcNAcb-Sp0	
335	GlcNAca1-4Galb1-4GlcNAcb1-3Galb1-4Glcb-Sp0	
336	GlcNAca1-4Galb1-4GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	
337	GlcNAca1-4Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
338	GlcNAca1-4Galb1-3GalNAc-Sp14	
339	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp12	
340	Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp12	
341	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6Manb1-4GlcNAcb1-4GlcNAc-Sp12	
342	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3Manb1-4GlcNAcb1-4GlcNAc-Sp12	
343	Galb1-4GlcNAcb1-2Mana1-3Manb1-4GlcNAcb1-4GlcNAc-Sp12	
344	Galb1-4GlcNAcb1-2Mana1-6Manb1-4GlcNAcb1-4GlcNAc-Sp12	
345	Mana1-6(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
346	GlcNAcb1-2Mana1-6(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp22	
347	Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp22	
348	Galb1-3GlcNAcb1-2Mana1-6(Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp22	9
349	(6S)GlcNAcb1-3Galb1-4GlcNAcb-Sp0	
350	KDNa2-3Galb1-4(Fuca1-3)GlcNAc-Sp0	
351	KDNa2-6Galb1-4GlcNAc-Sp0	
352	KDNa2-3Galb1-4Glc-Sp0	
353	KDNa2-3Galb1-3GalNAca-Sp14	
354	Fuca1-2Galb1-3GlcNAcb1-2Mana1-6(Fuca1-2Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp20	
355	Fuca1-2Galb1-4GlcNAcb1-2Mana1-6(Fuca1-2Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp20	
356	Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-6(Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAb-Sp20	

357	Gala1-3Galb1-4GlcNAcb1-2Mana1-6(Gala1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp20	
358	Galb1-4GlcNAcb1-2Mana1-6(Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
359	Fuca1-4(Galb1-3)GlcNAcb1-2Mana1-6(Fuca1-4(Galb1-3)GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp22	
360	Neu5Aca2-6GlcNAcb1-4GlcNAc-Sp21	
361	Neu5Aca2-6GlcNAcb1-4GlcNAcb1-4GlcNAc-Sp21	
362	Galb1-4(Fuca1-3)GlcNAcb1-6(Fuca1-2Galb1-4GlcNAcb1-3)Galb1-4Glc-Sp21	
363	Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-4(Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	**3**
364	GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb1-2Mana1-6(GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp20	
365	Gala1-3(Fuca1-2)Galb1-4GlcNAcb1-2Mana1-6(Gala1-3(Fuca1-2)Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp20	
366	Gala1-3Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-6(Gala1-3Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp20	
367	GalNAca1-3(Fuca1-2)Galb1-3GlcNAcb1-2Mana1-6(GalNAca1-3(Fuca1-2)Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp20	
368	Galα1-3(Fucα1-2)Galβ1-3GlcNAcβ1-2Manα1-6(Galα1-3(Fucα1-2)Galβ1-3GlcNAcβ1-2Manα1-3)Manβ1-4GlcNAcβ1-4GlcNAcβ-Sp20	
369	Fuca1-4(Fuca1-2Galb1-3)GlcNAcb1-2Mana1-3(Fuca1-4(Fuca1-2Galb1-3)GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp19	
370	Neu5Aca2-3Galb1-4GlcNAcb1-3GalNAc-Sp14	
371	Neu5Aca2-6Galb1-4GlcNAcb1-3GalNAc-Sp14	
372	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-3GalNAca-Sp14	
373	GalNAcb1-4GlcNAcb1-2Mana1-6(GalNAcb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp12	
374	Galb1-3GalNAca1-3(Fuca1-2)Galb1-4Glc-Sp0	
375	Galb1-3GalNAca1-3(Fuca1-2)Galb1-4GlcNAc-Sp0	
376	Galb1-3GlcNAcb1-3Galb1-4GlcNAcb1-6(Galb1-3GlcNAcb1-3)Galb1-4Glcb-Sp0	
377	Galb1-4(Fuca1-3)GlcNAcb1-6(Galb1-3GlcNAcb1-3)Galb1-4Glc-Sp21	
378	Galb1-4GlcNAcb1-6(Fuca1-4(Fuca1-2Galb1-3)GlcNAcb1-3)Galb1-4Glc-Sp21	
379	Galb1-4(Fuca1-3)GlcNAcb1-6(Fuca1-4(Fuca1-2Galb1-3)GlcNAcb1-3)Galb1-4Glc-Sp21	
380	Galb1-3GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb1-6(Galb1-3GlcNAcb1-3)Galb1-4Glc-Sp21	**7**
381	Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-2)Mana1-6(Galb1-4GlcNAcb1-4(Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	
382	GlcNAcb1-2Mana1-6(GlcNAcb1-4(GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	**1**
383	Fuca1-2Galb1-3GalNAca1-3(Fuca1-2)Galb1-4Glcb-Sp0	
384	Fuca1-2Galb1-3GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb-Sp0	
385	Galb1-3GlcNAcb1-3GalNAca-Sp14	
386	GalNAcb1-4(Neu5Aca2-3)Galb1-4GlcNAcb1-3GalNAca-Sp14	
387	GalNAca1-3(Fuca1-2)Galb1-3GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb-Sp0	
388	Gala1-3Galb1-3GlcNAcb1-2Mana1-6(Gala1-3Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp19	
389	Gala1-3Galb1-3(Fuca1-4)GlcNAcb1-2Mana1-6(Gala1-3Galb1-3(Fuca1-4)GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp19	
390	GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp12	
391	Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp12	
392	Neu5Aca2-3Galb1-3GlcNAcb1-3GalNAca-Sp14	
393	Fuca1-2Galb1-4GlcNAcb1-3GalNAca-Sp14	
394	Galb1-4(Fuca1-3)GlcNAcb1-3GalNAca-Sp14	
395	GalNAca1-3GalNAcb1-3Gala1-4Galb1-4GlcNAcb-Sp0	
396	Gala1-4Galb1-3GlcNAcb1-2Mana1-6(Gala1-4Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp19	

397	Gala1-4Galb1-4GlcNAcb1-2Mana1-6(Gala1-4Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp24	
398	Gala1-3Galb1-4GlcNAcb1-3GalNAca-Sp14	
399	Galb1-3GlcNAcb1-6Galb1-4GlcNAcb-Sp0	
400	Galb1-3GlcNAca1-6Galb1-4GlcNAcb-Sp0	
401	GalNAcb1-3Gala1-6Galb1-4Glcb-Sp8	
402	Gala1-3(Fuca1-2)Galb1-4(Fuca1-3)Glcb-Sp21	
403	Galb1-4GlcNAcb1-6(Neu5Aca2-6Galb1-3GlcNAcb1-3)Galb1-4Glc-Sp21	
404	Galb1-3GalNAcb1-4(Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp0	
405	Neu5Aca2-3Galb1-3GalNAcb1-4(Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp0	
406	Gala1-3(Fuca1-2)Galb1-4GlcNAcb1-3GalNAca-Sp14	
407	GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb1-3GalNAca-Sp14	
408	GalNAca1-3GalNAcb1-3Gala1-4Galb1-4Glcb-Sp0	
409	Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-3GalNAca-Sp14	
410	Gala1-3(Fuca1-2)Galb1-4(Fuca1-3)GlcNAcb1-3GalNAc-Sp14	
411	GalNAca1-3(Fuca1-2)Galb1-4(Fuca1-3)GlcNAcb1-3GalNAc-Sp14	
412	Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-6(Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp22	
413	Fuca1-2Galb1-4GlcNAcb1-2Mana1-6(Fuca1-2Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp22	
414	GlcNAcb1-2(GlcNAcb1-6)Mana1-6(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp19	
415	Fuca1-2Galb1-3GlcNAcb1-3GalNAc-Sp14	
416	Gala1-3(Fuca1-2)Galb1-3GlcNAcb1-3GalNAc-Sp14	
417	GalNAca1-3(Fuca1-2)Galb1-3GlcNAcb1-3GalNAc-Sp14	
418	Gala1-3Galb1-3GlcNAcb1-3GalNAc-Sp14	
419	Fuca1-2Galb1-3GlcNAcb1-2Mana1-6(Fuca1-2Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp22	
420	Gala1-3(Fuca1-2)Galb1-4GlcNAcb1-2Mana1-6(Gala1-3(Fuca1-2)Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp22	
421	Galb1-3GlcNAcb1-6(Galb1-3GlcNAcb1-2)Mana1-6(Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp19	
422	Galb1-4GlcNAcb1-6(Fuca1-2Galb1-3GlcNAcb1-3)Galb1-4Glc-Sp21	
423	Fuca1-3GlcNAcb1-6(Galb1-4GlcNAcb1-3)Galb1-4Glc-Sp21	
424	GlcNAcb1-2Mana1-6(GlcNAcb1-4)(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	
425	GlcNAcb1-2Mana1-6(GlcNAcb1-4)(GlcNAcb1-4(GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	
426	GlcNAcb1-6(GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	
427	GlcNAcb1-6(GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(GlcNAcb1-4(GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	
428	Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-4)(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	
429	Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-4)(Galb1-4GlcNAcb1-4(Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	
430	Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	
431	Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Galb1-4GlcNAcb1-4(Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	
432	Galb1-4Galb-Sp10	
433	Galb1-6Galb-Sp10	
434	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb-Sp8	
435	GalNAcb1-6GalNAcb-Sp8	
436	(6S)Galb1-3GlcNAcb-Sp0	
437	(6S)Galb1-3(6S)GlcNAc-Sp0	

438	Fuca1-2Galb1-4 GlcNAcb1-2Mana1-6(Fuca1-2Galb1-4GlcNAcb1-2(Fuca1-2Galb1-4GlcNAcb1-4)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
439	Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-6(Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-4(Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
440	Galb1-4(Fuca1-3)GlcNAcb1-6GalNAc-Sp14	
441	Galb1-4GlcNAcb1-2Mana-Sp0	
442	Fuca1-2Galb1-4GlcNAcb1-6(Fuca1-2Galb1-4GlcNAcb1-3)GalNAc-Sp14	
443	Gala1-3(Fuca1-2)Galb1-4GlcNAcb1-6(Gala1-3(Fuca1-2)Galb1-4GlcNAcb1-3)GalNAc-Sp14	
444	GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb1-6(GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb1-3)GalNAc-Sp14	
445	Neu5Aca2-8Neu5Aca2-3Galb1-3GalNAcb1-4(Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp0	
446	GalNAcb1-4Galb1-4Glcb-Sp0	
447	GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb1-2Mana1-6(GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp22	
448	Gala1-3(Fuca1-2)Galb1-3GlcNAcb1-2Mana1-6(Gala1-3(Fuca1-2)Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp22	
449	Neu5Aca2-6Galb1-4GlcNAcb1-6(Fuca1-2Galb1-3GlcNAcb1-3)Galb1-4Glc-Sp21	
450	GalNAca1-3(Fuca1-2)Galb1-3GlcNAcb1-2Mana1-6(GalNAca1-3(Fuca1-2)Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp22	
451	Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-2)Mana1-6(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp19	
452	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	
453	Neu5Aca2-3Galb1-4GlcNAcb1-4Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-4GlcNAcb1-4(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	
454	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	
455	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-4GlcNAcb1-4(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	
456	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-4)(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	
457	Neu5Aca2-6Galb1-4GlcNAcb1-4Mana1-6(GlcNAcb1-4)(Neu5Aca2-6Galb1-4GlcNAcb1-4(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	
458	Neu5Aca2-6Galb1-4GlcNAcb1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	
459	Neu5Aca2-6Galb1-4GlcNAcb1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Neu5Aca2-6Galb1-4GlcNAcb1-4(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	
460	Gala1-3(Fuca1-2)Galb1-3GalNAca-Sp8	
461	Gala1-3(Fuca1-2)Galb1-3GalNAcb-Sp8	
462	Glca1-6Glca1-6Glca1-6Glcb-Sp10	
463	Glca1-4Glca1-4Glca1-4Glcb-Sp10	
464	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-4GlcNAcb1-3)GalNAca-Sp14	
465	Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-6(Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	6
466	Fuca1-2Galb1-3(Fuca1-4)GlcNAcb1-2Mana1-6(Fuca1-2Galb1-3(Fuca1-4)GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp19	
467	GlcNAcb1-6(GlcNAcb1-2)Mana1-6(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	
468	Galb1-3GlcNAcb1-2Mana1-6(GlcNAcb1-4)(Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	
469	Neu5Aca2-6Galb1-4GlcNAcb1-6(Galb1-3GlcNAcb1-3)Galb1-4Glcb-Sp21	
470	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana-Sp0	
471	Neu5Aca2-3Galb1-4GlcNAcb1-6GalNAca-Sp14	
472	Neu5Aca2-6Galb1-4GlcNAcb1-6GalNAca-Sp14	

473	Neu5Aca2-6Galb1-4 GlcNAcb1-6(Neu5Aca2-6Galb1-4GlcNAcb1-3)GalNAca-Sp14	
474	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	
475	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	5
476	Mana1-6(Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp19	
477	Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-2)Mana1-6(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	
478	Neu5Aca2-3Galb1-3GlcNAcb1-2Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21	
479	Neu5Aca2-6Galb1-4GlcNAcb1-6(Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-3)Galb1-4Glc-Sp21	
480	Galb1-3GlcNAcb1-6GalNAca-Sp14	
481	Gala1-3Galb1-3GlcNAcb1-6GalNAca-Sp14	
482	Galb1-3(Fuca1-4)GlcNAcb1-6GalNAca-Sp14	
483	Neu5Aca2-3Galb1-3GlcNAcb1-6GalNAca-Sp14	
484	(3S)Galb1-3(Fuca1-4)GlcNAcb-Sp0	
485	Galb1-4(Fuca1-3)GlcNAcb1-6(Neu5Aca2-6(Neu5Aca2-3Galb1-3)GlcNAcb1-3)Galb1-4Glc-Sp21	
486	Fuca1-2Galb1-4GlcNAcb1-6GalNAca-Sp14	
487	Gala1-3Galb1-4GlcNAcb1-6GalNAca-Sp14	
488	Galb1-4(Fuca1-3)GlcNAcb1-2Mana-Sp0	
489	Fuca1-2(6S)Galb1-3GlcNAcb-Sp0	
490	Gala1-3(Fuca1-2)Galb1-4GlcNAcb1-6GalNAca-Sp14	
491	Fuca1-2Galb1-4GlcNAcb1-2Mana-Sp0	
492	Fuca1-2Galb1-3(6S)GlcNAcb-Sp0	
493	Fuca1-2(6S)Galb1-3(6S)GlcNAcb-Sp0	
494	Neu5Aca2-6GalNAcb1-4(6S)GlcNAcb-Sp8	
495	GalNAcb1-4(Fuca1-3)(6S)GlcNAcb-Sp8	
496	(3S)GalNAcb1-4(Fuca1-3)GlcNAcb-Sp8	
497	Fuca1-2Galb1-3GlcNAcb1-6(Fuca1-2Galb1-3GlcNAcb1-3)GalNAca-Sp14	
498	GalNAca1-3(Fuca1-2)Galb1-3GlcNAcb1-6GalNAca-Sp14	
499	GlcNAcb1-6(GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(GlcNAcb1-4(GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAc-Sp21	
500	Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)Galb1-4GlcNAcb1-4(Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAc-Sp21	
501	Galb1-3GlcNAca1-3Galb1-4GlcNAcb-Sp8	
502	Galb1-3(6S)GlcNAcb-Sp8	
503	(6S)(4S)GalNAcb1-4GlcNAc-Sp8	
504	(6S)GalNAcb1-4GlcNAc-Sp8	
505	(3S)GalNAcb1-4(3S)GlcNAc-Sp8	
506	GalNAcb1-4(6S)GlcNAc-Sp8	
507	(3S)GalNAcb1-4GlcNAc-Sp8	
508	(4S)GalNAcb-Sp10	
509	Galb1-4(6P)GlcNAcb-Sp0	
510	(6P)Galb1-4GlcNAcb-SP0	
511	GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb1-6GalNAc-Sp14	
512	Neu5Aca2-6Galb1-4GlcNAcb1-2Man-Sp0	
513	Gala1-3Galb1-4GlcNAcb1-2Mana-Sp0	
514	Gala1-3(Fuca1-2)Galb1-4GlcNAcb1-2Mana-Sp0	
515	GalNAca1-3(Fuca1-2)Galb1-4 GlcNAcb1-2Mana-Sp0	
516	Galb1-3GlcNAcb1-2Mana-Sp0	
517	Gala1-3(Fuca1-2)Galb1-3GlcNAcb1-6GalNAc-Sp14	
518	Neu5Aca2-3Galb1-3GlcNAcb1-2Mana-Sp0	
519	Gala1-3Galb1-3GlcNAcb1-2Mana-Sp0	
520	GalNAcb1-4GlcNAcb1-2Mana-Sp0	
521	Neu5Aca2-3Galb1-3GalNAcb1-4Galb1-4Glcb-Sp0	

522	GlcNAcb1-2 Mana1-6(GlcNAcb1-4)(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAc-Sp21	
523	Galb1-4GlcNAcb1-2 Mana1-6(GlcNAcb1-4)(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAc-Sp21	
524	Galb1-4GlcNAcb1-2 Mana1-6(Galb1-4GlcNAcb1-4)(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAc-Sp21	
525	Fuca1-4(Galb1-3)GlcNAcb1-2 Mana-Sp0	
526	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-2Mana-Sp0	
527	GlcNAcb1-3Galb1-4GlcNAcb1-6(GlcNAcb1-3)Galb1-4GlcNAc-Sp0	
528	GalNAca1-3(Fuca1-2)Galb1-3GalNAcb1-3Gala1-4Galb1-4Glc-Sp21	
529	Gala1-3(Fuca1-2)Galb1-3GalNAcb1-3Gala1-4Galb1-4Glc-Sp21	
530	Galb1-3GalNAcb1-3Gal-Sp21	
531	GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
532	GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp25	
533	Galβ1-4GlcNAcβ1-3Galβ1-4GlcNAcβ1-2Manα1-6(Galβ1-4GlcNAcβ1-3Galβ1-4GlcNAcβ1-2Manα1-3)Manβ1-4GlcNAcβ1-4GlcNAcβ-Sp12	
534	Fuca1-2Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(Fuca1-2Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp24	
535	GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
536	GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp25	
537	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
538	Galb1-3GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(Galb1-3GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp25	
539	Neu5Gca2-8Neu5Gca2-3Galb1-4GlcNAc-Sp0	
540	Neu5Aca2-8Neu5Gca2-3Galb1-4GlcNAc-Sp0	
541	Neu5Gca2-8Neu5Aca2-3Galb1-4GlcNAc-Sp0	
542	Neu5Gca2-8Neu5Gca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAc-Sp0	
543	Neu5Gca2-8Neu5Gca2-6Galb1-4GlcNAc-Sp0	
544	Neu5Aca2-8Neu5Aca2-3Galb1-4GlcNAc-Sp0	
545	GlcNAcb1-3Galb1-4GlcNAcb1-6(GlcNAcb1-3Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-3Galb1-4GlcNAcb1-2Man a1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp24	
546	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2)Mana1-6(Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Mana1-4GlcNAcb1-4GlcNAc-Sp24	
547	Gala1-3Galb1-4GlcNAcb1-2Mana1-6(Gala1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp24	
548	GlcNAcb1-3Galb1-4GlcNAcb1-6(GlcNAcb1-3Galb1-3)GalNAca-Sp14	
549	GalNAcb1-3GlcNAcb-Sp0	
550	GalNAcb1-4GlcNAcb1-3GalNAcb1-4GlcNAcb-Sp0	
551	GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp25	
552	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp25	
553	GlcNAb1-3Galb1-3GalNAc-Sp14	
554	Galb1-3GlcNAcb1-6(Galb1-3)GalNAc-Sp14	
555	(3S)GlcAb1-3Galb1-4GlcNAcb1-3Galb1-4Glc-Sp0	
556	(3S)GlcAb1-3Galb1-4GlcNAcb1-2Mana-Sp0	
557	Galb1-3GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-6(Galb1-3GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAb1-2)Mana1-6(Galb1-3GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	

558	Galb1-3GlcNAcb1-3Galb1-4GlcNAcb1-6(Galb1-3GlcNAcb1-3Galb1-4GlcNAb1-2)Mana1-6(Galb1-3GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	
559	Neu5Aca2-8Neu5Aca2-3Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glc-Sp21	
560	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	
561	GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	
562	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-3Galb1-4GlcNAb1-2)Mana1-6(Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	
563	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAb1-2)Mana1-6(Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	
564	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3GalNAca-Sp14	
565	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-6(Galb1-3)GalNAca-Sp14	
566	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3)GalNAca-Sp14	
567	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3GalNAca-Sp14	
568	GlcNAcb1-3Galb1-4GlcNAcb1-3GalNAca-Sp14	
569	GlcNAcb1-3Galb1-4GlcNAcb1-6(Galb1-3)GalNAca-Sp14	
570	GlcNAcb1-3Galb1-4GlcNAcb1-6(GlcNAcb1-3Galb1-4GlcNAcb1-3)GalNAca-Sp14	
571	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3)GalNAca-Sp14	
572	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3GalNAca-Sp14	
573	GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3GalNAca-Sp14	
574	Galb1-4GlcNAcb1-3Galb1-3GalNAca-Sp14	
575	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-6(Galb1-3)GalNAca-Sp14	
576	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-6(Galb1-3)GalNAca-Sp14	
577	Neu5Aca2-6Galb1-4GlcNAcb1-6(Galb1-3)GalNAca-Sp14	
578	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
579	GlcNAcb1-6(Neu5Aca2-3Galb1-3)GalNAca-Sp14	
580	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-6(Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3)GalNAca-Sp14	
581	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
582	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
583	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	
584	GlcNAcb1-3Fuca-Sp21	
585	Galb1-3GalNAcb1-4(Neu5Aca2-8Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp21	

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